

Quantity One[®]

User Guide for Version 4
Windows and Macintosh

Quantity One User Guide

Bio-Rad Technical Service Department

Phone: (800) 424-6723, option 2, option 3

(510) 741-6576

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com (U.S.)

LSG.TechServ.Intl@Bio-Rad.com (International)

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Preface

1. About This Document

This user guide is designed to be used as a reference in your everyday use of Quantity One®. It provides detailed information about the tools and commands of Quantity One for the Windows and Macintosh platforms. Any platform differences in procedures and commands are noted in the text.

This guide assumes you have a working knowledge of your computer operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

Example	Indicates
File > Open	Choosing the Open command under the File menu.
Dragging	Positioning the cursor on an object and holding down the left mouse button while you move the mouse.
Control+s	Holding down the Control key while typing the letter s.
Right-click / Left-click / Double-click	Clicking the right mouse button / Clicking the left mouse button / Clicking the left mouse button twice.

Some of the illustrations of menus and dialog boxes found in this manual are taken from the Windows version of the software, and some are taken from the Macintosh version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

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2. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of Quantity One are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of Quantity One by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with Quantity One) to make software feature requests.

1. Introduction

1.1. Introduction to Quantity One

Quantity One[®] is a powerful, flexible software package for imaging and analyzing 1-D electrophoresis gels, dot blots, slot blots, and colonies.

The software runs in a Windows or Macintosh environment and has an easy-to-use graphical interface with standard pull-down menus, toolbars, and keyboard commands.

Quantity One can quantitate and analyze a wide variety of biological data, including radioactive, chemiluminescent, fluorescent, and color-stained samples acquired from densitometers, phosphor imagers, fluorescent imagers, and gel documentation systems.

An image of a sample is captured using the controls in the imaging device window and displayed on your computer monitor. Image processing and analysis operations are performed using commands from the menus and toolbars.

Images can be magnified, annotated, rotated, and resized. They can be printed using standard and video printers.

The lane-based functions can be used to calculate molecular weights, isoelectric points, VNTRs, Rf values, and other values. The software can measure total and average quantities, determine relative and actual amounts of protein, and count colonies in a Petri dish.

The software can cope with distortions in the shape of lanes and bands. Lanes can be adjusted along their lengths to compensate for any curvature or smiling of gels.

Irregularly shaped, non-lane-based data can be accurately quantitated using the Volume tools.

Data captured by the imaging device are stored in a file under a user-defined name. Files can be shared among all the Discovery Series[™] software, or

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images can be easily converted into TIFF format for compatibility with other Macintosh and Windows applications.

1.2. Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by Quantity One are light and / or radiation detectors that convert signals from biological samples into digital data. Quantity One then displays the digital data on your computer screen, in the form of gray scale or color images.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

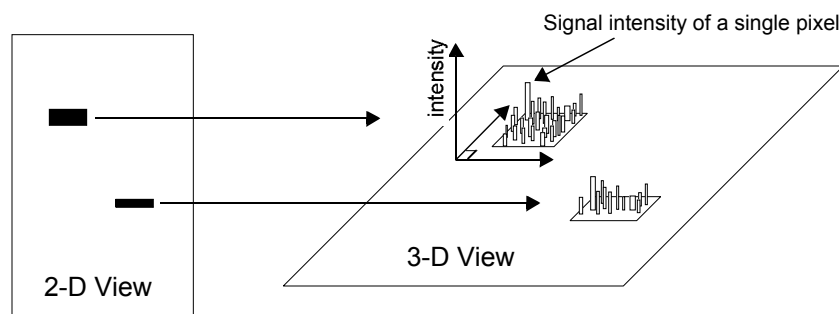


Fig.1-1. Representation of the pixels in two digitally imaged bands in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700 and 710 densitometers, the Gel Doc with a white light source, or the

Introduction

Fluor-S and Fluor-S MAX MultiImagers with white light illumination. Signal intensity is expressed in counts when using the Personal FX or FX, or in the case of the Gel Doc, Fluor-S, or Fluor-S MAX when using the UV light source.

1.3. Gel Quality

Quantity One is very tolerant of an assortment of electrophoretic artifacts. Lanes need not be perfectly straight or parallel. Bands need not be perfectly resolved.

However, for accurate lane-based quantitation, we suggest that bands be reasonably flat and horizontal. Lane-based quantitation involves calculating the average intensity of pixels across the band width and integrating over the band height. For the automatic band finder to function optimally, bands should be well-resolved.

Dots that appear as halos, rings, or craters, or that are of unequal diameter, may be incorrectly quantified using the automatic functions.

1.4. Steps Involved in Quantity One

The following steps are involved in using Quantity One.

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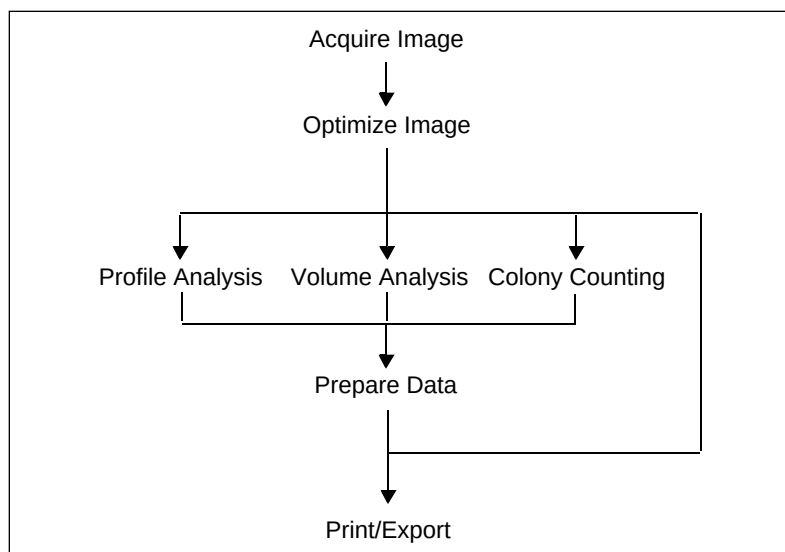


Fig.1-2. Steps Involved in Using Quantity One.

Acquire Image

Before you can use Quantity One to analyze a biological image, you need to capture the image into your computer. This may be done with one of the several Bio-Rad instruments supported by this software: the Molecular Imager FX and Personal Molecular Imager FX; the GS-700 and GS-710 Imaging Densitometers; the Gel Doc 1000/2000 gel documentation system; and the Fluor-S and Fluor-S MAX MultiImagers. The resulting images are stored in files on your hard disk, network file server, or removable storage media.

Optimize Image

Once you have acquired an image of your sample, you may need to reduce any noise or background density caused by film fogging or the opacity of your carrier medium. A variety of functions exist to minimize background noise while maintaining data integrity.

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Analyze Image

Once a “clean” image is available, you can use Quantity One to gather and analyze your data. In the case of 1-D gels, the software has tools for identifying lanes and defining, quantitating, and matching bands. Other tools allow you to measure and compare the quantities of individual bands or count the number of colonies in a Petri dish.

Qualitative and quantitative data can be displayed in tabular and graphical formats.

Print/Export

When your imaging and /or analysis is complete, you can print out your results in the form of simple images, images with overlays, reports, tables, and graphs. You can export your images and data to other systems for further analysis.

1.5. Computer Requirements for Running Quantity One

Quantity One will run on a PC as a Windows 95/98/NT 4.0 32-bit application or on a Macintosh as a Macintosh PowerPC application.

The amount of computer memory required for using Quantity One is mainly determined by the size of the images you will scan and analyze. Some images—particularly those scanned at high resolution—can be quite large, and your hard disk may quickly run out of space. For this reason, we recommend that you archive images on a network file server or some kind of removable storage media after your analysis.

PC

The following are the **minimum** system requirements for installing and running Quantity One on a PC:

Operating system: Windows 95/98/NT 4.0.

Processor: Pentium 166.

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RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or a network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Adaptec SCSI card with EZ-SCSI software.
Printer:	Optional.

Macintosh

Note: The default amount of memory assigned to Quantity One on the Macintosh is 64 MB. If the total RAM in your Macintosh is 64 MB or less (the minimum recommended amount is 128 MB), you should reduce the amount of memory assigned to Quantity One to 10 MB less than your total RAM. With the Quantity One application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

The following are the **minimum** system requirements for installing and running Quantity One on a Macintosh:

Operating system:	System 7.2 or higher.
Processor / Model:	PowerPC Macintosh 9500.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Macintosh has built-in SCSI.
Printer:	Optional.

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1.6. Installing Quantity One

1.6.a. Installing Quantity One on a PC

Before attempting to install and use Quantity One on a PC, you should be familiar with your Windows 95/98 or Windows NT 4.0 operating system.

Quantity One can be installed from a CD-ROM or you can download the installation program from the Internet. Make sure that Windows is up and running on your computer before attempting to install Quantity One. We recommend that you check your hard disk with a disk utility program (such as Norton Disk Doctor) before loading Quantity One.

Installing Quantity One from a CD-ROM

Insert the Discovery Series™ CD-ROM into the CD-ROM drive on your computer.

Downloading from the Internet

You can download Quantity One from the Internet using Bio-Rad's FTP site. Go to <ftp://ftp.genetics.bio-rad.com/Public/DiscoverySeries/Windows/QuantityOne> and click on Quantity One.exe. This will download the installer onto your computer. Then you can double-click on the Setup.exe icon on your desktop to begin running the installer.



Fig.1-3. Setup.exe icon (Windows).

The Installation Program

The Quantity One installation program will guide you through a series of screens. The installer will create a default directory tree under Program Files on your hard disk called Bio-Rad/The Discovery Series/Bin (you can select your own directory if you wish). The main Quantity One program will be

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placed in the Bin directory. Additional directories for storing user profiles and sample images will also be created

The installer will place a Quantity One icon on your desktop and create a Discovery Series folder in the Programs folder on your Windows Start menu.

Finally, after installation is complete, the installer will ask you if you want to open Quantity One.

1.6.b. Installing Quantity One on a Macintosh

Quantity One can be installed on a Macintosh from a CD-ROM or you can download the installation program from the Internet.

Installing Quantity One from a CD-ROM

Insert the Discovery Series CD-ROM into the CD-ROM drive on your Macintosh.



Fig.1-4. Quantity One installation program icon (Macintosh).

Downloading the Installation Program from the Internet

You can download Quantity One from the Internet using Bio-Rad's FTP site. Go to [ftp://ftp.genetics.bio-rad.com/Public/DiscoverySeries/Macintosh/Quantity One](ftp://ftp.genetics.bio-rad.com/Public/DiscoverySeries/Macintosh/QuantityOne) and click on Install Q1. This will download the installer and place it on your desktop. Then you can double-click on the installer icon to begin running the installer.

The Installation Program

The Quantity One installer will guide you through a series of screens. Once installation is complete, the Quantity One folder will appear on your desktop.

Introduction

1.7. Starting Quantity One

A hardware protection key (HPK) has been provided with your Quantity One software.

Most computers will not need an HPK to run Quantity One. However, if you attempt to start Quantity One and receive an “Unable to obtain authorization” message, turn off your PC or Macintosh, attach the HPK, and restart the software.

Note: Some parallel port devices such as zip drives may be incompatible with HPKs. Please check with your peripherals vendor.

1.7.a. Attaching the Hardware Protection Key

Before attaching the HPK, you should first attempt to start Quantity One without the HPK. If Quantity One opens successfully, you can skip this section.

PC

The code for the PC HPK is EYYCY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your computer. If you have a printer attached to your computer's parallel port, please turn that off as well.

The HPK attaches to the parallel port on the back of your PC. If a printer cable is attached to this port, disconnect it. After you have attached the HPK, you can attach the printer cable to the key itself and restart your computer and printer.

You will also need to install the system driver that allows your Quantity One software to “read” the HPK.

Note: Windows NT users must be in the local administrator group to install the HPK driver.

To install the driver, click on your Windows Start menu and select Programs > The Discovery Series. (This directory will not exist until you run the Quantity

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One installer.) Select Install HASP HPK driver to begin installation. After the driver is installed, a message box will appear confirming the installation.

Macintosh

The code for the Macintosh HPK is QCDIY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your Macintosh.

The Macintosh HPK must be inserted in the Apple Desktop Bus (ADB) path. The ADB is located on the back of your Macintosh.



Fig.1-5. Apple Desktop Bus icon on back of Macintosh.

The HPK can be inserted at any point in the ADB path—between the computer and the keyboard, between the keyboard and the mouse, between the keyboard and the monitor, etc. After you have attached the HPK, you can restart your computer.

1.7.b. Opening Quantity One

PC

The installation program creates a Quantity One application icon on your desktop. To start the program, double-click on this icon.



Fig.1-6. Quantity One application icon.

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The installer also creates a Discovery Series folder in the Programs folder on your Windows Start menu. You can start Quantity One by double-clicking on the icon in this folder.

Macintosh

After installation, the Quantity One folder will be open on your desktop. To start the program, double-click on the application icon inside the folder.

1.8. Registering Quantity One

When you open Quantity One, the Main Registration Screen will pop up welcoming you to Quantity One.

Note: If you receive a message saying “Unable to obtain authorization,” attach the HPK if you haven’t already (refer to the previous section). If you have already attached your HPK, make sure that it is attached correctly and (if you have a PC) your HPK driver is installed. If you continue to have problems, contact Bio-Rad technical service at the numbers listed in section 1.9., Contacting Bio-Rad, below.

The type of screen you see will depend on whether you have an HPK attached or not.

Without HPK

If you do not have an HPK attached, you can register to receive a 30-day free trial of the software.

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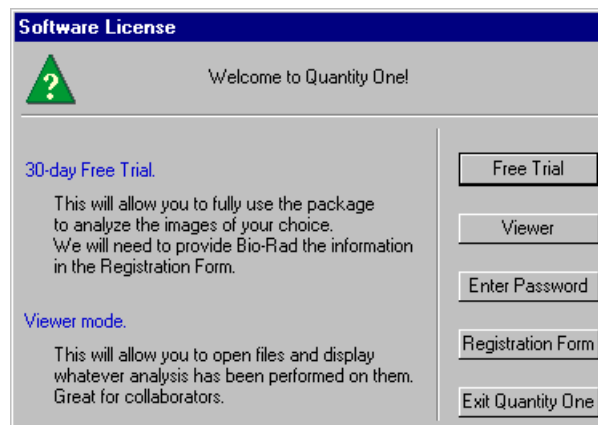


Fig.1-7. Main Registration Screen without HPK

Click on the Free Trial button. This will open the Software License Registration Form.

With HPK

If you have an HPK attached, the Main Registration Screen will reflect the fact that you receive a 30-day temporary license with your HPK ("Your license will expire on _____").

Introduction

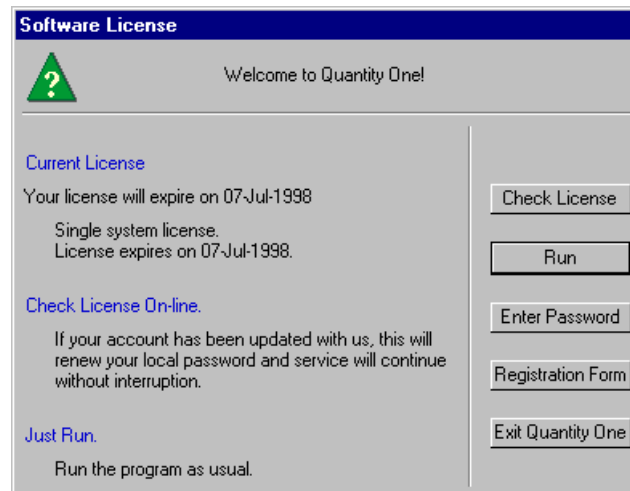


Fig.1-8. Main Registration Screen with HPK.

Click on the Run button to begin using the software.

Some time during the 30-day trial period, click on the Registration Form button to register your software. If your 30-day period has expired, a Free Trial button will appear when you open the software. Click on this button to register your software.

This will open the Software License Registration Form.

1.8.a. Software License Registration Form

To obtain your software license, you will need to fill out in the information in the Software License Registration Form.

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The screenshot shows a window titled "Software License Registration Form". Inside, there is instructional text at the top: "Please fill out the form below to authorize full use of your software." followed by instructions for users with and without Internet access. Below this is a numbered list of three options for registration: faxing, calling, or emailing. The form itself is divided into several sections with input fields. The "Software User" section includes fields for "Dr./Mr./Ms.", "First Name", "Last Name", "Phone #", "Fax #", and "E-mail". The "Company or Institution" section includes fields for "Company/Institution", "Principal Investigator", "Division/Department", "Street Address", "City", "State or Province", and "Postal Code". The "Purchase Information" section includes fields for "Bio-Rad Representative", "Purchase Order Number", "Software Serial Number", and a dropdown for "What imaging device will you use?" (currently showing "Gel Doc 2000"). The "Software and System" section is a read-only area showing "Software: Quantity One", "Version: 4.0", "Operating System: Windows NT", and "System ID: 4SVXQLCHB9". At the bottom are three buttons: "Submit via Internet", "Print", and "Close".

Software License Registration Form	
Please fill out the form below to authorize full use of your software. If you have Internet access, please send the form directly to Bio-Rad by clicking on the Submit via Internet button. This will immediately update your registration. If you do not have Internet access, you can print out this form and send it to Bio-Rad. NOTE: There will be a delay while we update your registration. You can either: 1. Fax the form to Bio-Rad at 1-510-741-5801. We will then e-mail or fax you your password. 2. Call 1-800-424-6723, option 1 (inside U.S.) or 1-510-741-6996 (Intl.) and ask for Software Registration. 3. E-mail the contents of this form to LSG.Software.Registration@Bio-Rad.com	
Software User	Company or Institution
Dr./Mr./Ms.	Company/Institution
First Name	Principal Investigator
Last Name	Division/Department
Phone #	Street Address
Fax #	
E-mail	City State or Province
	Postal Code Country
Purchase Information	Software and System
Bio-Rad Representative	Software: Quantity One
Purchase Order Number	Version: 4.0
Software Serial Number	Operating System: Windows NT
What imaging device will you use? Gel Doc 2000	System ID: 4SVXQLCHB9
Submit via Internet Print Close	

Fig.1-9. Software License Registration Form.

Note: If you do not yet have a Purchase Order Number or Software Serial Number, you may leave these fields blank to receive a trial license.

Submitting the Registration Form by Internet

If you have Internet access on your computer (the same computer on which you loaded Quantity One), you can register quickly and easily.

In the Software License Registration Form, click on the Submit via Internet button.

Your information will be submitted automatically over the Internet, and a password will be generated automatically and sent back to your computer. Simply continue to run Quantity One as before.

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This password will be good for 30 days. During this period, if you have already submitted your Software Serial Number, click on Check License in the Main Registration Screen to update your license. (To access the Main Registration Screen from within Quantity One, select Help > Register.)

If you have not yet submitted your Software Serial Number, open the form again, enter the serial number, and resubmit it over the Internet. In 1–2 days, click on the Check License button to update your license.

Submitting the Registration Form by Fax or E-mail

If you do not have Internet access, click on the Print button in the Software License Registration Form and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the Registration Form.

If you include your Software Serial Number in the form, Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

If you do not include your Software Serial Number in the form, Bio-Rad will contact you in 2–3 days with a temporary 30-day license. Then, some time during the 30-day period, enter the serial number in the form and fax the revised form to Bio-Rad. Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

1.8.b. Entering a Password

If you submit your registration information over the Internet, you do not need to enter a password; it is done automatically.

If you fax your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on Enter Password in the Main Registration Screen. If you are not currently in the Main Registration Screen, select Register from the Help menu in Quantity One.

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Enter Password

To register, please contact Bio-Rad during US/Pacific business hours:
Phone: (800) 424-6723 opt 2 opt 3 (in the U.S.)
(510) 741-6576
Fax: (510) 741-5082
e-mail: LSG_TechServ_US@Bio-Rad.com (in the U.S.)
LSG_TechServ_Intl@Bio-Rad.com (International)

Please provide the information in the completed registration form.

Current license
No license.

System ID
357316D0

Password
Quantity One OK Not OK

Enter Cancel

Fig.1-10. Enter Password screen.

In the Enter Password screen, type in your password next to the Quantity One prompt.

Once you have typed in the correct password, the OK light next to the password field will change to green and the Enter button will activate. Click on Enter to run Quantity One.

1.9. Contacting Bio-Rad

If you have problems with the HPK / password system, please contact Bio-Rad technical service. Technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone: (800) 424-6723, option 2, option 3
(510) 741-6576

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com (in the U.S.)
LSG.TechServ.Intl@Bio-Rad.com (International)

For software registration, phone:

(800) 424-6723, option 1, or (510) 741-6996

2. General Operation

2.1. The Graphical Interface

2.1.a. Menu Bar

Quantity One has a standard menu bar with eleven pulldown menus that contain all the major features and functions available in the software.

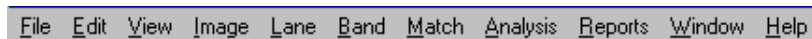


Fig.2-1. Menu bar.

The pulldown menus contain groups of related functions:

- File—General file commands (Open, Save, Print), imaging device acquisition windows (Gel Doc, GS-700, GS-710, Fluor-S, Fluor-S MAX, Personal FX, FX).
- Edit—Annotation tools, preferences, miscellaneous editing tools.
- View—General viewing tools (Zoom Box, Grab), quick analysis tools (Density in Box, Plot Density), Transform function.
- Image—Image processing tools (Crop, Flip, Subtract Background).
- Lane—Lane-finding tools, array tools.
- Band—Band-finding tools, contour tools.
- Match—Standards, band matching tools.
- Analysis—Volume tools, colony counting, miscellaneous analysis functions (Differential Display, VNTR).
- Reports—Analysis reports (Lanes, Matches, Volumes, VNTR), Phylogenetic Tree, Similarity Matrix.
- Window—Tile windows commands, imitate zoom.
- Help—Quick Guides, on-line Help, software registration.

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Below the menu bar is the main toolbar, containing some of the most commonly-used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

2.1.b. Main Toolbar

The main toolbar in Quantity One appears below the menu. It includes buttons for the main file commands (Open, Save, Print) and essential viewing tools (Zoom Box, Grab, etc.), as well as buttons for opening the secondary toolbars and the most useful Quick Guides (Printing, Volumes, Molecular Weight, and Colony Counting).

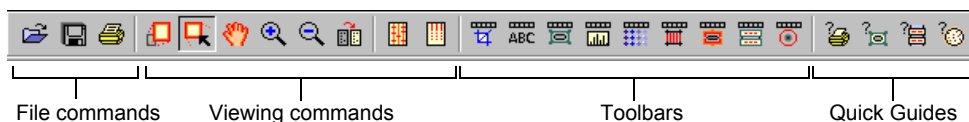


Fig.2-2. Main toolbar.

Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the Preferences dialog box under Edit > Preferences. You can also specify how long the Tool Help will remain displayed.

2.1.c. Status Boxes

There are two status boxes in Quantity One. These appear to the right of the main toolbar.

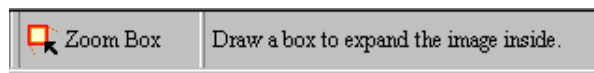


Fig.2-3. Status boxes.

General Operation

The first box displays any function that is assigned to the mouse (see section 2.2., Mouse-assignable Tools). If you select a command such as Zoom Box, the name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a button, a short explanation about that command will be displayed in this second status box.

2.1.d. Secondary Toolbars

Quantity One has nine secondary toolbars, which contain icons for groups of associated functions. You can open these toolbars from the main toolbar or from the View > Toolbars pulldown menu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

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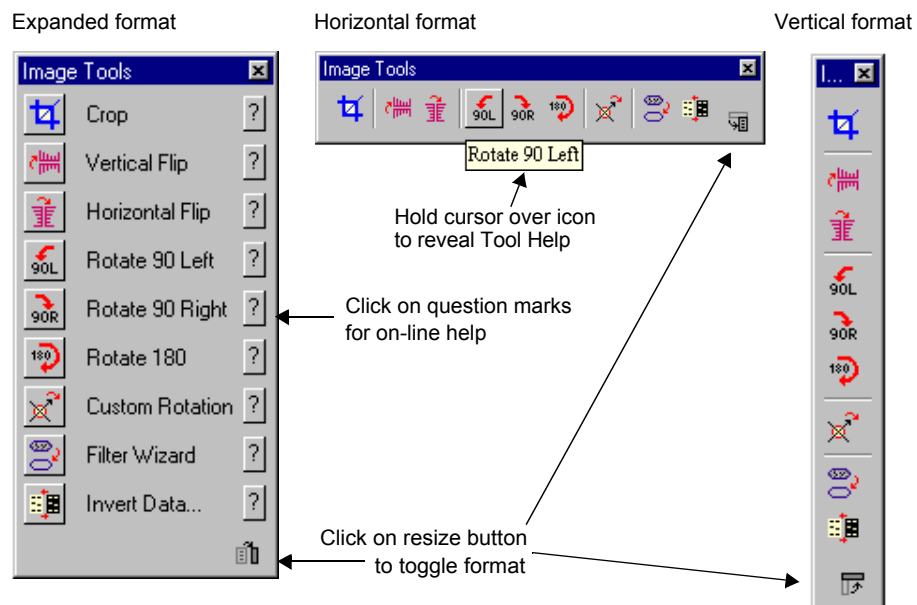


Fig.2-4. Secondary toolbar formats and features.

The expanded toolbar format shows the name of each of the commands. Click on the “?” icon next to the name to display on-line Help for that command.

2.1.e. Quick Guides

Quantity One has seven Quick Guides to guide you through the major applications of the software. The Quick Guides are listed under the Help menu; four of these are also available on the main toolbar.

General Operation

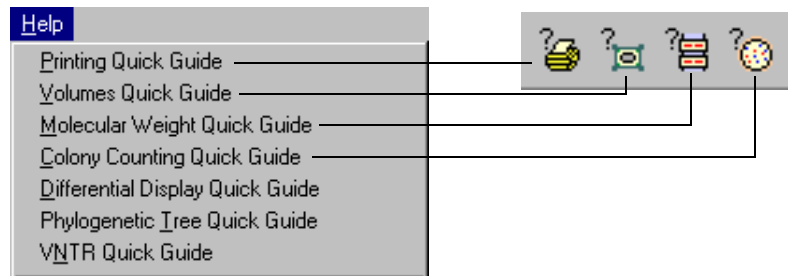


Fig.2-5. Quick Guides listed on Help menu and main toolbar.

The Quick Guides are similar in appearance to the secondary toolbars, but are application-specific.

Each Quick Guide contains all of the functions necessary for a particular application, from opening the image to outputting data.

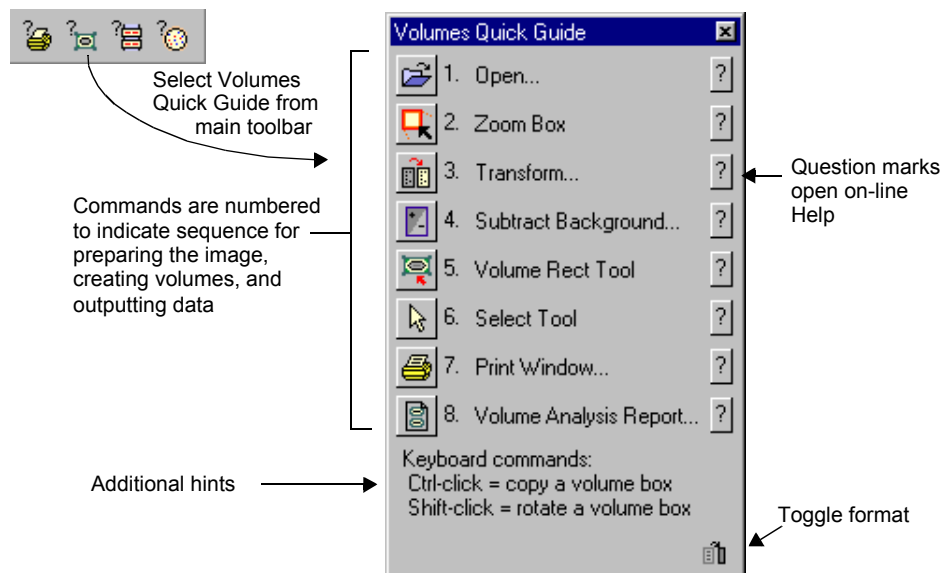


Fig.2-6. Example of a Quick Guide: Volumes

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Note: You will find that as you become more familiar with the application, you will skip operations that are not needed for your particular images.

In their expanded format, the Quick Guide commands are numbered as well as named, so that the order of operation is clear. Simply follow the steps and the Quick Guide will lead you through the analysis.

As with the secondary toolbars, you can click on the “?” next to the name of a function to display the Help text.

2.2. Mouse-assignable Tools

Quantity One has a number of commands that don’t perform actions right away, but instead “assign” a function to your mouse (e.g., Zoom Box, Density at Cursor, Lane Background). You first select one of these tools from a menu or toolbar, then execute the command by clicking or dragging on the image.

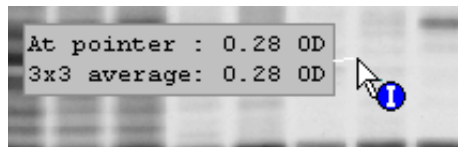


Fig.2-7. Example of a mouse-assignable tool: Density at Cursor

When a “mouse-assignable” function is selected, the cursor appearance changes. The name and icon of the function appear in the status box next to the main toolbar (see section 2.1.c).

Note: To *deassign* a function from the mouse, click on the toolbar button of the assigned function or click in the status box displaying the assigned function. You can also double-click on the Hide Overlays button.

2.3. Keyboard Shortcuts

Many of the functions in Quantity One may be executed using keyboard shortcuts. These are listed in the Help menu under Keyboard Layout, as well as in Appendix A of this manual.

General Operation

The pulldown menus also list the shortcut keys for the menu commands.

2.4. File Commands and Functions

This section describes the basic file commands and functions of Quantity One. These are selected from the File menu.



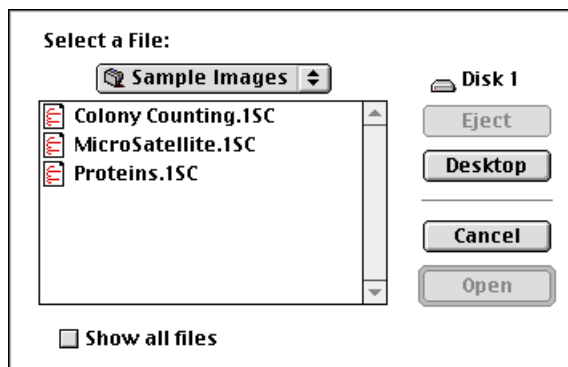
Fig.2-8. File menu.

Open

To open a previously saved image, select Open from the File menu or main toolbar. This opens the Open dialog box.

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Macintosh version:



Windows version:

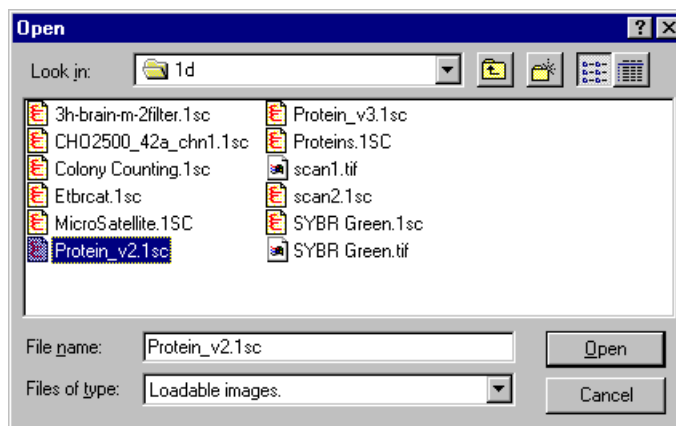


Fig.2-9. Open dialog box.

In the dialog box, open a file by selecting the file name and clicking on the Open button. To look for files in other directories, use the directory pulldown menu at the top of the dialog box.

An image created in the Windows version of Quantity One can be opened in the Macintosh version, and visa versa. However, you must add a .1sc extension to your Macintosh files to open them in Windows.

Note: This version of Quantity One will open any image created with an earlier version of Quantity One.

General Operation

Quantity One comes with a selection of sample images. In Windows, these may be found in the Discovery Series / Sample Images / 1D directory. On the Macintosh, they are stored in the Sample Images folder in the Quantity One folder.

Opening TIFF Images

The Open command can also be used to import TIFF images from other software applications.

There are many types of TIFF formats that exist on the market. Not all are supported by the Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with the Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard™, Microtek™, and Sharp™ each make scanners that produce compatible 8-bit grayscale images.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager (storage phosphor) systems use 16-bit pixel values to describe intensity of scale. Molecular Dynamics™ imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

Note: Quantity One can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.
2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in the Discovery Series, although most scanners that are capable of producing

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24-bit and indexed color images will be able to produce grayscale scans as well.

3. **Compressed Files.** The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with the Discovery Series should be formatted with the compression option turned off.

Close

To close an image or scan, select File > Close. If the file has changed since it was last saved, a message box will ask you if you want to save the changes before closing.

Close All

File > Close All closes all open images. If you have made changes to any of the images, a message box will open for each unsaved image giving you the option of saving those changes.

Save

File > Save will save a new image or an old image to which you have made changes.

In Windows, new Quantity One images will be given a .1sc extension when they are first saved.

Save As

File > Save As can be used to save a new image, rename an old image, or save a copy of a image in a different directory.

In Windows, new Quantity One images will be given a .1sc extension when they are first saved.

Save All

Save All on the main toolbar or File menu saves all images that are currently open.

General Operation

In Windows, new Quantity One images will be given a .1sc extension when they are first saved.

Revert to Saved

File > Revert to Saved will reload the last saved version of the image you are working on. This is a quick way to undo any alterations you may have made to an image since you last saved it.

Any changes you have made since last saving the file will be lost. (Also, all open dialog boxes will be closed.) A message box will warn you before completing the command.

File Info

File > File Info displays general information about your image, including the scan date, scan area, number of pixels in the scan, data range, and the size of the scan file. There is also a field where you can type in a file description or comments.

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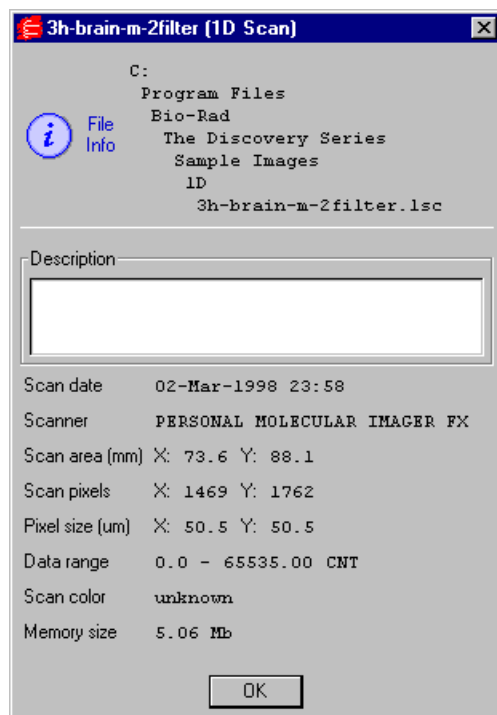


Fig.2-10. File Info box.

Reduce File Size

Image files can be quite large, and computer systems do not have unlimited memory or storage space. If you are having difficulty loading or storing a particular image, you may want to reduce the size of the file by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 10.7.a., Cropping Images.)

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus memory size.

General Operation

Note: Reducing the file size of an image will result in some loss of resolution. In most cases this will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select File > Reduce File Size to open the Reduce File Size dialog box.

The Reduce File Size dialog box shows you the size of the pixels in the image (Pixel Size: X by Y microns), the number of pixels in the image (Pixel Count: X by Y pixels), and the memory size of the image.

As you increase the size of the pixels, the pixel count will decrease, as will the memory size. You can increase the pixel size in either dimension (see the following figure for an example).

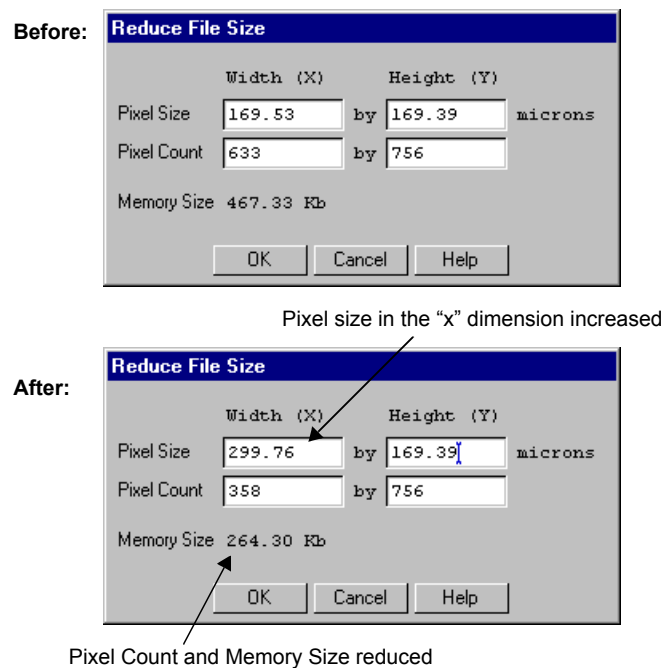


Fig.2-11. Reduce File Size dialog box, before and after pixel size increase.

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Note: Since with most 1-D gels you are more concerned with resolving bands in the vertical direction than the horizontal direction, you may want to reduce the file size by making rectangular pixels. That is, keeping the pixel size in the “y” dimension the same, while increasing the size in the “x” dimension. This lets you decrease the total number of pixels and therefore file size without sacrificing detail.

When you are finished, click on the OK button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

Reducing the file size is an irreversible process. For that reason, we suggest you make a copy of the image and reduce its file size. That way, if you lose too much resolution, you can simply delete the copy and try again. Once you are happy with the reduced image, don't forget to delete the original. The goal is to save space!

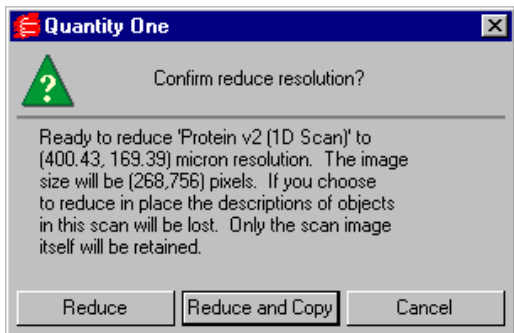


Fig.2-12. Confirm Reduce File Size pop-up box.

If you choose to make a copy of the image, you will be asked to enter a name for the new copy before the operation is performed.

Imaging Device Acquisition Windows

The File menu contains a list of Bio-Rad imaging devices supported by the Discovery Series software. These are:

General Operation

1. Gel Doc 1000 / 2000
2. GS-700 Imaging Densitometer
3. GS-710 Imaging Densitometer
4. Fluor-S MultiImager
5. Fluor-S MAX MultiImager
6. Personal Molecular Imager FX
7. Molecular Imager FX

Selecting a name in the File menu will open the acquisition window that allows you to scan using that instrument.

See the individual chapters on the imaging devices for more details.

Exit

File > Exit quits the application.

2.5. Preferences

The Preferences dialog box (accessed under the Edit menu) allows you to customize basic features of your system, such as displays and toolbars.

After you have selected your preferences, click on OK to implement any changes you make.

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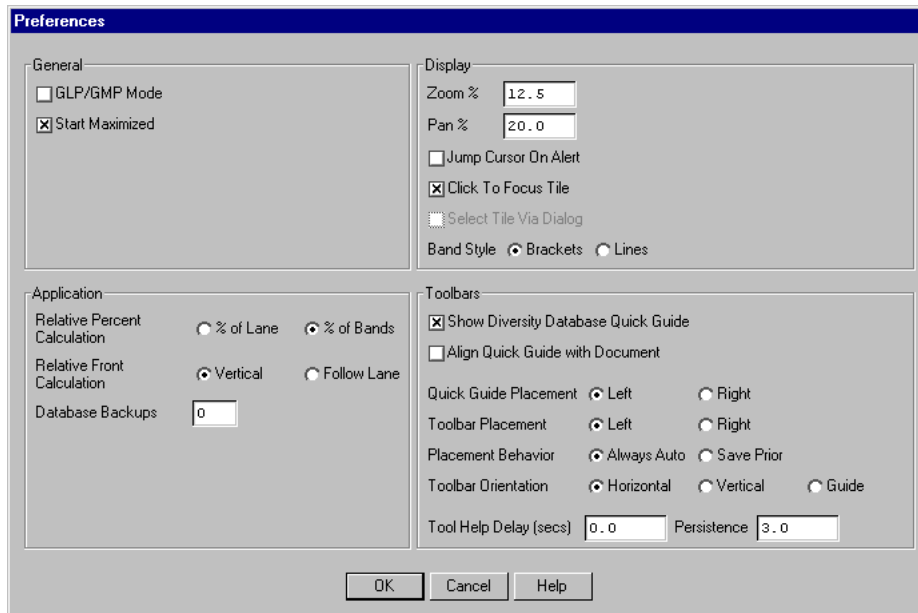


Fig.2-13. Preferences dialog box

2.5.a. General

GLP/GMP Mode

The GLP / GMP Mode checkbox allows you to prevent changes to an image that would change the raw image data. In GLP / GMP mode, the software will not allow the following operations to be performed (these are all located on the Image menu):

- Subtract Background
- Custom Rotation
- Filters
- Invert Data

If a user attempts to use any of these functions in GLP / GMP mode, he will receive a message that that function is not available.

General Operation

To set GLP / GMP mode, click on the checkbox. A dialog box will pop up asking you to enter a password.

After you enter a password and click on OK, another dialog will ask you to reenter the password for confirmation. Retype the password and click on OK again.

To disable GLP / GMP mode, click on the checkbox to deselect it. The dialog box will pop up asking you to enter the password. When you click on OK, GLP / GMP Mode will be disabled.

Start Maximized

In the Windows version, the Start Maximized checkbox determines whether Quantity One occupies the entire computer screen when first opened. If this is unchecked, Quantity One's menu and status bars will appear across the top of the screen and any toolbars will appear "floating" on the screen.

2.5.b. Display

The Zoom % field allows you to specify the percentage by which an image moves closer or farther away when you use the Zoom functions. This percentage is based on the size of the image.

Pan % determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

When an alert box pops up in Quantity One, you can set your cursor to automatically go to the OK button in the box by selecting Jump Cursor on Alert.

Bands in your gel image can be marked with brackets that define the top and bottom boundaries of the band, or they can be marked with a dash at the center of the band. Indicate your preference by clicking on the Brackets or Lines button next to the Band Style prompt. (This setting can also be changed in the Band Attributes dialog box.)

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2.5.c. Application

Relative Percent Calculation

The Relative Percent Calculation option allows you to define how Relative % will be determined for all reports, histograms, and band information functions in Quantity One: either as a percentage of the signal intensity of an entire lane or as a percentage of the signal intensity of the defined bands in a lane.

Selecting % of Lane means that the total lane intensity will equal 100 percent and the band that you select will be reported as a fraction thereof.

Selecting % of Bands means that the sum of the intensity of the defined bands in a lane will equal 100 percent, and that your band will be reported as a fraction of that sum.

If you create, adjust, or remove bands with Relative % defined as % of Bands, the Relative % of all the other bands in the lane will update to reflect changes in the total intensity of defined bands.

Relative Front Calculation

The Relative Front Calculation option lets you to define how relative fronts (Rf) will be calculated for lane and band analysis. This is useful if your gel image is curved or slanted.

Rf is calculated by dividing the distance a band has migrated from the gel's origin by the total distance between the origin and the dye front. These distances can be determined in one of two ways:

1. Drop a vertical line from the origin to the dye front (Vertical Projection);
2. Measure along the length of the lane (Follow Lane).

If a lane is straight and vertical, these methods will give identical results. However, the results will differ if a lane is curved or slanted. Choose your preferred method by clicking on one of the two buttons next to the Relative Front Calculation prompt.

General Operation

2.5.d. Toolbars

These settings determine the behavior and positioning of the secondary toolbars and Quick Guides.

If the Show Volumes Quick Guide checkbox is selected, the Volumes Quick Guide will open automatically when you open Quantity One.

If Align Quick Guide with Document is selected, the Quick Guides will pop up flush with the edge of your documents. Otherwise, they will appear flush with the edge of the screen.

Quick Guide Placement and Toolbar Placement determine which side of the screen the Quick Guides and toolbars will first open.

The Placement Behavior setting determines whether a Quick Guide or toolbar will always pop up in the same place and format (Always Auto), or whether they will pop up in the last location they were moved to and the last format selected (Save Prior).

The Toolbar Orientation option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open Quantity One.

Tool Help Delay allows you to specify the amount of time (in seconds) the cursor must remain over a toolbar icon before the Tool Help appears. First-time users may want to specify a short delay to learn the names of the toolbar functions, while experienced users can specify a longer delay once they are familiar with the icons.

Tool Help Persistence determines how long the Tool Help will linger on the screen after you move the cursor off a toolbar icon.

2.6. User Settings

If Quantity One is on a workstation that has multiple users, each user can have his or her own set of Quantity One preferences and settings.

In multiple-user situations, Quantity One's preferences and settings are associated with individual user names. On a PC with Windows 95/98/NT, your user name is the name you use to log onto the computer. On a

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Macintosh, your user name is the Owner Name under the File Sharing (OS 8.x) or Sharing Setup (OS 7.x) control panel.

Note: If you do not log onto your PC under Windows 95/98 or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.

3. Gel Doc 1000/2000



Fig.3-1. Gel Doc 1000/2000.

Before you can begin acquiring images using the Gel Doc[®] 1000/2000 gel documentation system, the system must be properly installed and connected with the host computer. See the Gel Doc hardware manual for installation, startup, and operating instructions.

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To use the Gel Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install Quantity One.

Make sure that your Gel Doc camera is turned on. If the camera is not turned on, the Gel Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

3.1. Opening the Gel Doc Acquisition Window

To acquire images using the Gel Doc, go to the File menu and select Gel Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

Gel Doc 1000/2000

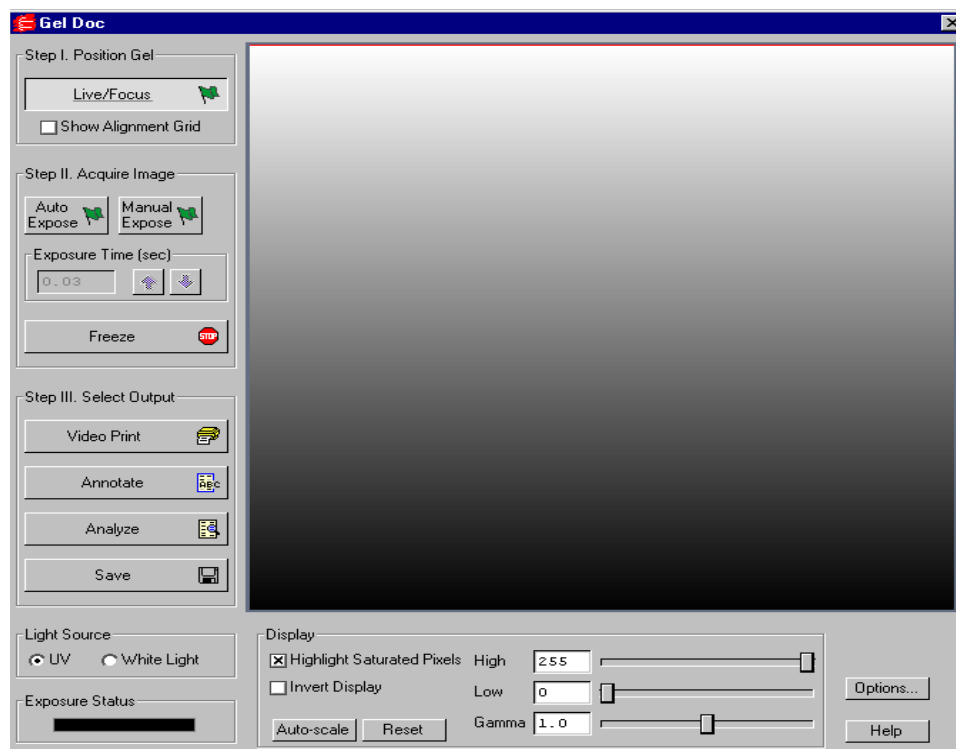


Fig.3-2. Gel Doc acquisition window

The Gel Doc video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Gel Doc:

1. Position and focus the gel or other object to be imaged.
2. Acquire the image.

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3. Select the output.

3.2. Step I. Position Gel

The Gel Doc window will open in “live” mode, giving you a live video display of your sample. (Frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.) In this mode, the Live/Focus button will appear selected.

You can use live mode to zoom and focus the camera on your sample area and position your sample within the area. You can also select the Show Alignment Grid checkbox to facilitate positioning.

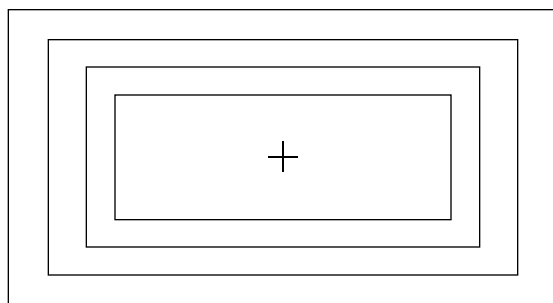


Fig.3-3. Gel Doc alignment grid.

The image should stay in focus while zooming. If focus is not maintained, refer to your Gel Doc hardware manual.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc box so that it is visible by the camera.

3.3. Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from live mode.

For UV light or chemiluminescent applications, or if your aperture is closed down in white light applications, this mode will probably not give you the best possible image. To illuminate faint areas of your sample, you will need to select an exposure time for the image.

“Exposure” refers to the integration of the image on the camera CCD over a set period of time. The effect is analogous to exposing photographic film to light over a period of time. The Gel Doc has two exposure modes: Auto and Manual.

To acquire an image, you will typically perform the following steps:

1. Select Auto Expose to determine an approximate exposure time.
2. After Auto Expose is finished, “fine tune” that exposure time using Manual Expose. You can also change the camera aperture at this time.
3. Click on Freeze or any of the other buttons in the window to stop the exposure process.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip the Auto Expose step and go directly to Manual Expose.

Auto Expose

Clicking on the Auto Expose button will cancel live mode and begin an automatic exposure. (The Auto Expose button will appear selected throughout the exposure.) If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

In Auto Expose mode, the image is continuously integrated on the camera CCD until it has reached a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

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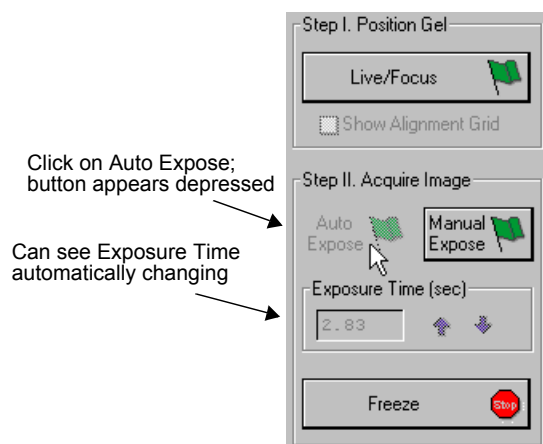


Fig.3-4. Acquiring an image, step 1: Select Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

Manual Expose is automatically activated after Auto Expose has deactivated. Or, if you know the approximate exposure time you want, you can simply click on the Manual Expose button.

Gel Doc 1000/2000

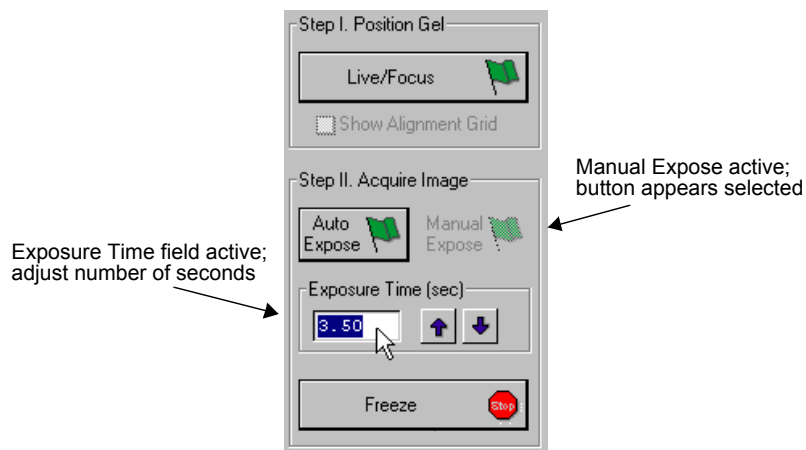


Fig.3-5. Acquiring an image, step 2: Adjust exposure time.

When Manual Expose is active (the button will appear selected), the last captured image will be displayed in the Gel Doc image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

You can adjust the exposure time by changing the number of seconds in the Exposure Time field. To do so, enter a number directly or use the arrow buttons next to the field.

Freeze

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Light Source, Display controls, etc.).

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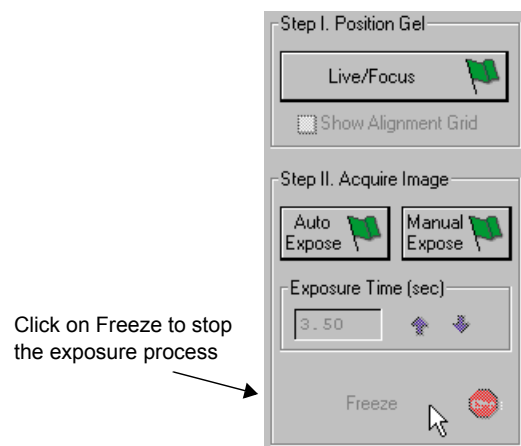


Fig.3-6. Acquiring an image, Step 3: Freeze the exposure.

3.4. Step III. Select Output

The Gel Doc window has several output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Annotation Tools toolbar will also pop up to allow you to annotate your image. (See section 10.11., Annotating Images, for details.)

Gel Doc 1000/2000

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in Quantity One.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications. (See Save As in section 2.4. for more information.)

3.5. Light Source

The Light Source option buttons allow you to set the type and scale of your Gel Doc data.

UV

Select this light source for fluorescent and chemiluminescent samples. The data will be measured in linear intensity units.

White Light

Select this light source for reflective and transmissive samples. The data will be measured in uncalibrated optical density (uOD) units.

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3.6. Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

3.7. Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc window.

These controls are similar to those in the Transform dialog box. (See section 10.6.)

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

Reset

Reset will return the image to its original, unmodified appearance.

3.8. Options

Clicking on the Options button opens the Options dialog box. Here you can specify certain settings for your Gel Doc system.

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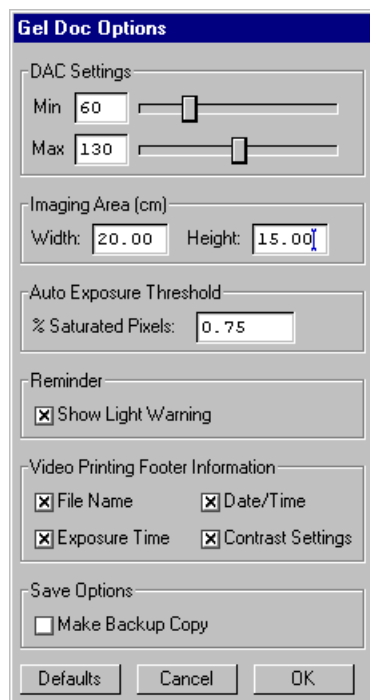


Fig.3-7. Available options in the Gel Doc acquisition window.

Clicking on OK implements any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

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With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

4. GS-700 Imaging Densitometer



Fig.4-1. GS-700 Imaging Densitometer

Before you can begin scanning images with the GS-700 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-700 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-700, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

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You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

4.1. Opening the GS-700 Acquisition Window

To acquire images using the GS-700, go to the File menu and select GS-700.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

GS-700 Imaging Densitometer

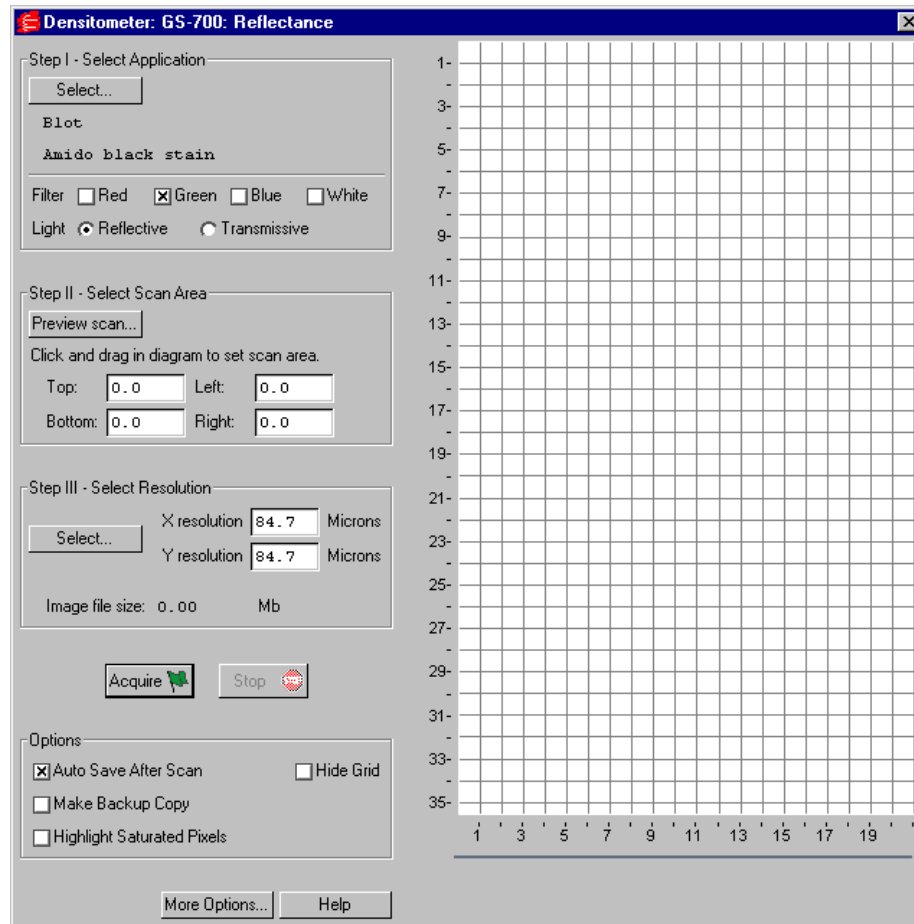


Fig.4-2. GS-700 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–35 top to bottom and 1–21 left to right if the light source is uncalibrated reflective, and 1–25 top to bottom and 1–20 left to right if the light source is uncalibrated transmissive (see below for details).

To hide the gridlines, click on the Hide Grid checkbox under Options.

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The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the GS-700:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Acquire the image.

4.2. Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

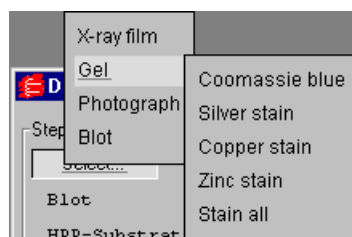


Fig.4-3. Example of the application tree in the GS-700 dialog box.

GS-700 Imaging Densitometer

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

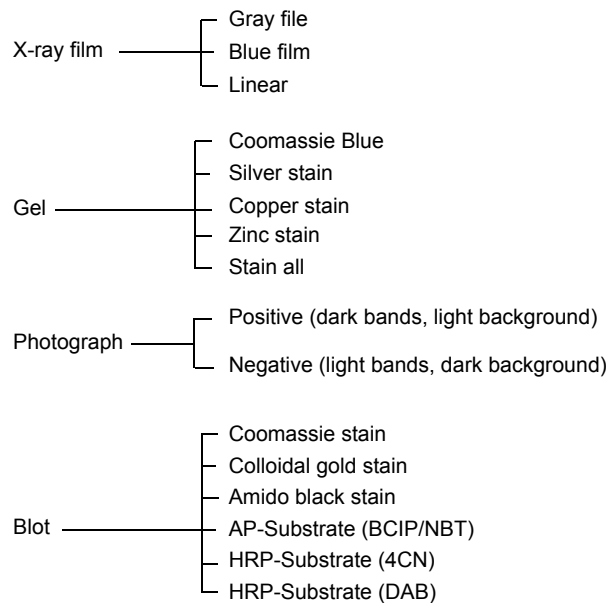


Fig.4-4. Applications available in the GS-700 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

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Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig.4-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, TLC plates, and photographs. The scanning dimensions in this mode are 21 cm x 35 cm (uncalibrated).
- Transmissive mode is for scanning transparent mediums such as films, gels, and slides. The scanning dimensions in this mode are 20 cm x 25 cm (uncalibrated).

4.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

GS-700 Imaging Densitometer

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

4.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

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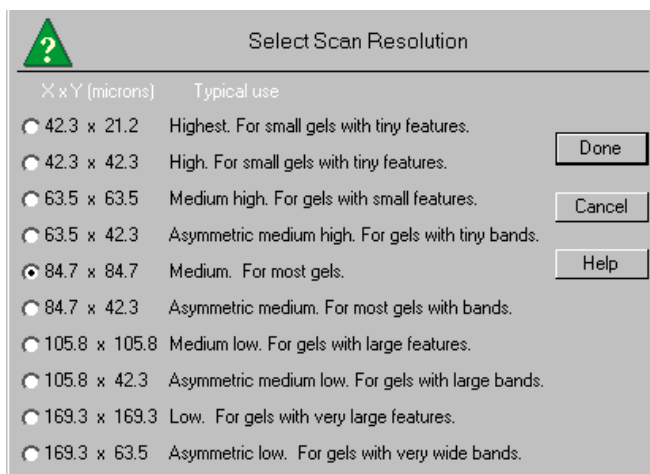


Fig.4-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section 4.7. for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

GS-700 Imaging Densitometer

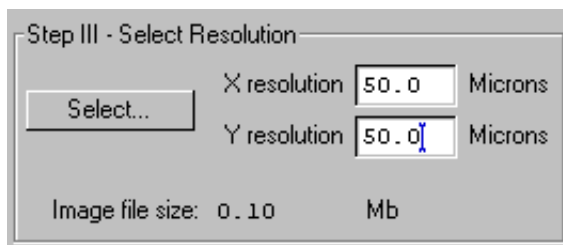


Fig.4-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

4.5. Acquire the Image

If you want to calibrate your scans, read the following section on calibration before scanning.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Quantity One image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

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Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

4.6. Calibration

If you have installed a calibration strip on the GS-700 scanner plate, you can automatically calibrate your scans. To order a calibration strip from Bio-Rad, contact your Bio-Rad sales representative.

See the calibration manual for details on installing the calibration strip.

To set the automatic calibration settings, click on the More Options button in the GS-700 acquisition window. This will open the Densitometer Options dialog box.

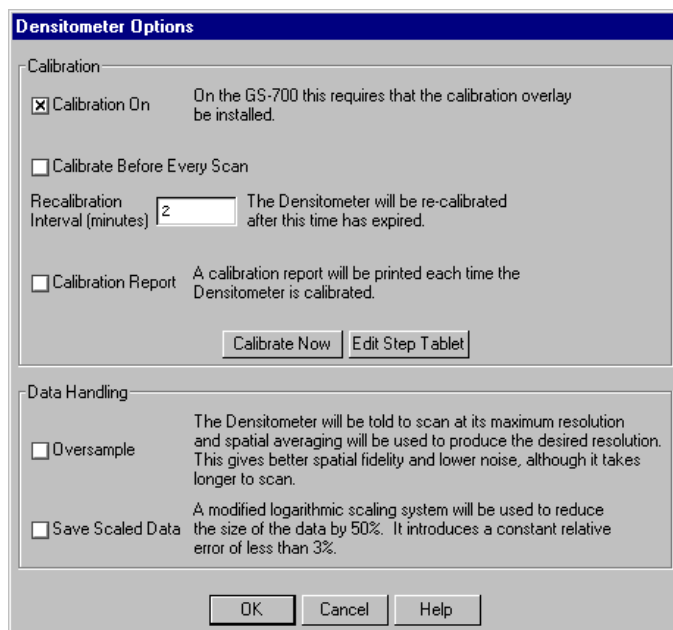


Fig.4-8. More Options in the GS-700 acquisition window.

To enable automatic calibration, click on the Calibration On checkbox.

GS-700 Imaging Densitometer

Note: Your calibration strip must be installed for this feature to work!

With Calibration On selected, the other calibration settings become active.

Calibration Strip Window

When calibration is turned on, a calibration strip window will appear below the main scanning window and the length of the main scanning window will be reduced to 29 cm reflective and 23 cm transmissive.

Every time the densitometer calibrates, an image of the calibration strip will appear in the calibration strip window.

4.6.a. Editing the Step Tablet

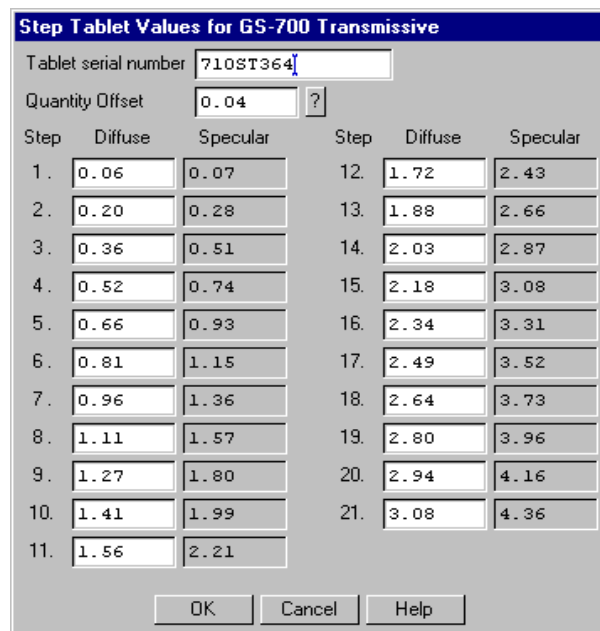
Before you can calibrate, you must enter the step tablet values for your calibration strip into the Step Tablet Values form.

In the package with your calibration strip, you will find the manufacturer's printout of the diffuse density values for each segment of your step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Scanning with incorrect step tablet values entered into the form can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-700 will open.

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The dialog box is titled "Step Tablet Values for GS-700 Transmissive". It contains a "Tablet serial number" field with the value "710ST364", a "Quantity Offset" field with the value "0.04" and a question mark icon, and a table of step values. The table has two columns of data, each with three sub-columns: Step, Diffuse, and Specular. The steps range from 1 to 21. At the bottom are "OK", "Cancel", and "Help" buttons.

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.06	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig.4-9. Step Tablet dialog box.

The dialog box will indicate whether the displayed form is for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

Note: In transmissive scanning, the scanner is calibrated against a transmissive step tablet. In reflective scanning, the scanner is calibrated against a reflective step tablet, unless the reflective calibration strip is uncalibrated. If the reflective calibration strip does not have step tablet values associated with it, you can use the default values in the form.

In the Tablet Serial Number field, you can type in the serial number for the step tablet values you are entering. (The reflective step tablet will probably not have a serial number.)

Your calibration strip will have a clear plastic cover. If the quantity offset value for the clear cover is included in your calibration strip package, you can

GS-700 Imaging Densitometer

enter this number in the Quantity Offset field. Otherwise, use the default value.

Finally, enter the values in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned. You only need to enter new values if you use a new step tablet.

When you finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

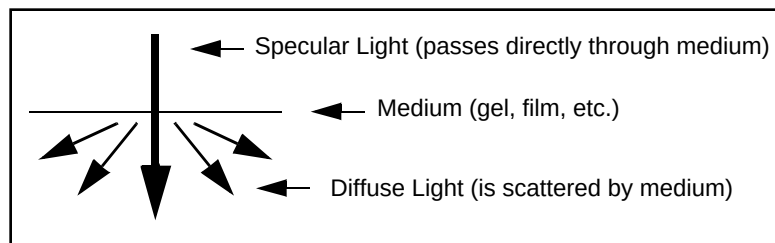


Fig.4-10. Specular and diffuse density

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Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \diamond \text{Diffuse OD}$$

4.6.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the densitometer to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: With calibration turned on, the scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

4.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-700 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

GS-700 Imaging Densitometer

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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5. GS-710 Imaging Densitometer



Fig.5-1. GS-710 Imaging Densitometer

Before you can begin scanning images with the GS-710 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-710 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-710, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

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You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

5.1. Opening the GS-710 Acquisition Window

To acquire images using the GS-710, go to the File menu and select GS-710.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

GS-710 Imaging Densitometer

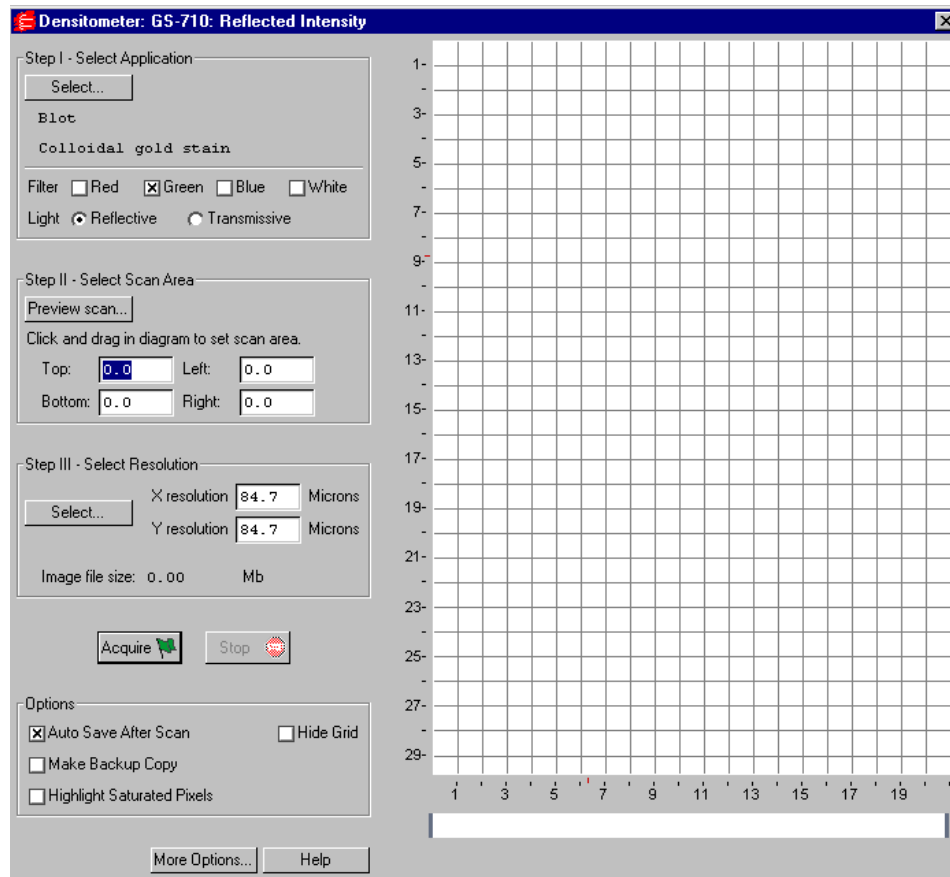


Fig.5-2. GS-710 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–29 top to bottom and 1–21 left to right if the light source is reflective, and 1–23 top to bottom and 1–20 left to right if the light source is transmissive (see below).

To hide the gridlines, click on the Hide Grid checkbox under Options.

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Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-710:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

5.2. Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

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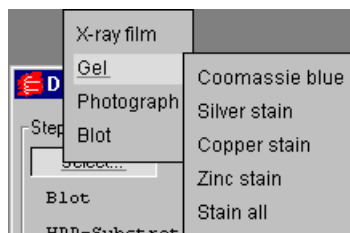


Fig.5-3. Example of the application tree in the GS-710 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

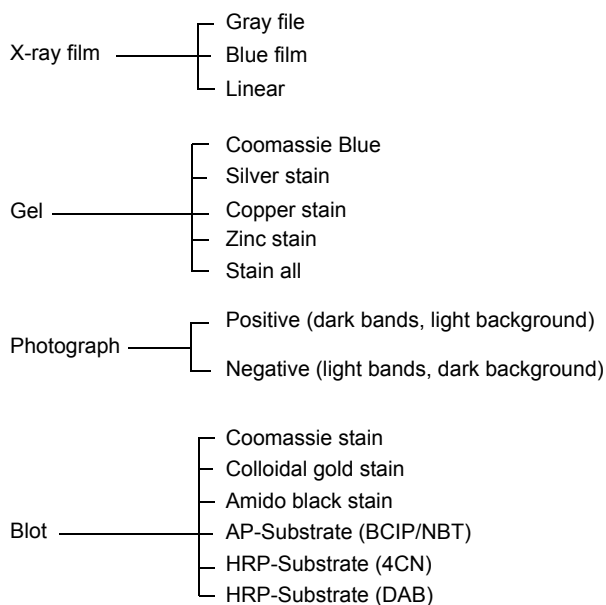


Fig.5-4. Applications available in the GS-710 acquisition window.

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Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig.5-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

5.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

GS-710 Imaging Densitometer

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

5.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

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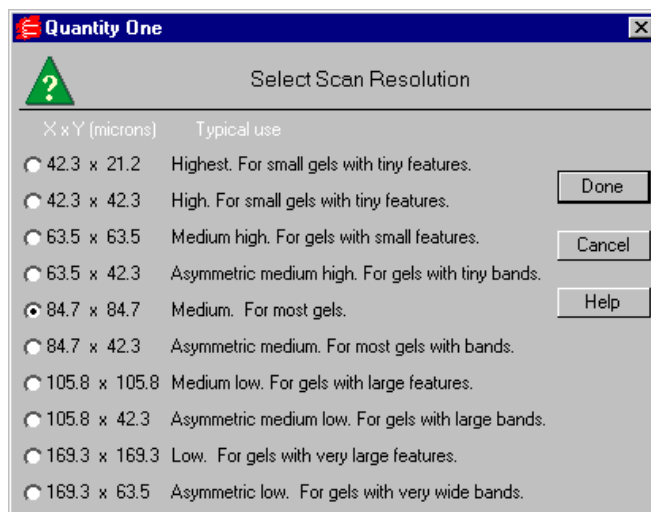


Fig.5-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section 5.7., Other Options, for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

GS-710 Imaging Densitometer

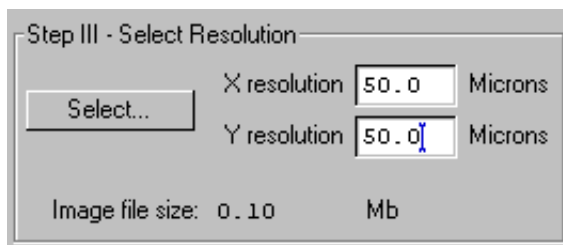


Fig.5-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

5.5. Calibration

Images scanned using the GS-710 are automatically calibrated. The scanner plate has a built-in calibration strip for both transmissive and reflective scanning.

- The transmissive calibration strip calibrates the densitometer from 0 to 3.0 O.D.
- The reflective calibration strip calibrates the densitometer to 2.0 O.D.

The first time you use the GS-710, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-710 acquisition window. This will open the Densitometer Options dialog box.

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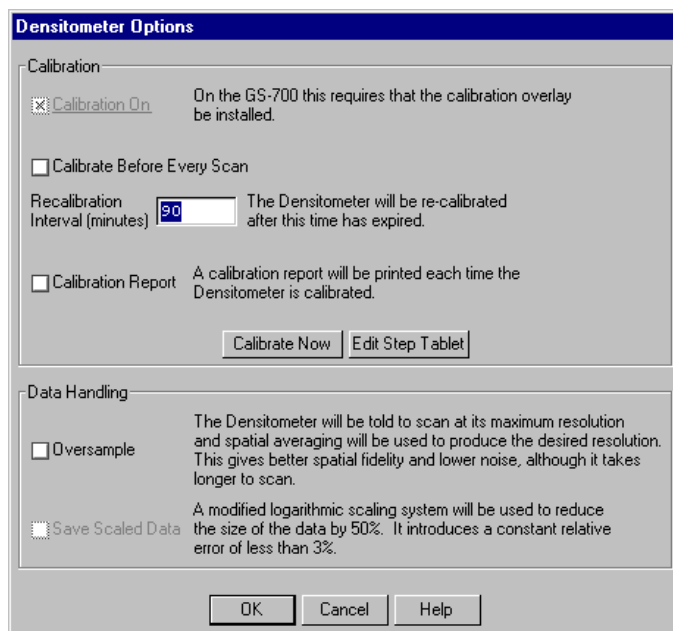


Fig.5-8. Densitometer Options dialog box.

5.5.a. Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

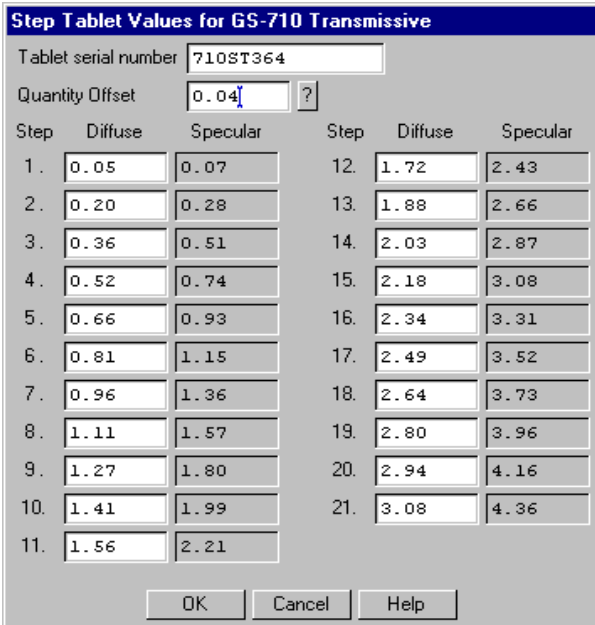
Note: In reflective scanning, the scanner is calibrated against the reflective step tablet. However, you do not need to enter the values for this step tablet; you can use the default values in the Step Tablet Values form.

Attached to the outside of your GS-710, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

GS-710 Imaging Densitometer

Note: Scanning in transmissive mode with incorrect step tablet values entered into the computer can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-710 will open.



The dialog box is titled "Step Tablet Values for GS-710 Transmissive". It contains a "Tablet serial number" field with the value "710ST364" and a "Quantity Offset" field with the value "0.04" and a question mark icon. Below these fields is a table with two columns of data, each with three sub-columns: "Step", "Diffuse", and "Specular". The table contains 21 rows of data. At the bottom of the dialog box are three buttons: "OK", "Cancel", and "Help".

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.05	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig.5-9. Step Tablet dialog box.

The dialog box will indicate whether the values are for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

If you are in reflective mode, you do not need to change the default step tablet values.

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If you are in transmissive mode, you do need to enter the correct step tablet values that were shipped with your GS-710. You can type the serial number for these values into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-710. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

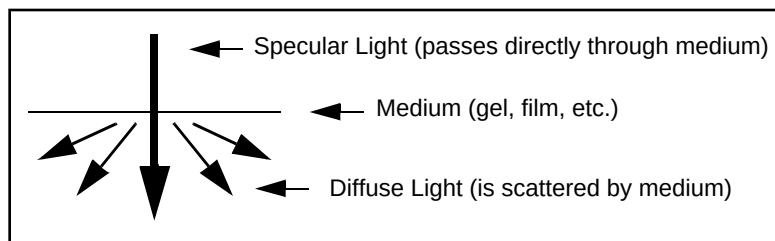


Fig.5-10. Specular and diffuse density

GS-710 Imaging Densitometer

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \diamond \text{Diffuse OD}$$

5.5.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-710 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

5.6. Acquire the Image

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Quantity One image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

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Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

5.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-710 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

GS-710 Imaging Densitometer

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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6. Fluor-S Multimager

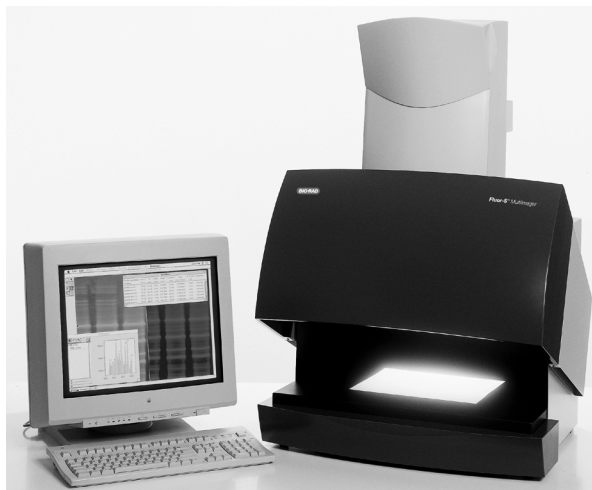


Fig.6-1. Fluor-S Multimager.

Before you can begin acquiring images using the Fluor-S[®] MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S is green before attempting to capture an image. If you are using a PC, the Fluor-S should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

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Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

6.1. Opening the Fluor-S Acquisition Window

To acquire images using the Fluor-S, go to the File menu and select Fluor-S.... The acquisition window for the imager will open, displaying a control panel and a video display window.

Fluor-S Multimager

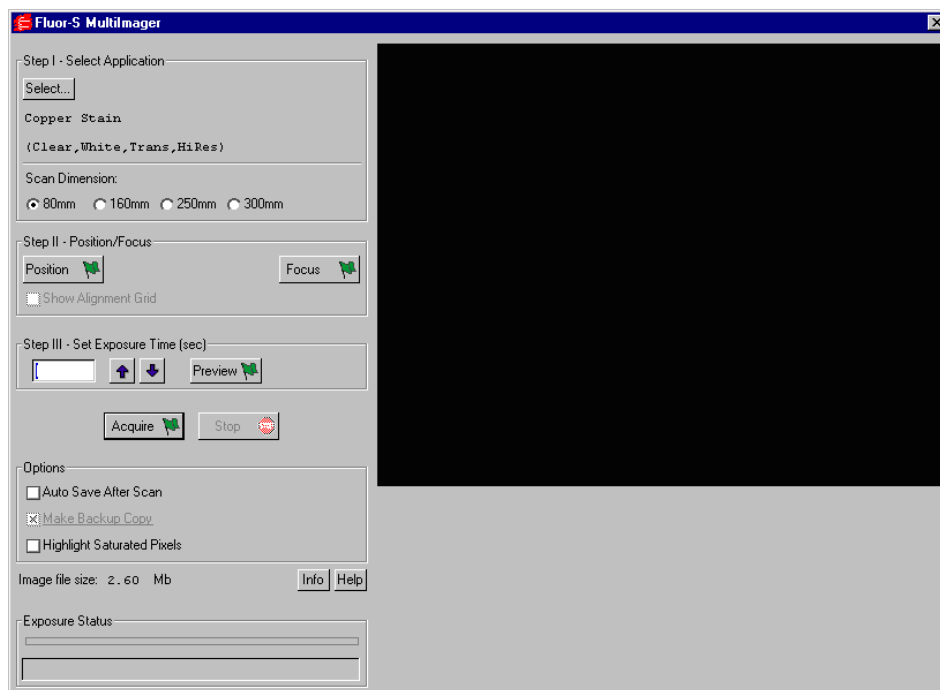


Fig.6-2. Fluor-S acquisition window

When the Fluor-S window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

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6.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

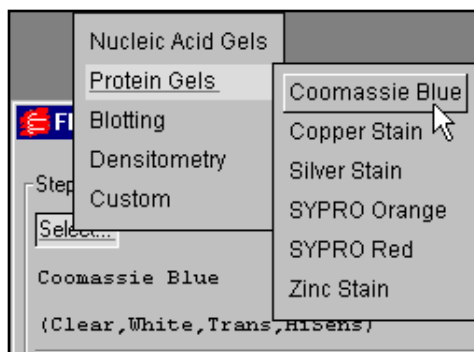


Fig.6-3. The application tree in the Fluor-S acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Resolution or High Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Fluor-S Multimager

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

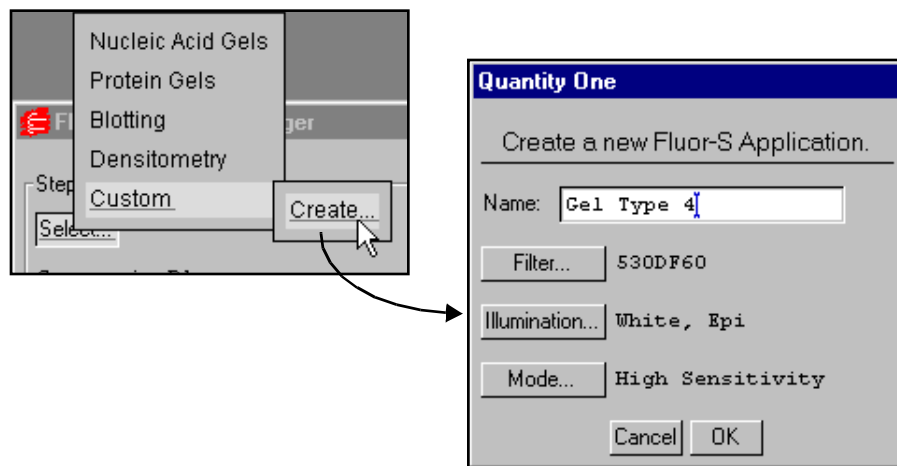


Fig.6-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

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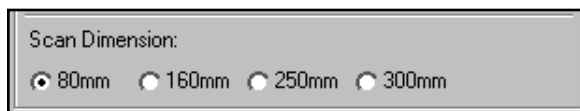


Fig.6-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Resolution Versus High Sensitivity

High Resolution and High Sensitivity are mutually exclusive options. High Resolution is the normal operating mode for the Fluor-S. High Sensitivity provides optimal sensitivity for low-light applications such as chemiluminescence.

In High Resolution mode, images are captured at the maximum resolution of the camera.

In High Sensitivity mode, the pixels in the camera are “binned” (i.e., four pixels are combined into one) to increase the amount of signal per pixel. However, combining the pixels results in a reduction in the resolution of the image.

6.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S box automatically turns on. To turn the light off while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The first step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S will begin capturing a “live” image and updating it every second.

Fluor-S Multimager

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

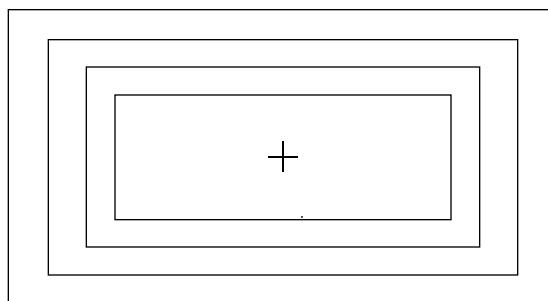


Fig.6-6. Fluor-S alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting. (See the following table on recommended f-stops.)

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

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6.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of the image on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application. The table provides recommended exposure times for various applications.

Table 1: Recommended f-stops and Exposure Times

Application	f-stop ¹ Zoom lens	f-stop ¹ 50mm lens	Exposure time ²
Ethidium bromide	f-4	f-2.8	5–30 sec.
SYBR Green	f-4	f-2.8	5–30 sec.
SYPRO Orange	f-4	f-2.8	5–30 sec.
Radiant Red	f-4	f-2.8	5–30 sec.
Fluorescein	f-4	f-2.8	30 sec.–2 min.
Texas Red	f-4	f-2.8	30 sec.–2 min.
Colorimetric Stains (e.g., Coomassie or Silver)	f-11	f-11	1–5 sec.
X-ray Film	f-11	f-11	1–5 sec.
Photographs	f-11	f-11	1–5 sec.
Chemi ³		f-1.8	1–10 min.
¹ For sharper focusing, close the f-stop down 1–2 stops from full open while focusing. ² Increase exposure time two fold for each step increase in f-stop. ³ For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.			

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For

Fluor-S Multimager

chemiluminescent samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig.6-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

6.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Quantity One will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

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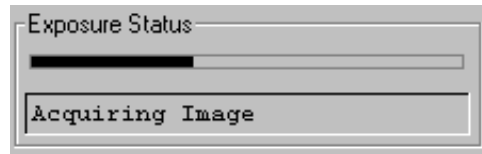


Fig.6-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Quantity One's menu and toolbar functions.

6.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

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Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field.

The dimensions of your imaging area are determined by how much you are zooming in or out on your sample. The dimensions determine the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. It is also necessary if you are comparing the quantities of objects (e.g., using the Volume Tools) between images.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create. This size is determined by whether you selected High Resolution or High Sensitivity when you selected an application.

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If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

7. Fluor-S MAX Multimager

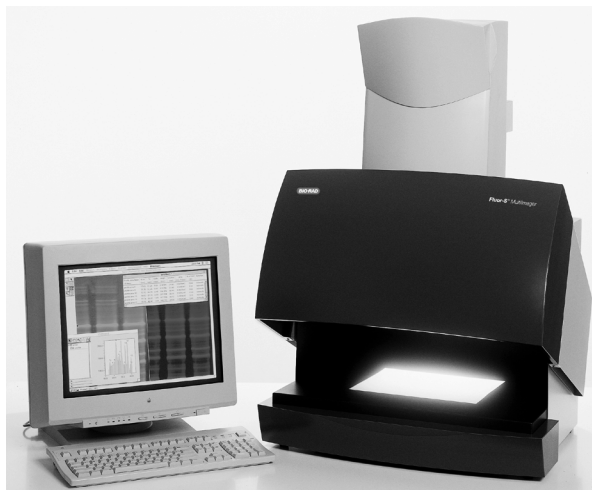


Fig.7-1. Fluor-S MAX Multimager.

Before you can begin acquiring images using the Fluor-S[®] MAX MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S MAX hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S MAX is green before attempting to capture an image. If you are using a PC, the Fluor-S MAX should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S MAX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S MAX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

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Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

7.1. Opening the Fluor-S MAX Acquisition Window

To acquire images using the Fluor-S MAX, go to the File menu and select Fluor-S MAX.... The acquisition window for the imager will open, displaying a control panel and a video display window.

Fluor-S MAX Multimager

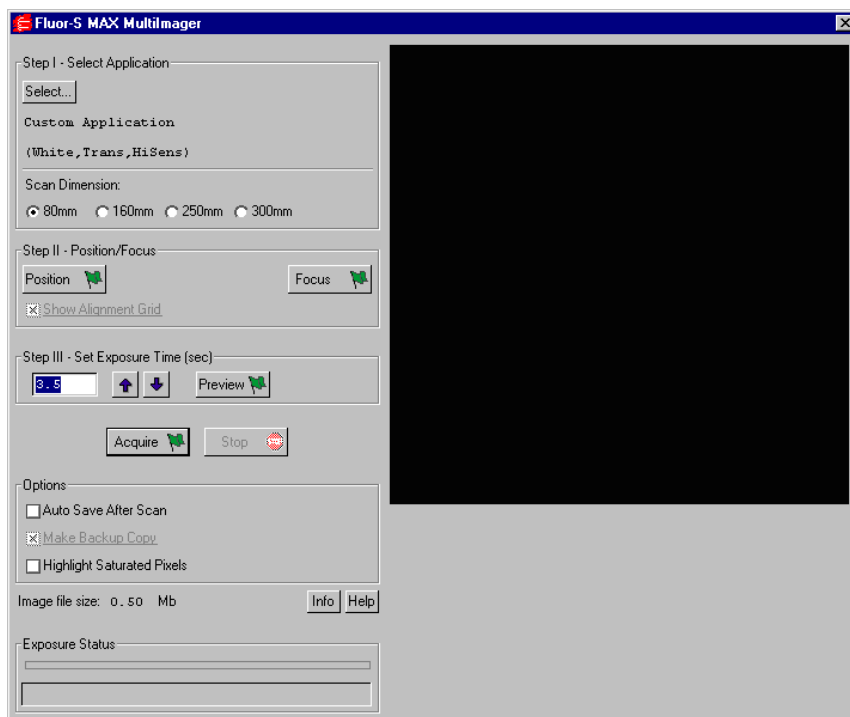


Fig.7-2. Fluor-S MAX acquisition window.

When the Fluor-S MAX window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S MAX:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

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7.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

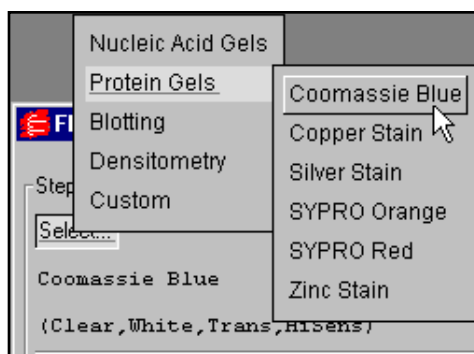


Fig.7-3. The application tree in the Fluor-S MAX acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Sensitivity or Ultra Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S MAX for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Fluor-S MAX Multimager

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

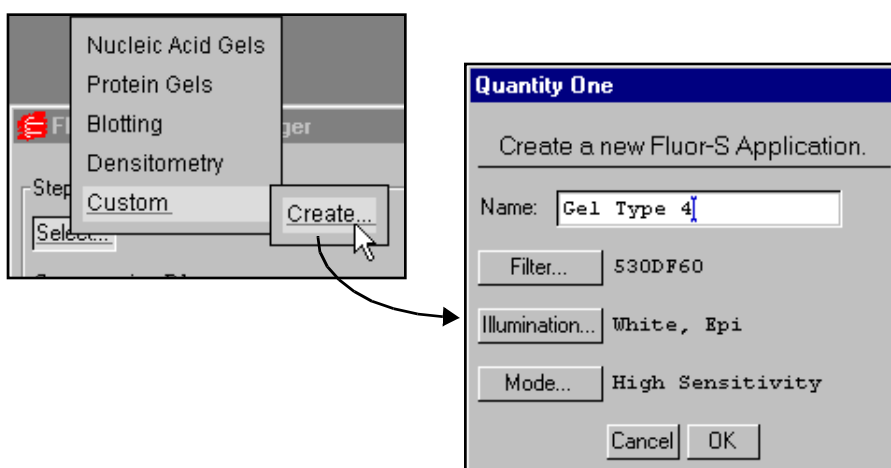


Fig.7-4. Creating a new custom application.

To select the filter, type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

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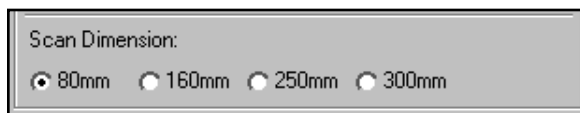


Fig.7-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Sensitivity and Ultra Sensitivity

High Sensitivity and Ultra Sensitivity are different camera modes. In Ultra Sensitivity mode, the Fluor-S MAX camera is cooled to -35.0 degrees C. In High Sensitivity, the camera is cooled to -20.0 degrees C.

High Sensitivity is the normal operating mode of the Fluor-S MAX. Ultra Sensitivity provides optimal sensitivity for low-light applications. It may be selected for the chemiluminescence and chemifluorescence applications, or for a custom application.

Note: When you change from High to Ultra Sensitivity or visa versa, there will be a delay of several minutes while the Fluor-S MAX camera cools down or warms up. If you attempt to acquire an image during this period, you will be notified of the changing temperature. If you do not want to wait, you can cancel the mode change.

7.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S MAX box automatically turns off. This is because if you focus with the camera at maximum aperture (see note below), leaving the light on would make it difficult to view the image. To turn the light on while positioning or focusing, hold down the SHIFT key when clicking on the button.

Fluor-S MAX Multimager

Position

The next step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S MAX will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

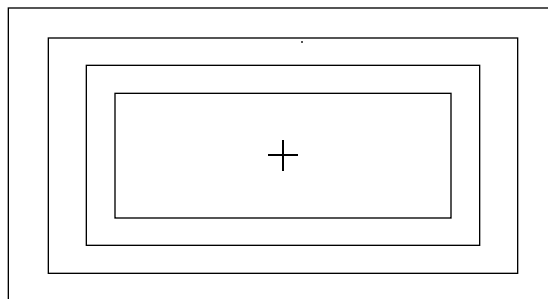


Fig.7-6. Fluor-S MAX alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

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When you are finished focusing, click on the Stop button.

7.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemi samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig.7-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

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You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

7.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Quantity One will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

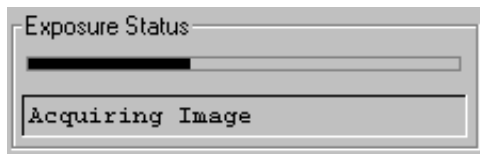


Fig.7-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

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You can then analyze the image using Quantity One's menu and toolbar functions.

7.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

The imaging area is the area of a sample that is captured by the camera and displayed in the video display window. To specify the size of your imaging

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area, click on the Info button and enter each dimension in the appropriate field. When you adjust one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a sample that is 4.5 x 4.5 cm, then you would enter 4.5 for the width and height of your imaging area.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) between images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

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8. Personal Molecular Imager FX



Fig.8-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

Note: The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card

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installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

8.1. Opening the Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.

Personal Molecular Imager FX

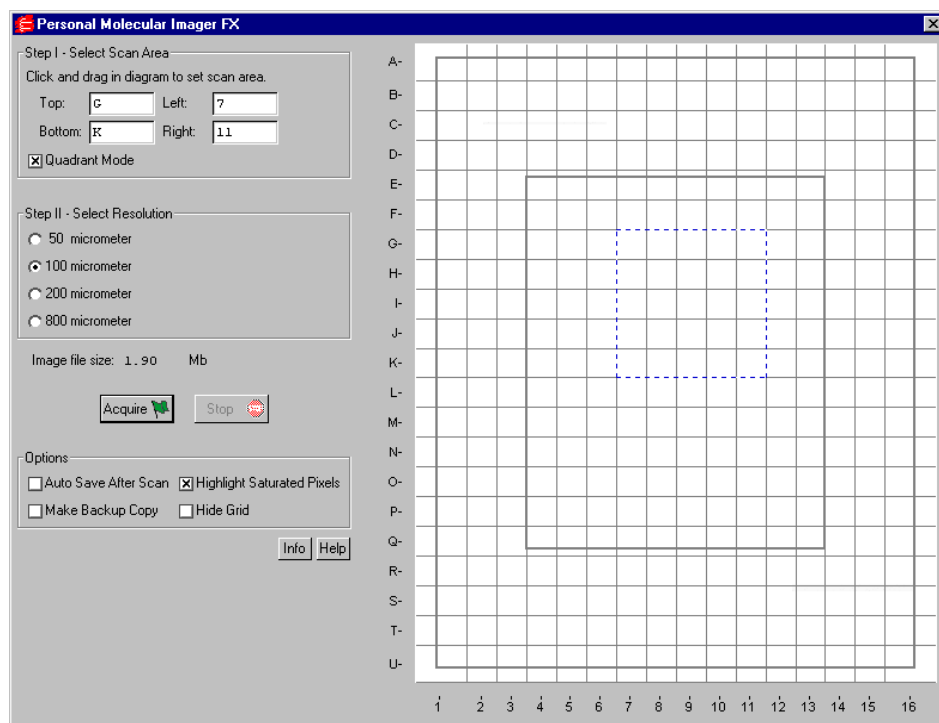


Fig.8-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the Personal FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

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1. Select the scan area
2. Select the resolution
3. Acquire the image

8.2. Step I. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

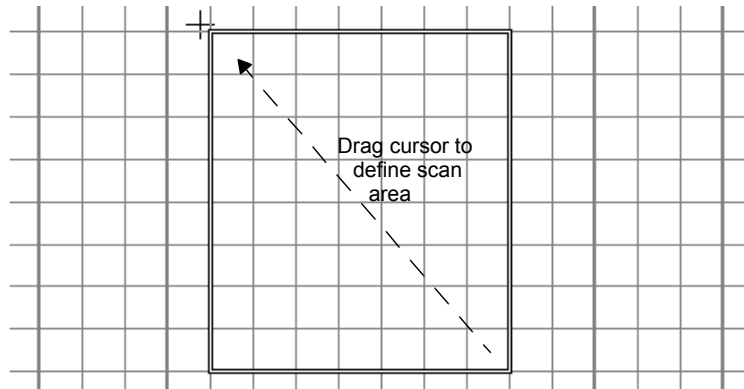


Fig.8-3. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

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You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

8.3. Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

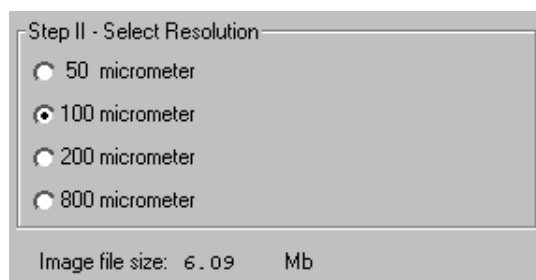


Fig.8-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

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File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

8.4. Acquire the Image

Once you have selected your scan area and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Quantity One's menu and toolbar functions.

8.5. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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9. Molecular Imager FX



Fig.9-1. Molecular Imager FX

Before you can begin acquiring images using the Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the FX hardware manual for installation, startup, and operating instructions.

Note: The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If

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you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Quantity One can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

9.1. Opening the FX Control Panel

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.

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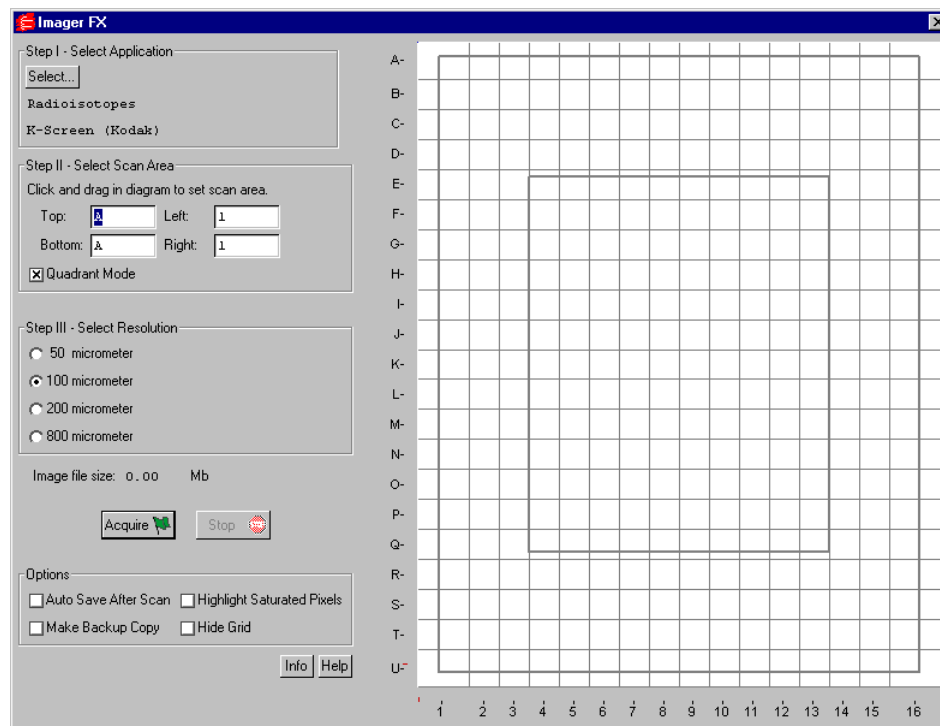


Fig.9-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

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1. Select the application.
2. Select the scan area.
3. Select the resolution.
4. Acquire the image.

9.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application.

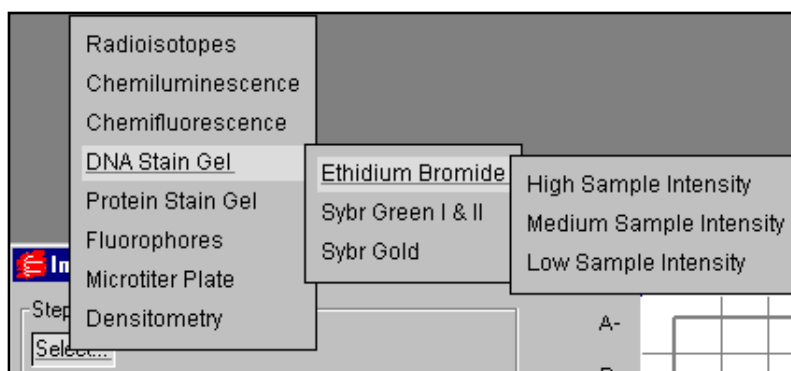


Fig.9-3. Example of an application tree: Ethidium Bromide gel.

The applications and associated settings are listed in a tree that expands from left to right. When you select an application, the software automatically sets the appropriate filter(s) in the FX for that particular application.

First you select the category of your application, next you select your particular application, and finally (if appropriate) you select the intensity of your samples.

To exit the tree without selecting, press the ESC key.

Your selection will be displayed below the Select button.

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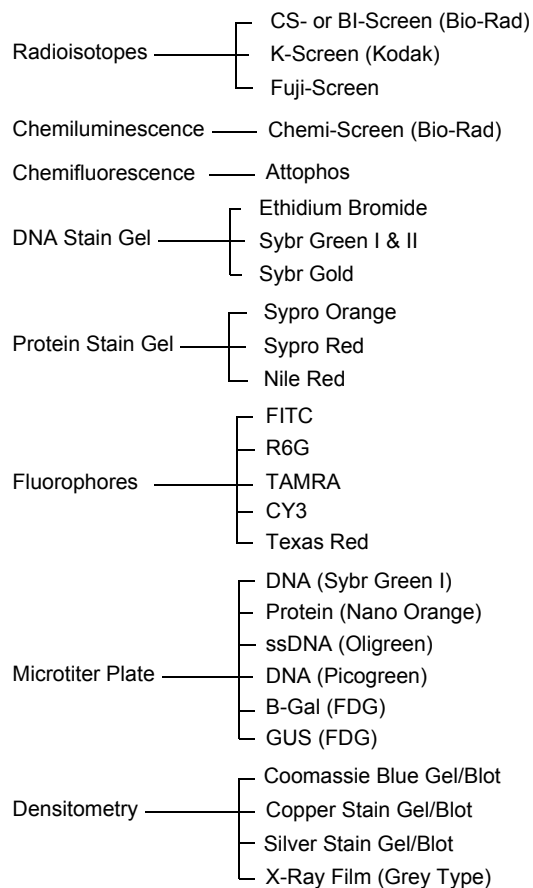


Fig.9-4. Applications available in the FX acquisition window.

Sample Intensity

When selecting any application except radioisotopes, you will need to specify a sample intensity. This is simply a rough estimate of how much sample is apparent in your gel or other object.

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Do not worry if you are not sure about the level of intensity of your sample—if you make a mistake, you can always change the setting and capture another image.

For example, if you specify a low sample intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to medium sample intensity and rescan. If you specify a high sample intensity and the resulting image is too faint, select medium or low sample intensity and rescan.

9.3. Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

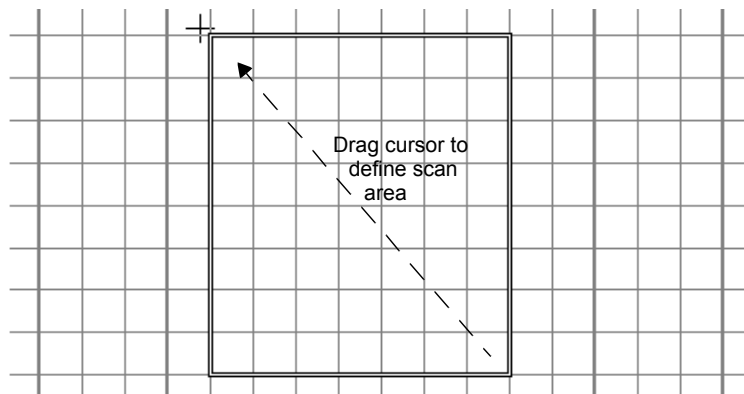


Fig.9-5. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.

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- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

9.4. Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

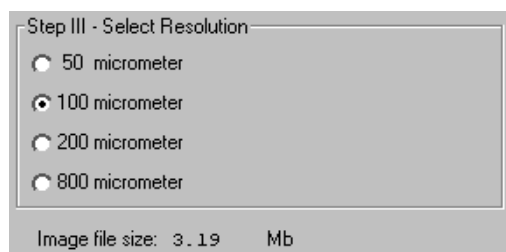


Fig.9-6. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.

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- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

9.5. Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date,

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time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Quantity One's menu and toolbar functions.

9.6. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

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Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

10. Viewing and Editing Images

This chapter describes Quantity One's viewing tools, which you can use to magnify your images and clarify your data. This chapter also describes the tools for cropping, flipping, and rotating images, eliminating and filtering noise, and annotating images.

These tools may be found on the View, Image, and Edit menus.

Note: The following chapters contain instructions for analyzing X-ray films, wet and dry gels, blots, and photographs. For the sake of simplicity, these are all referred to as “gels.”

10.1. Magnifying and Positioning Tools

The magnifying and positioning tools in Quantity One are located on the View menu and Window menu; some of these functions are also found on the Main toolbar.

These commands will only change how the image is displayed on your computer screen. *They will not change the underlying data.*

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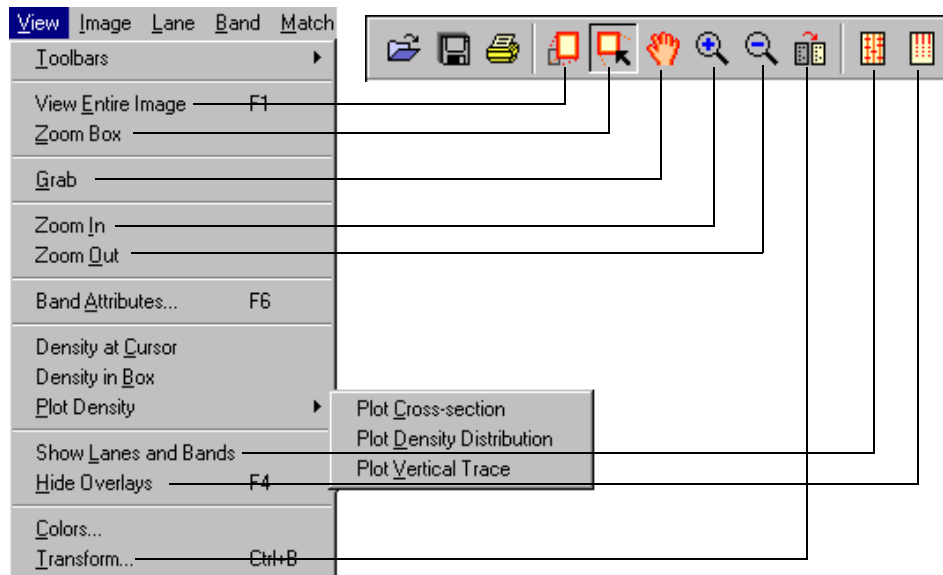


Fig.10-1. Viewing functions on View menu and Main toolbar.

Zoom Box

Zoom Box allows you to select a small area of the image to magnify so that it fills the entire image window.

First, click on the Zoom Box button on your Main toolbar or select View > Zoom Box. Your cursor arrow will change to a cross. Then drag the cursor on the image to enclose the area you want to magnify. The area of the image you selected will be magnified to fill the entire window.

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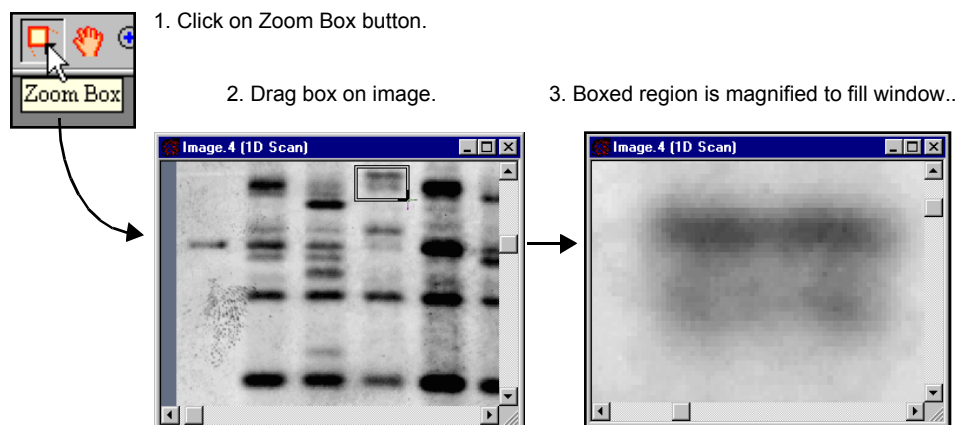


Fig.10-2. Zoom Box tool.

Zoom In/Zoom Out

These tools work like standard magnifying tools in other applications.

Click on the Zoom In or Zoom Out button on your Main toolbar (or select from the View menu). Your cursor will change to a magnifying glass. Click on an area of the image to zoom in or zoom out a defined amount, determined by the setting in the Edit > Preferences dialog box.

Grab

This tool allows you to change the position of your image in the image window. Select View > Grab or click on the Grab button. Your cursor will change to a “hand” symbol. Dragging the Grab symbol on the image will move the image in any direction.

You can also move your image inside the image window by using the ARROW keys.

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View Entire Image

If you have magnified part of an image or moved part of an image out of view, View Entire Image returns to the original, full view of the image.

Centering an Image

You can center the image window on any point in an image quickly and easily using the F3 key command. This is useful if you are comparing the same region on two gel images and want to center both image windows on the same point.

Position your cursor on the point on the image that you want at the center of the image window, then press the F3 key. The image will shift so that that point is at the center of the image window.

Imitate Zoom

If you are comparing two or more images and want to zoom in on the same area on all of them, use the Imitate Zoom command.

First zoom in on one of the images. Then, with that image window still selected (the title bar will be blue), select Imitate Zoom from the Window menu.

The zoom factor and region of the selected image will be applied to all the images.

Note: Imitate zoom will only work on comparable images with similar dimensions.

Tiling Windows

If you have more than one image open, the Tile commands on the Window menu allow you to arrange your images neatly on the screen.

Tile will resize your windows and arrange them on the screen left to right and top to bottom.

Tile Vertical will resize your windows and arrange them side-by-side on the screen.

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Tile Horizontal will resize your windows and stack them top-to-bottom on the screen

10.2. Band Attributes

If you have identified bands in your image, you can display information about them using the Band Attributes command.

Select View > Band Attributes to open the dialog box.

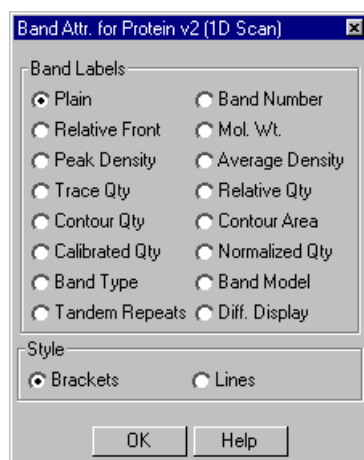


Fig.10-3. Band Attributes dialog box.

Click on any of the buttons in the Band Labels group to display that particular attribute next to each defined band on your image. (If you select Plain, no attribute will be displayed.)

The options in the Style group allow you to choose how your defined bands are marked—as brackets surrounding each band, or as a single line at the center of each band.

The brackets format is described in more detail in section 12.1., How Quantity One Identifies and Quantitates Bands.

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Lines are usually easier to read than brackets in images with closely packed bands, while brackets are better for displaying the boundaries of bands.

10.3. Density Tools

The Density Tools (under the View menu and on the Density Tools toolbar) are designed to provide a quick measure of the signal intensity of the data in your image.

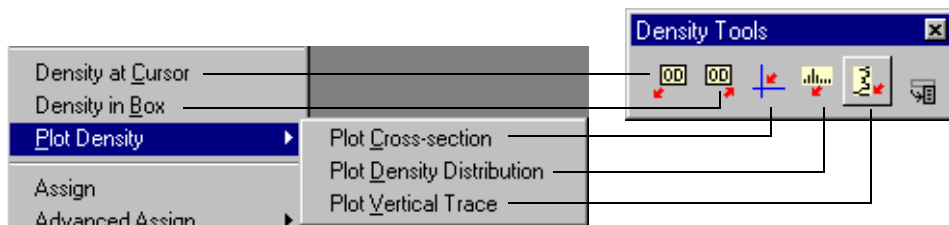


Fig.10-4. Density tools on the menu and toolbar.

Density at Cursor

Density at Cursor displays the signal intensity of the pixel on the image where you click your mouse. It also shows the average intensity for a 3 x 3 pixel box centered on that point.

To use this tool, click on the Density at Cursor button on the Density toolbar or select View > Density at Cursor. Then click on the point of interest on the image.

Density in Box

Density in Box displays the average and total intensity within a boxed region on the image. Select Density in Box from the Density toolbar or View menu, then define the region you want to measure by dragging your cursor across the image.

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Plot Density Distribution

Plot Density Distribution displays a histogram of the signal intensity distribution for the part of the image displayed in the image window. The average intensity is marked in yellow on the histogram.

The histogram will appear along the right side of the image. Use the Zoom functions to display the histogram for magnified regions of your image.

Plot Cross-section

Plot Cross-section displays an intensity trace of a cross-section of the image centered on the point where you click your mouse. The horizontal trace is displayed along the top of the image, and the vertical trace is displayed along the side of the image.

The intensity at the point you clicked on is displayed, as is the maximum intensity along the lines of the cross-section. Dragging the mouse with this function selected will continuously update the display.

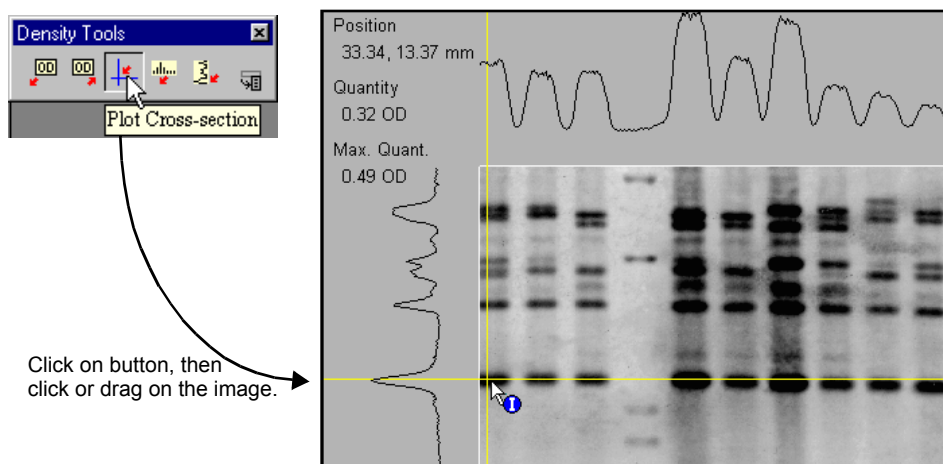


Fig.10-5. Plot Cross-section tool.

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Plot Vertical Trace

Plot Vertical Trace plots an intensity trace of a vertical cross-section of the image centered on the point where you click your mouse. Dragging the mouse with this function selected will continuously update the display.

Note: These traces will look different than the traces for functions like Plot Lane or Plot Band, because the sampling width for the Density traces is only one pixel.

10.4. Showing and Hiding Overlays

As you create plots and traces and identify bands or spots on an image, you create a variety of image overlays. Hide Overlays on the Main toolbar and View menu is useful for concealing any overlays you have created. It can also be used to erase plots, traces, and information boxes that are created by other functions.

Note: Clicking once on Hide Overlays will conceal the overlays. Clicking twice will deassign any function that has been assigned to the mouse.

View > Show Lanes and Bands reveals the lane and band overlays that have been added to an image.

10.5. Colors

View > Colors opens a dialog box in which you can adjust the colors of the image, windows, buttons, etc.

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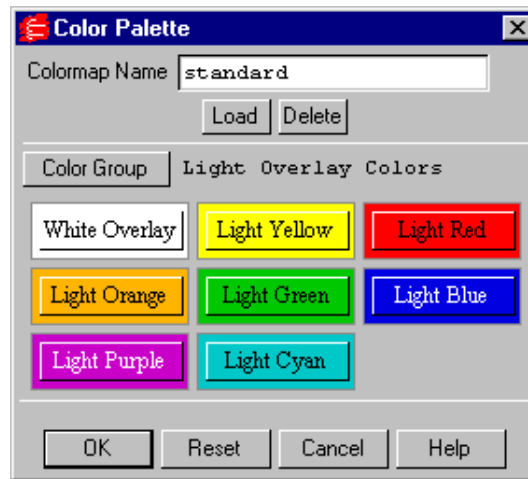


Fig.10-6. Colors dialog box.

Selecting a Color Group

Within the dialog box, the Color Group button allows you to select the colors of a particular group of objects (e.g., pop-up boxes, image colors, etc.). Click on the button to open the list of objects you can change.

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Fig.10-7. List of Color Groups.

Click on a color group in the list to select it.

Changing a Color

After you have selected the color group to change, click on the specific color button to change. The Color Edit dialog box will open, allowing you to adjust the red-green-blue (RGB) values of the color you selected.

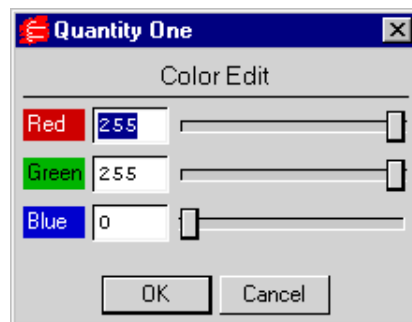


Fig.10-8. Color Edit dialog box.

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Drag the sliders or enter a value in the fields. The color of the button will change with your adjustments.

Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The Colormap Name field displays the name of a defined set of colors and color groups. There are several standard colormaps included with Quantity One, or you can create your own.

To select a predefined colormap, click on the Load button.

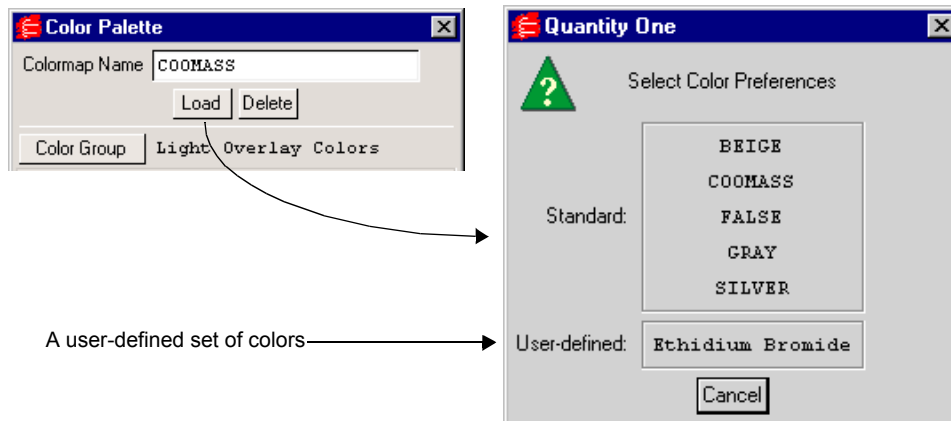


Fig.10-9. Selecting a Colormap.

From the list displayed, click on the set of colors you want to apply.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on OK to apply your changes.

To remove a colormap, click on the Delete button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

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If you change your mind about applying any changes you make, click on the Cancel button. If you want to return to the Standard colormap, click on the Reset button. All colors will be returned to their default values.

10.6. Transform

If features in your image are indistinct or fine details appear to be lost in background noise, you can use the Transform functions to optimize your displayed image. To open the Transform dialog box, select View > Transform.

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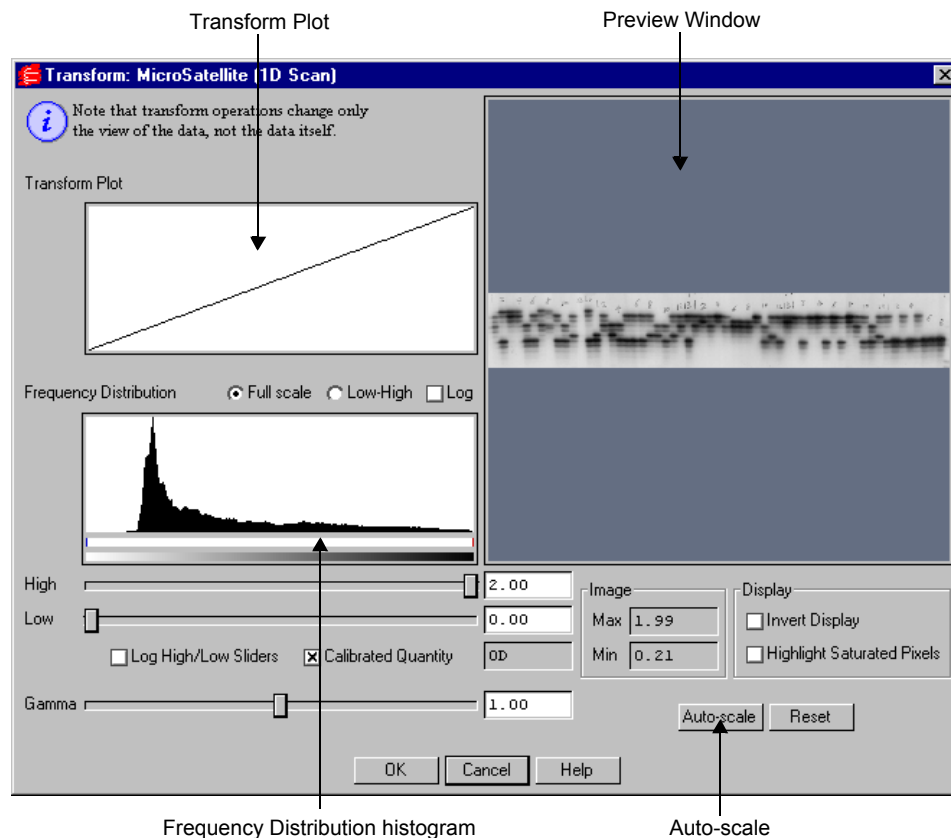


Fig.10-10. Transform dialog box.

The Transform dialog box contains a Preview Window, a Frequency Distribution histogram, a Transform Plot, and three main methods of optimizing your image: Auto-scale, High and Low sliders, and a Gamma slider. You can use these controls to adjust the way the software transforms raw data into visual data.

Note: Any changes made with the Transform controls will only affect how the image is displayed on your screen. *They will not affect the underlying data.*

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10.6.a. Subwindows in the Transform Dialog

Preview Window

The Preview Window in the Transform dialog shows a smaller view of the same image that is displayed in the main image window. Changes in the Transform controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

Frequency Distribution Histogram

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left (“gray”) end of the histogram due to background noise.

Transform Plot

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of your computer screen.

10.6.b. Transform Controls

Auto-scale

Clicking on the Auto-scale button will optimize your image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). This enhances minor variations in the image, making fine details easier to see. You can then “fine-tune” the display using the High, Low, and Gamma sliders described below.

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High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

As you drag the sliders, the slider markers on the Frequency Distribution histogram will move. Everything to the left of the Low marker will be remapped to minimum intensity, while everything to the right of the High marker will be remapped to maximum intensity. Using the histogram, you can position the markers at either end of the data range in your image, and use the low slider to cut off the "spike" of background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Log High/Low Sliders changes the feedback from the slider handles, so that when you drag them, the slider markers move a shorter distance in the histogram. This allows for finer adjustments when your data is in a narrow range.

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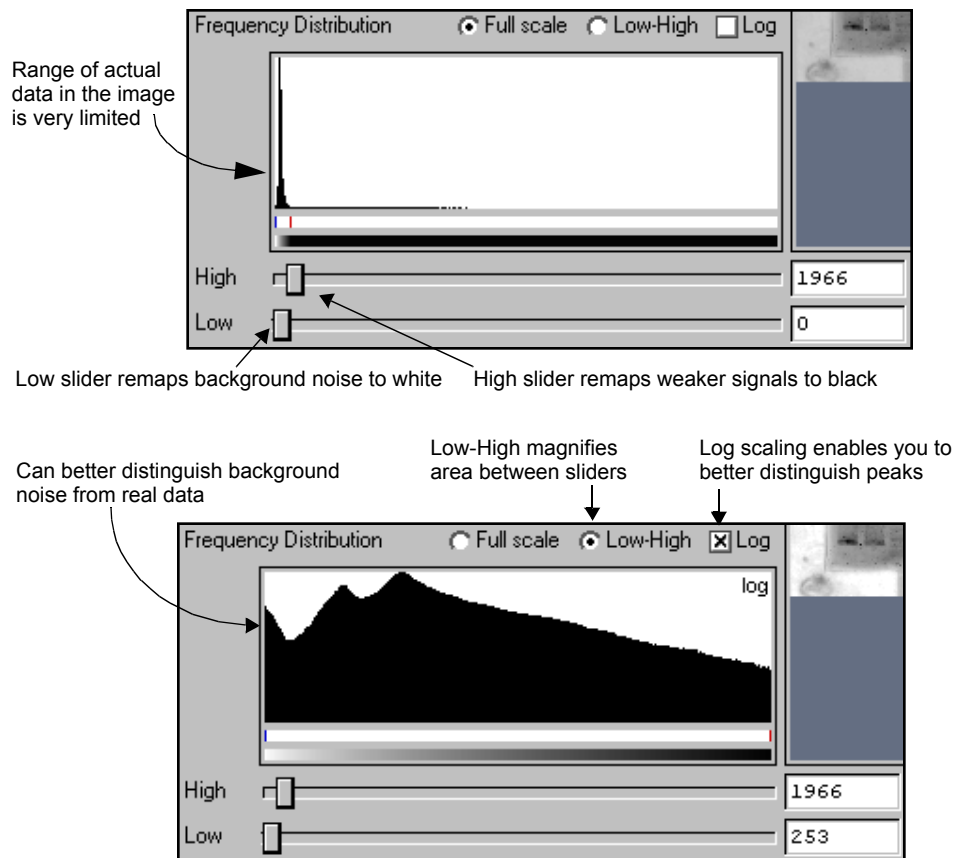


Fig.10-11. Two views of the Frequency Distribution histogram.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle expands or compresses the contrast range at the dark or light end of the range, and this is reflected in the Transform Plot and Preview Window.

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10.6.c. Other Features

Full Scale/Low-High

The Full Scale/Low-High option buttons adjust how the range of data in the image is displayed in the Frequency Distribution histogram and Transform Plot. They do not change how the data is displayed in the image window.

Selecting Full Scale adjusts the Frequency Distribution and Transform Plot displays so they show the full intensity range of the image.

Selecting From Low to High magnifies the range between the Low and High sliders. This makes it easier to view your data if it does not occupy the full intensity range of the image.

Log

The Log checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

Image Max/Min and Units

Image Max and Min display the range of signal intensity in the image.

The image units are determined by the type of scanner used to create the image. In the case of densitometers, you can select Calibrated Quantity to display your image units in O.D.s.

Invert Display

The Invert Display checkbox flips light bands on a dark background to dark bands on a light background, and visa versa. Once, again, the actual data will not change—only the image.

Highlight Saturated Pixels

When the Highlight Saturated Pixels checkbox is selected, areas of the image with saturated signal intensity are highlighted in red.

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Reset

If at any time you want to return to an unmodified view of the scan data, click on Reset.

10.7. Resizing and Reorienting Images

Frequently, your images will require some editing prior to analysis. This section describes the features that allow you to change the size and orientation of your images.

Quantity One's image editing tools are located on the Image menu and the Image toolbar.

Note: Many of these image editing commands are irreversible. A pop-up alert box will ask you to confirm each edit that will change your image irreversibly.

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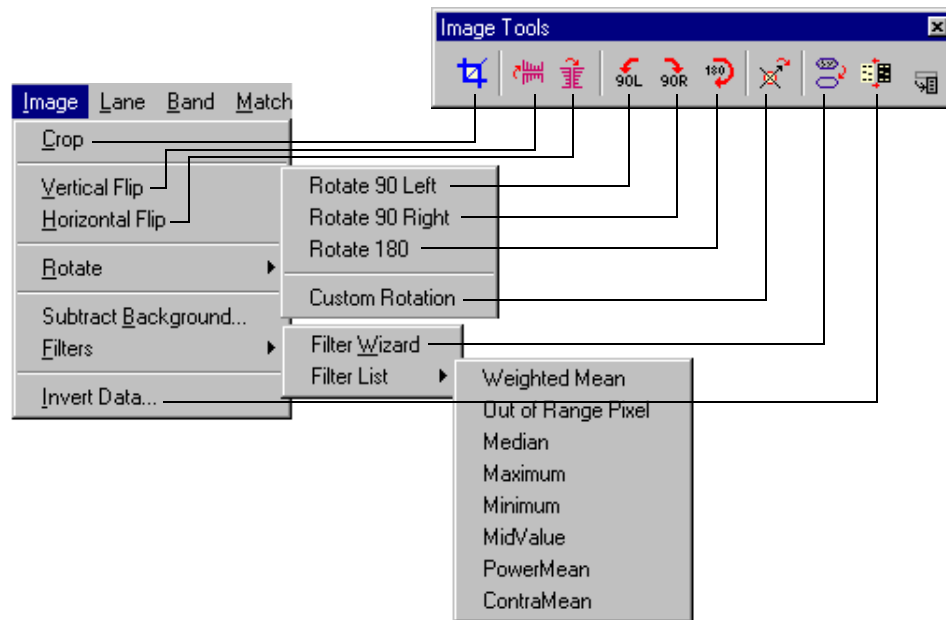


Fig.10-12. Image menu and toolbar.

10.7.a. Cropping Images

To eliminate unwanted parts of an image, you can use the Crop tool. This is also an easy way to reduce the memory requirements of an image.

To crop, click on the Crop button on the Image toolbar or select Image > Crop. Your cursor appearance will change to a Crop symbol.

Define the region to be cropped by dragging the cursor across the image, creating a box. Everything outside the box will be deleted.

Information about the physical dimensions of the crop area (given in millimeters and number of pixels) is listed at the bottom of the crop box, as is information about the memory size of the image inside the crop area.

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1. To *reposition* the crop box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.
2. To *resize* the box, position your cursor on a box border or corner. The cursor appearance will change to a bidirectional arrow. You can then drag that border or corner in or out, resizing the box.
3. To *redraw* the box, position your cursor outside the box. The cursor appearance will change back to the Crop tool, and you can draw another box, replacing the one you just drew.

Once you are satisfied with your crop box, position your cursor within the box slightly off-center. The cursor appearance will change to a scissors symbol. You can then click on the mouse to perform the crop.

A pop-up box will ask you whether you want to: (1) crop the original image, (2) save a copy of the area inside the crop box as a separate image, keeping the original image intact, or (3) cancel out of the cropping operation.

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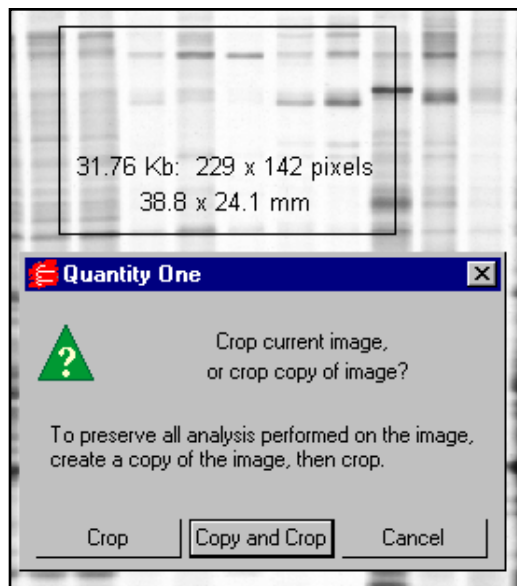


Fig.10-13. Crop box and pop-up Crop dialog.

If you select the Copy and Crop button, a dialog box will be displayed in which you can enter the name of the cropped image and its version number. Click on the OK button once you have finished.

10.7.b. Flipping and Rotating Images

If your image is not properly oriented, you can flip and/or rotate the image.

Note: These actions will erase any overlays you have created or analysis you have performed on an image. You will be asked to confirm the flip / rotation before the command is executed.

Flipping

To flip the image right-to-left, select Horizontal Flip from the Image menu or toolbar. To flip the image top-to-bottom, select Vertical Flip.

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90° Rotations

Select Rotate 90 Left, Rotate 90 Right, or Rotate 180 from the Image > Rotate menu or Image toolbar to perform the specified rotation. You will be asked to confirm your choice before the command is executed.

Note: Any analysis performed on the image cannot be rotated and will be lost.

Custom Rotation

If you need to rotate your image in increments other than 90°, you can use the Custom Rotation command.

Select Custom Rotation from the Image > Rotate menu or Image toolbar. A green “plus” sign will appear next to your cursor. Click on the image you want to rotate and a circular overlay with an orange arrow will appear. A small dialog box also will open, indicating the angle of rotation in degrees and radians.

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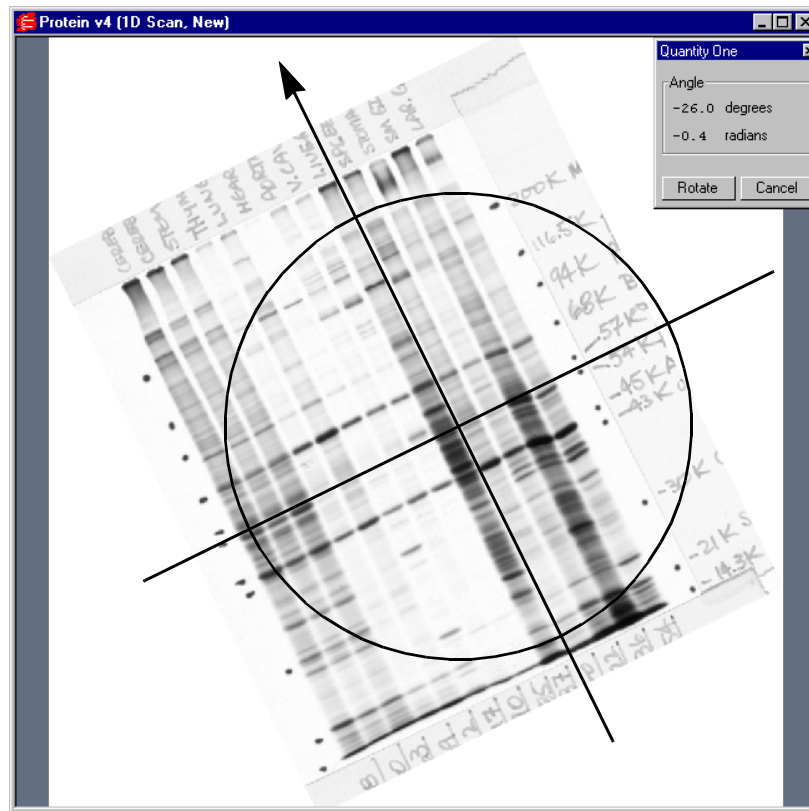


Fig.10-14. Custom rotation; the arrow points in the direction of the new top of the image.

To perform the rotation, position your cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the dialog box will change. Position the arrow so that it points in the direction of the new top of the image. You can “fine-tune” your rotation as much as you like.

Note: If you want to center your arrow on a particular point on the image (e.g., to align along a particular lane), you can use the F3 key command. Position your cursor on the point on the image you want to center on, and press the F3 key. The image will move so that the center of the arrow and your cursor point are aligned.

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To complete the rotation, click on the Rotate button in the small dialog box. Another, smaller image window will open containing the rotated image. You will then have the option of renaming your new image and changing the version number.

If you are not satisfied with your rotated image, simply delete it and start over.

Note: Because an image is composed of square or rectangular pixels, Custom Rotation has to perform some minor smoothing on the image to turn it at a non-90° angle. In addition, any analysis performed on the image cannot be rotated and will be lost.

10.8. Subtracting Background Noise from Entire Images

Quantity One has a number of tools for subtracting background noise from images to improve the clarity of your data. This section describes background subtraction for the entire image.

Quantity One also has background subtraction tools for individual lanes (see section 11.2.) and for volumes analysis (see section 14.3.).

Whole-image background subtraction is useful for reducing background noise resulting from the opacity of the carrier medium (film, gel matrix, or blot matrix) or film fogging.

Note: Whole-image background subtraction is an irreversible process. You will be asked if you want to make a copy of the image before completing the operation.

To open the dialog box for performing whole-image background subtraction, select Image > Subtract Background.

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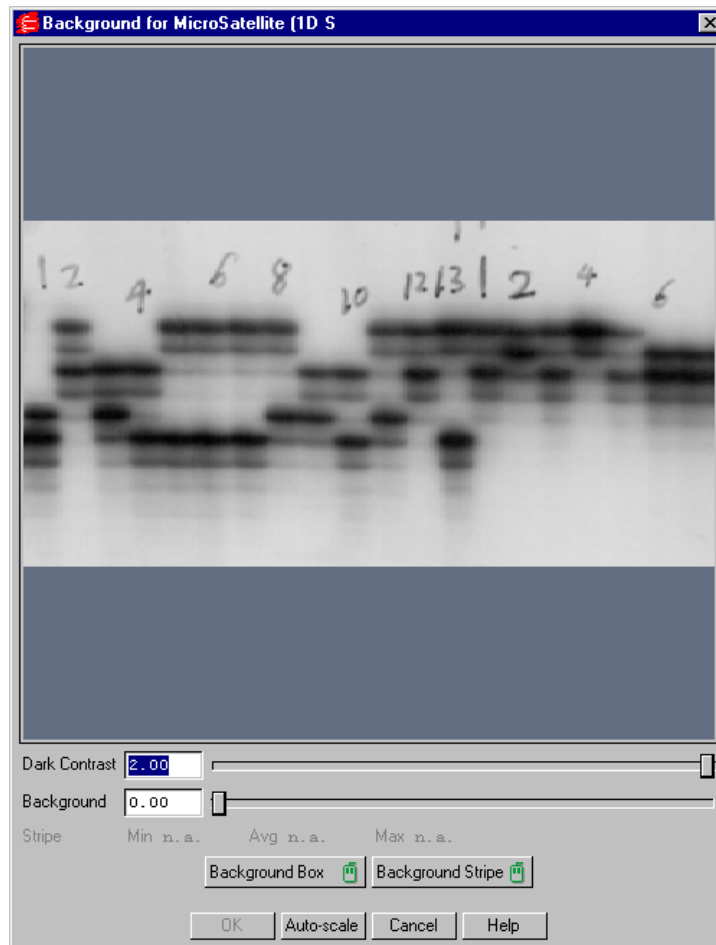


Fig.10-15. Subtract Background dialog box.

The Subtract Background dialog box has a Preview Window, which contains a smaller view of the same image that is displayed in the main image window. Changes in the subtract background controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

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You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

A description of each of the available features in the Subtract Background dialog is given below.

Auto-scale

Clicking on the Auto-scale button will automatically adjust your Dark Contrast and Background settings to optimal levels. You can then “fine tune” these settings using the other controls.

Dark Contrast Slider

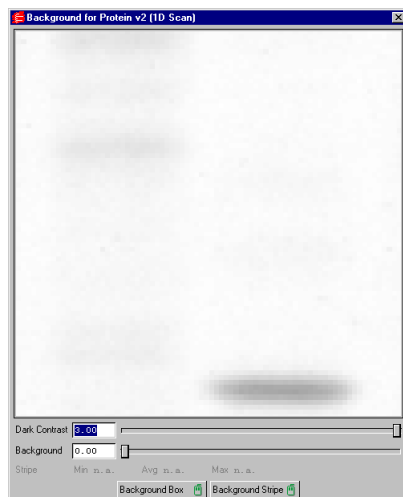
Adjusting the Dark Contrast prior to background subtraction allows you to make fine distinctions in the amount of background noise present in your image. Faint artifacts that may not be obvious to the unaided eye are revealed by lowering the Dark Contrast level to close to the subtraction levels.

The Dark Contrast slider is similar to the High slider in the Transform dialog box. Dragging the slider handle to the left will make faint signals appear stronger. You can also move the slider incrementally by clicking on the slider bar, or you can type a value into the field next to the slider.

Note: Adjusting the Dark Contrast slider by itself does not eliminate background noise; therefore, the OK button will not activate if you only adjust this slider. If you only want to adjust the display contrast, and *not* subtract background, use the View > Transform function.

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Example A: without adjustment



Example B: with adjustment

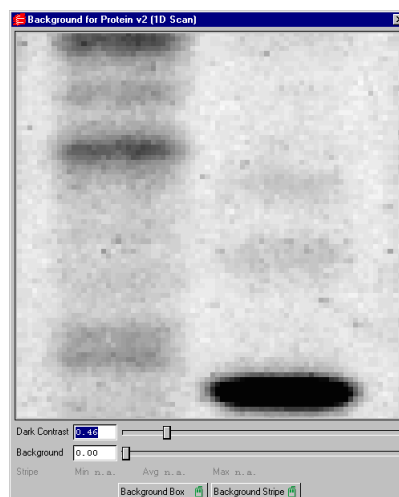


Fig.10-16. Images without and with Dark Contrast adjustment. Note that in Example A, with no Dark Contrast adjustment (3.00 O.D.), the surrounding medium appears free of background noise and image artifacts, but in Example B, with adjustment (0.46 O.D.), artifacts in the image become immediately obvious, indicating that the subtraction levels were not sufficient to remove all background artifacts from the gel image.

Background Slider

To manually adjust the background subtraction level for your image, drag the Background slider to the right. You can also move the slider incrementally by clicking on the slider bar, or you can type a specific background value into the field next to the bar.

Objects with signal intensities lower than the subtraction level will be eliminated from the image when you click on OK.

Background Box

This method of subtraction is best suited for images with uniform backgrounds.

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Click on the Background Box button, then drag on the image, defining an area that is representative of the background for your entire gel image. The average intensity of the pixels within the box will be used as the background level to be subtracted from your image.

Background Stripe

This function is useful when the image background is horizontally uniform but changes from the top to the bottom of the image, such as on a gradient gel.

To subtract a background gradient, click on Background Stripe, then drag on a background region to create a lane-like box down the length of the image. The average intensity of the pixels in every horizontal pixel row within the stripe will be subtracted from that entire pixel row. This way, if your image has more background noise at the bottom than at the top, more background will be removed from the lower regions of the image.

Note: Make sure that your background stripe box stretches down the entire length of the part of the image you are interested in. If your background stripe is shorter than your image length, the software will take the last background value at top end of the stripe and subtract it from the upper part of the image, and take the last background value at the bottom of the stripe and subtract it from the lower part of the image.

Completing the Subtraction

When you are happy with the background subtraction shown in the preview image, click on the OK button at the bottom of the dialog box.

Because whole-image background subtraction is an irreversible process, a pop-up box will give you the option of subtracting from the original image, creating a copy of the image to subtract from, or cancelling out of this operation.

If you choose to copy and subtract, you will be asked to enter a new name and /or version number for the new copy before the operation is performed. Once this information has been entered, the background subtraction will be performed.

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10.9. Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (like data) relatively unaffected. Quantity One has a wide range of filters for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

Note: Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of your image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

10.9.a. Using the Filter Wizard

The Filter Wizard is designed to guide you through the filter selection process. First, you will identify the type of noise in your image. Next, you will select the size of the filter to use on that noise. Finally, you will filter the image.

To open the Wizard, select Image > Filters > Filter Wizard or click on the Filter button on the Image Tools toolbar.

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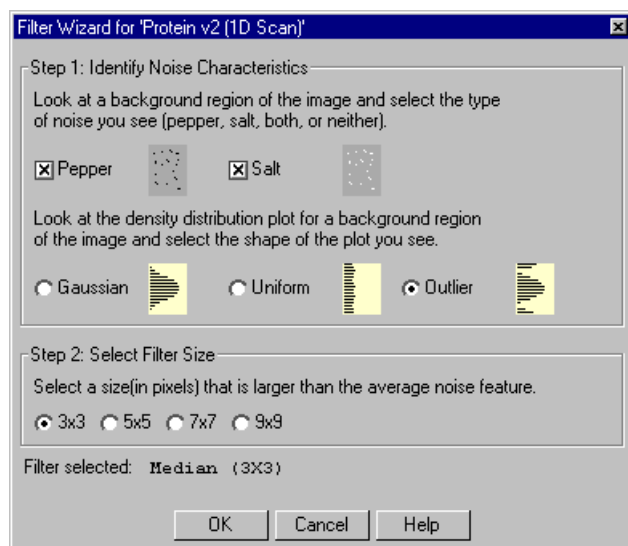


Fig.10-17. Filter Wizard dialog box.

When the Wizard dialog box opens, a vertical plot of the density distribution will pop up next to the image, showing the data range in the image and the amount of signal at each point in the range. (This is the View > Plot Density > Plot Vertical Trace histogram.)

This histogram can be used in conjunction with the Wizard to identify the type of noise in your image.

Step I: Identify Noise Characteristics

The first step in the Wizard is to identify the type of noise in your image. The Wizard allows you to select from a variety of different noise types and combinations.

To determine the type of noise in your image, it is a good idea to first magnify a background region of the image (i.e., a region where there is no data) using the Zoom Box command on the Main toolbar. Note that as you magnify a region, the density distribution histogram changes to reflect only the signal within the region you have magnified.

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After examining the magnified region, select one, both, or neither of the check boxes described below.

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range. This type of noise is common in electronic cameras with malfunctioning pixels. It can also be caused by dust or lint in the imaging optics or scratches on photographic film. Salt is a type of outlier noise (see below).
- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range. Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of your noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.
- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to Step 2.

Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from 3 × 3 pixels to 9 × 9 pixels. To select an appropriate size, magnify a background region of your image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than your data features.

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Note: A smaller filter will alter your image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

Step 3: Begin Filtering

After you have completed your selections, the filter name and size will be displayed at the bottom of the Filter Wizard dialog box.

To begin filtering, click on the OK button. Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to Copy and Filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

10.9.b. Selecting a Filter Directly

If you already know the type and size of filter you want, you can select it directly by selecting Image > Filters > Filter List. The pulldown menu includes all the filters available in the Discovery Series.

The types of filters in the Discovery Series are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt-and-pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt-and-pepper noise, this filter calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.

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- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being processed with the maximum value of the pixels within the filtering window.
- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.
- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.

To begin filtering, select a filter type from the pull-down list. A pop-up box will ask you to select a filter size.

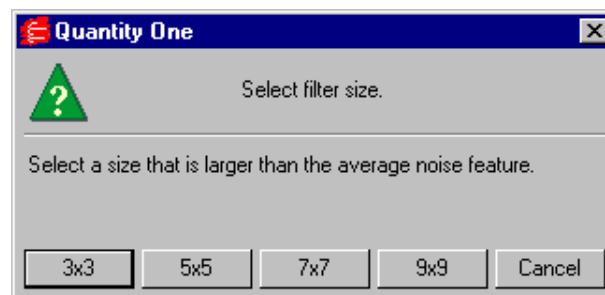


Fig.10-18. Selecting a filter size.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. (See the previous section for guidance on selecting a size.)

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Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to copy and filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

10.10. Invert Data

The Invert checkbox in the Transform dialog box (section 4.6.b) inverts only the appearance of the image. In some cases, however, you may want to invert the actual image data.

If your image has light bands or spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you will need to use the Invert Data function before you can analyze the image.

This function is reversible, so if you change your mind, you can always switch it back.

To invert your image data, select Image > Invert Data. You may need to use the Transform function to adjust the appearance of your inverted image.

10.11. Annotating Images

If you want to create and display textual notes directly on your image, select Annotation Tools from the Edit menu or Main toolbar.

This will open the Annotation Tools toolbar.

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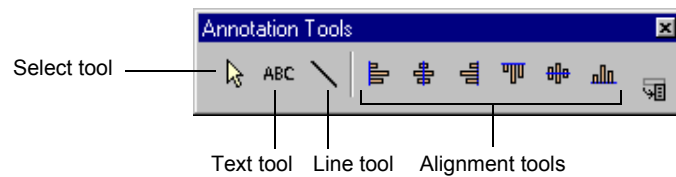


Fig.10-19. Annotation Tools toolbar.

To create an annotation, click on the Text Tool, then click on the image at the spot where you want the annotation to appear. This opens the Text Annotation Properties dialog box.

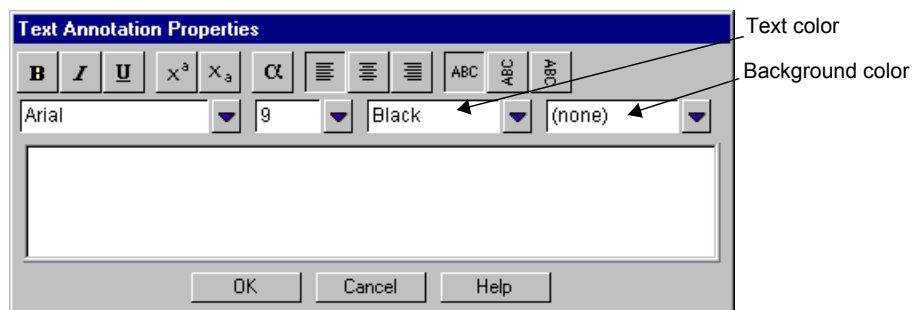


Fig.10-20. Text Annotation Properties dialog box.

To enter your annotation, simply begin typing in the main text field. The buttons in the dialog allow you to select the properties of your text, including format, alignment, and justification.

The pull-down boxes (from left to right) allow you to select the font style, font size, color of the text, and color of the background within the text box.

Once you have typed your annotation text, click on OK.

When you exit the Text Annotation Properties dialog by clicking OK, the annotation you just created will appear on the image at the spot where you originally clicked.

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Editing, Positioning, and Copying Individual Annotations

You must have either the Select Tool or the Text Tool assigned to your mouse before performing the following commands. To assign one of these functions, click on the appropriate button on the Annotation Tools toolbar.

- To edit an annotation, double-click on it to open the Text Annotation Properties dialog. The existing text will be displayed and can be edited.
- To move an annotation, position your cursor on it and drag.
- To copy, hold down the CONTROL key while you drag the annotation.
- To delete, click on the annotation once to select it, then press the DELETE key.

Line Tool

You can use the Line Tool on the Annotation Tools toolbar to draw a line between an annotation and an image feature, or between any two points of interest on your image.

Click on the Line Tool button, then drag on your image to create the line.

- To resize or adjust a line, position your cursor on one end of the line (marked by a circle) and drag.
- To move a line, position your cursor on the middle of the line and drag.
- To copy, hold down the CONTROL key while you drag the line.
- To add arrowheads to a line, double-click on the middle of the line. A dialog box will pop up with options to add arrowheads to one or both ends of the line.
- To delete a line, click on it once to select it, then press the DELETE key.

Working with Multiple Annotations/Lines

To edit multiple annotations and / or lines, first click on the Select Tool button on your Annotation Tools toolbar. Then either drag a box around the objects you want to select or hold down the SHIFT key while you click on individual objects.

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Note: When dragging a box around objects to be selected, make sure that the box entirely surrounds all the objects.

Each object in the selected group will have a green border.

You can drag the selected objects as a group, or click on the Alignment buttons on the Annotation Tools to align them.

Holding down the CONTROL key while you drag will duplicate all of the objects you have selected. Pressing the DELETE key will delete them.

You can create as many annotations and / or lines as you want.

Viewing Previously Created Annotations

To display previously created annotations after opening an image, select Annotation Tools from the Image menu.

If you have concealed all your overlays using Hide Overlays, clicking on any of the buttons on the Annotation Tools toolbar will display the hidden annotations.

10.12. Erasing All Analysis from an Image

If you want to delete all analysis that has been performed on an image (including lanes, bands, volumes, annotations, etc.), you can use the Clear Analysis command.

Select Clear Analysis from the Edit menu. Since this process is irreversible, Quantity One will ask you for confirmation before completing the command.

10.13. Sort and Recalculate

If the numbering of your lanes or bands (see Chapters 11 and 12) is incorrect because of an addition or deletion, you can renumber them using the Sort and Recalculate function.

Select Sort and Recalculate from the Edit menu. The lanes and bands in your image will be displayed, correctly numbered and labeled.

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11. Lanes and Arrays

Before you can use many of Quantity One's analysis functions, you must first define lanes and bands on your gel image. The next three chapters describe the tools for finding lanes and bands in an image, creating standards, and matching bands among lanes.

This chapter focuses on lanes, as well as the tools for defining dot and slot arrays.

11.1. Defining Lanes

You can define lanes individually or as part of a frame. The functions for doing this are under the Lane menu and on the Lane toolbar.

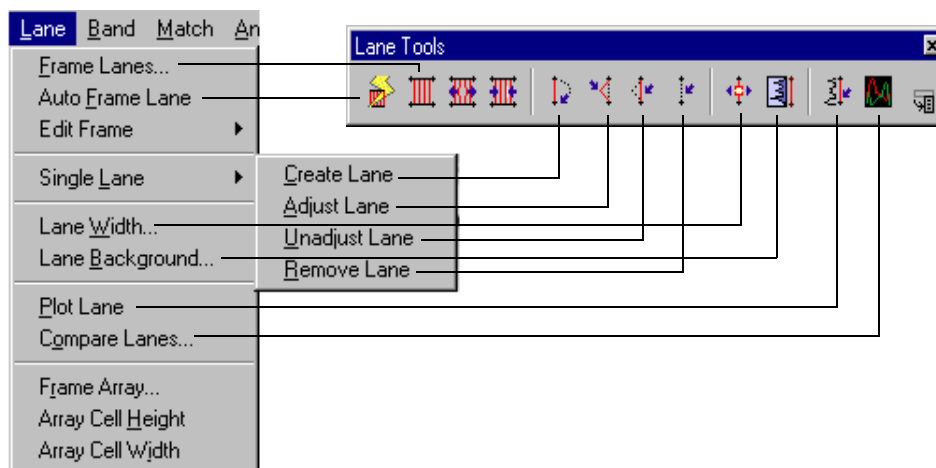


Fig.11-1. Lane menu and toolbar.

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11.1.a. Creating a Lane Frame

The fastest way to define all of the lanes in your image is to create a lane frame using the Auto Frame Lanes command. If Auto Frame Lanes does not work well on your images, you can create and place a lane frame manually.

Auto Frame Lanes

Note: Auto Frame Lanes works best with images with large numbers of clearly defined lanes and bands. Also, the lanes should be reasonably vertical.

Select Lane > Auto Frame Lanes or click on the button on the Lane Tools toolbar. The lane-finding program will automatically detect the lanes in your image and place a grid over them.

If Auto Frame Lanes detects too few or too many lanes, you can add or delete lanes using the single lane commands described below.

If Auto Frame Lanes does not work on the image, you will be prompted to create a lane frame manually.

If you are not satisfied with the frame created by Auto Frame Lanes, select Edit > Clear Analysis and use the manual Frame Lanes command.

Creating a Lane Frame Manually

Lane > Frame Lanes opens a dialog box in which you can type in the number of lanes in your gel.

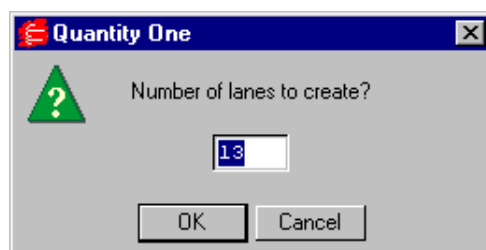


Fig.11-2. Frame Lanes dialog box.

Lanes and Arrays

Click on the OK button to complete the action.

Features of a Lane Frame

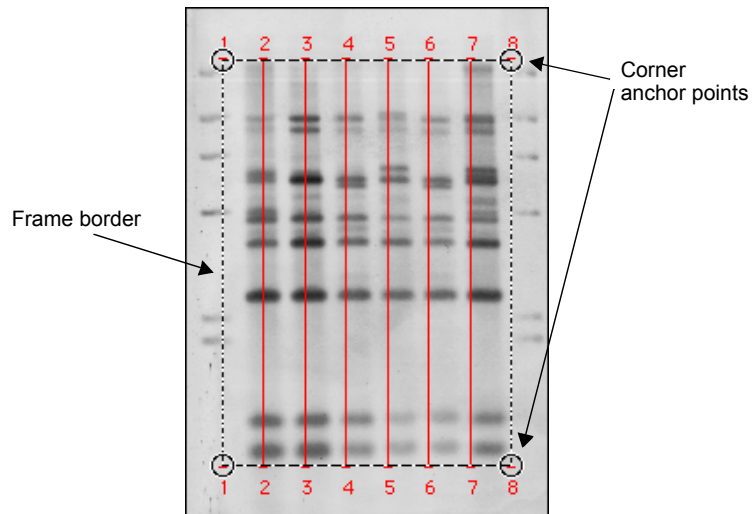


Fig.11-3. Features of a lane frame.

The lane frame overlay contains individual lanes numbered sequentially from left to right. The borders of the frame are marked with dashed lines, the interior lanes are solid red lines, and each anchor point (corner) is marked with a circle.

11.1.b. Editing the Frame

If your frame does not follow the lanes on your image to your satisfaction, you can adjust it using the frame editing commands. These are located on the Lane > Edit Frame submenu.

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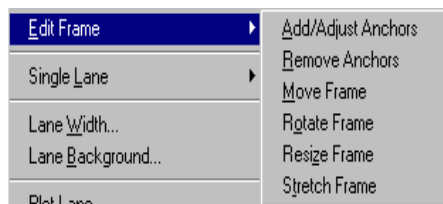


Fig.11-4. Edit Frame submenu.

Adjusting the Entire Frame

The following commands will adjust the entire frame:

- To stretch the frame (e.g., if you want to include additional bands at the top or bottom of the image), select Stretch Frame from the submenu and drag an anchor point in or out. The opposite anchor point will remain fixed while the frame expands or contracts.
- To move the entire frame to a new position, select Move Frame from the Lane > Edit Frame submenu and drag an anchor point. The entire frame will move.
- To rotate the frame, select Rotate Frame from the submenu and drag an anchor point. The entire frame will rotate.
- To resize the frame, select Resize Frame from the submenu and drag an anchor point in or out. The frame will expand or contract from the center.

Adjusting/Adding Individual Anchors

After you have created a frame, the Adjust Anchors function will automatically be assigned to your mouse.

To adjust an individual corner anchor point of a frame, select Adjust Anchors (if it is not already assigned to your mouse) and drag the anchor point. This will move both the anchor point and attached frame lines.

If the lanes are not straight or if the dye front smiles, you can create additional anchor points within the frame to change the shapes of individual lines.

Lanes and Arrays

Still using Adjust Anchors, click anywhere on the dashed lines of the frame border. This creates additional anchor points, both where you clicked and on the other side of the frame. This also creates another dashed line across the frame

You can then drag the new anchor points just as you did the corner anchor points, thereby “bending” the frame. Other internal anchor points can be placed along the new dashed line if further adjustments to the lane frame are required.

You can create as many additional anchors as necessary for the lines of the frame to correspond to the lanes on the gel image. If you change your mind about the position of an anchor, you can reposition it by simply dragging it.

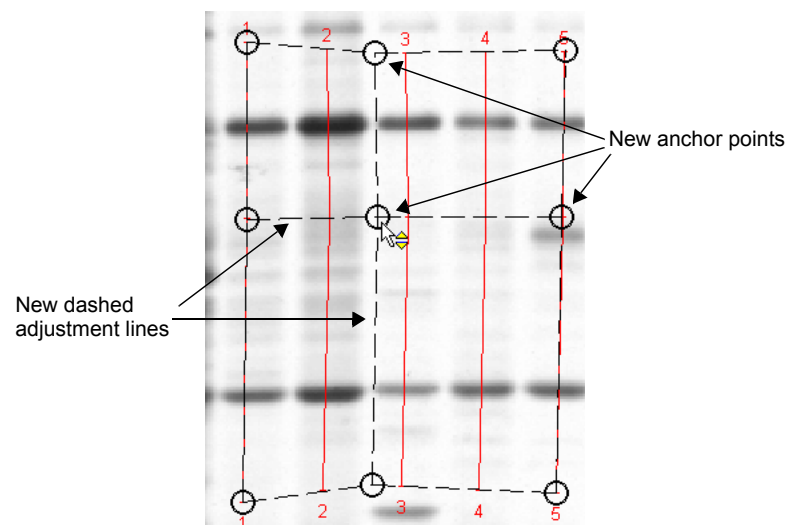


Fig.11-5. Adjusting the anchor points of the lane frame.

Removing/Unadjusting Anchor Points

If you decide to remove an anchor point, select Unadjust Anchors from the Edit Frame submenu or toolbar and click on the anchor you want to remove. The anchor will disappear and the lanes will “unadjust,” reflecting the removal of the anchor.

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11.1.c. Defining a Single Lane

You can define individual lanes using the single lane functions. These are located on the Single Lane submenu of the Lane menu or on the Lane Tools toolbar.

Note: You can use the single lane commands on lanes within a frame; however, the lane will be detached from the frame before the command is executed.

To mark an individual lane, select Create Lane, then drag a line from the top to the bottom of the desired lane.

Repeat this procedure to manually mark all the lanes you want to identify on your gel image.

Note: If while defining lanes your lane numbering gets out of sequence, click on Edit > Sort and Recalculate to renumber the lanes.

11.1.d. Adjusting Single Lanes

You can adjust the position of any individual lane line you create by using the Adjust Lane command.

Select Adjust Lane, then either drag one of the existing anchor points or click anywhere on the lane to create a new anchor point and drag it to the desired position. Repeat this procedure as many times as necessary until the lane line accurately follows the lane on the gel image.

11.1.e. Unadjusting Single Lanes

You can undo any of your lane adjustments with the Unadjust Lane command. Select Unadjust Lane from the menu or toolbar, then click on an anchor point you created to remove it. If you remove the anchor points at either end of the lane line, you will delete the line from the image.

11.1.f. Deleting Lanes

You can delete both single lane lines and lanes from a lane frame.

Select Remove Lane from the Lane menu or toolbar and click on the lane you want to remove. A pop-up box will ask you to confirm the deletion.

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Note: If you delete a lane from a group of lanes or a frame, click on Edit > Sort and Recalculate to renumber the remaining lanes.

11.1.g. Lane Sampling Width

The Lane Sampling Width command allows you to adjust the width of a single lane (in millimeters). Only the pixels within the sampling width are used for lane profiling and data quantitation purposes.

Select Lane Sampling Width from the Lane menu or toolbar, then click on a lane. This will open the Sampling Width dialog box, displaying the current width of the lane you clicked on.

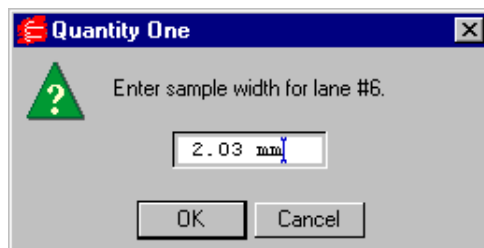


Fig.11-6. Sampling Width dialog box.

Enter a new width in millimeters and click on the OK button.

You can adjust the sampling width of all the lanes in your image from within the Detect Bands dialog box (see section 11.2.a). See section 12.1., How Quantity One Identifies and Quantitates Bands, for a full discussion of the effect of sampling width on band quantitation.

11.1.h. Creating a Lane Profile

After you have defined a lane, you can create an intensity profile of that lane using the Plot Lane function. A lane profile provides a quick visualization of the intensity of your sample data, and is also useful for determining the amount of background noise in your image.

Select Plot Lane from the Lane menu or toolbar, then click on a lane. A profile window will pop up.

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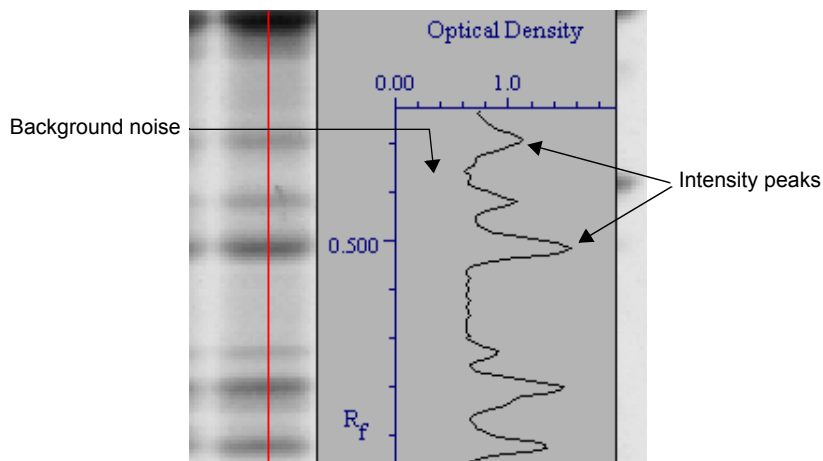


Fig.11-7. Profile of a defined lane.

To profile another lane, simply click on the lane with the Plot Lane command still assigned to the mouse.

The profile is generated by calculating the average intensity of the pixels in every horizontal scan line along the defined lane.

The profile shows the data objects in the image, represented by the intensity peaks in the profile, as well as the background noise in the image, represented by the region underneath the intensity peaks. The number of pixels used in this calculation depends on the lane sampling width (see previous section).

To close the lane profile window, click on Hide Overlays on the main toolbar.

11.2. Lane-Based Background Subtraction

After defining your lanes, we strongly recommend that you perform lane-based background subtraction. This is the best method for removing background noise from your lanes.

Select Lane Background from the Lane menu or toolbar. When you click on a lane, a lane profile will be displayed and the Lane Background dialog box will open.

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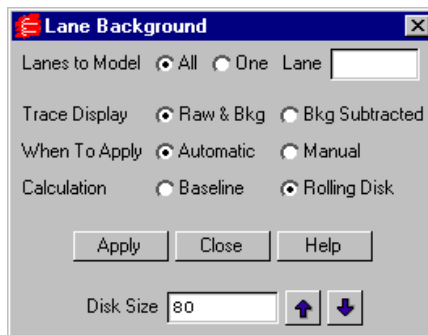


Fig.11-8. Lane Background dialog box.

The following information can be specified in the dialog.

Lanes to Model

If you want to apply the background subtraction to all the lanes of your image, click on the All button. Otherwise, select One and enter the number of the lane you want to perform the operation on.

Trace Display

The profile trace of the lane can be displayed in one of two ways. Selecting the Raw & Bkg option (with the functions Automatic and Rolling Disk activated) will display the original “raw” trace of the lane in white with an orange line representing the background noise beneath the peaks of the trace.

Clicking on the Bkg Subtracted button will display the lane trace as if all the background noise were actually removed. This display better enables you to visually compare the relative intensities of the bands in the lane.

When to Apply

As you change either the calculation method or the rolling disk size, the new settings can be applied to the image automatically every time a change is made by selecting the Automatic option.

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Alternatively, you can apply each new setting manually by selecting the Manual option and clicking on the Apply button whenever the settings are changed.

Calculation Method

If you select Baseline, background noise will not be subtracted from your lanes. Selecting Rolling Disk will allow you to use the Rolling Disk method of background subtraction.

“Rolling disk” refers to a hypothetical disk that follows the contour of the lane’s profile trace, removing different intensities along the length of the lane.

The size of the disk determines how much background will be subtracted. A large disk will follow the profile trace less closely, touching fewer points along the trace and identifying less background. A smaller disk will more closely follow the profile trace, thus identifying more background.

If you select Rolling Disk, the dialog box will expand to reveal a field showing the size of the rolling disk. The Disk Size is the disk radius in pixels.

You can type in a numerical value for the disk size or you can use the arrows to change the value in 10 percent increments.

A disk radius that is too large will result in slow execution and poor removal of background. A disk radius that is too small may subtract nonbackground intensity. Typical values range from 50 to 150.

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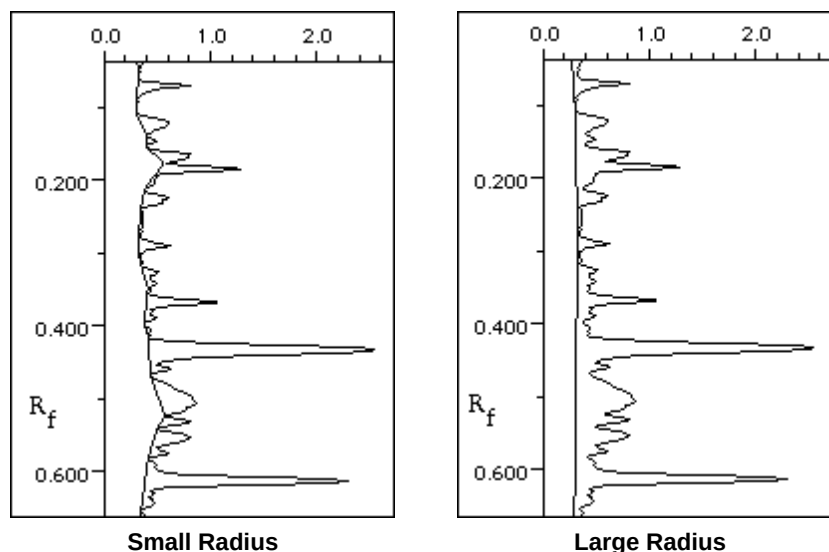


Fig.11-9. Examples of the background trace for small and large rolling disks. The small disk follows the profile trace more closely, resulting in more background subtraction.

Try different disk sizes to find the one that gives you the desired background noise trace.

When you apply the Rolling Disk background subtraction, the lane trace display will change but the image will not reflect the change in background intensity.

11.3. Compare Lanes

The Compare Lanes command opens a multicolored graph that allows you to superimpose the intensity profiles of any number of lanes present on any number of open images. Once a lane is displayed, Compare Lanes can zoom in on particular regions of interest and display them in greater detail.

Select Compare Lanes from either the Lane menu or toolbar, then click on the first lane you want to display.

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Note: Your image must have defined lanes for this command to work.

After the Compare Lanes window has opened, you can add lanes by clicking on the lanes you want to include.

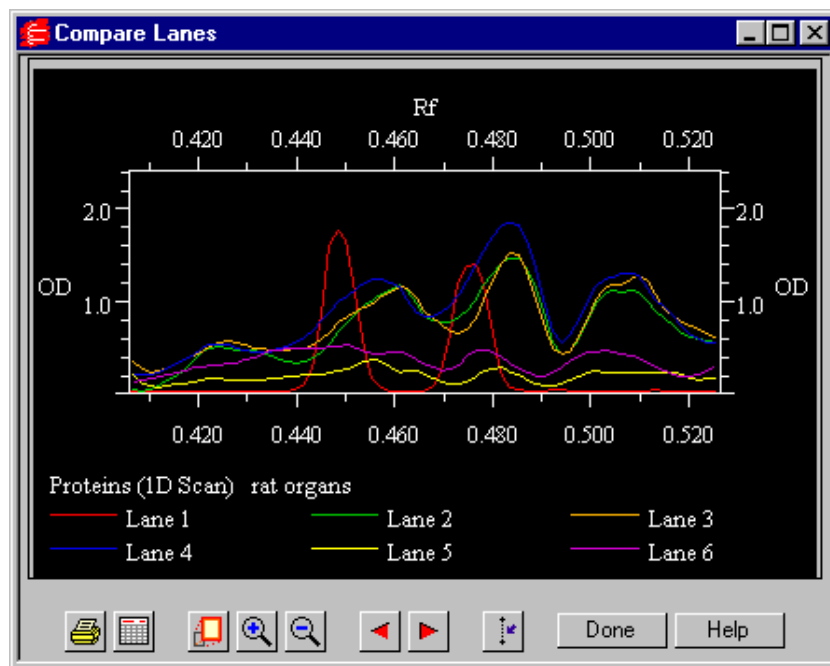


Fig.11-10. The Compare Lanes dialog box.

Each lane you select will be profiled in one of eight colors, and will be identified by color in a legend underneath the graph. If you select more than eight lanes to display, the colors will repeat themselves, but each lane will still be identified underneath the trace display. There is no absolute limit to the number of lanes that can be displayed simultaneously.

The X axis of the graph is Rf values and the Y axis is pixel intensity values. Compare Lanes automatically “best fits” lanes within the display window to maximize the range of intensity values included in the graph. Rf values are displayed from 0.0 to 1.0.

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Magnifying Within Compare Lanes

The Zoom In and Zoom Out buttons in the dialog can be used to zoom in and out within the display to magnify regions of interest. Dragging the cursor horizontally across the graph changes the cursor appearance to a bidirectional arrow and magnifies the range you defined by dragging.

Note: The magnifying functions in Compare Lanes only magnify the profile in the direction of the X axis. Therefore, the profile will appear to “stretch” without increasing in height.

The Full View button returns the graph to its complete, unadjusted appearance.

If you have zoomed in on part of the graph, the Left and Right scroll buttons in the dialog can be used to pan left or right.

Function Buttons

The Remove Lane button displays a dialog box that allows you to remove any lane profile from the graph. If you remove a lane, the color designations of the remaining lanes will change. Check the lane legend for an updated color code.

Click on the Print button to print a copy of the Compare Lanes display.

Click on the Export button to export the data in your graph to Excel or 1-2-3 spreadsheets.

11.4. Arrays

The array functions in Quantity One allow you to frame the cells in your array and specify their dimensions. You can then quantitate them using the Quantity Standards function described in section 13.3.

11.4.a. Defining an Array

If you are working with a dot array or a slot array, you already know the number of cells in your array and how they are arranged. Rather than

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creating lanes and detecting bands, you can skip these steps by telling Quantity One the number of columns and rows in your arrays.

Frame Arrays (on the Lane menu or Array Tools toolbar) lets you create such a frame in a single step.

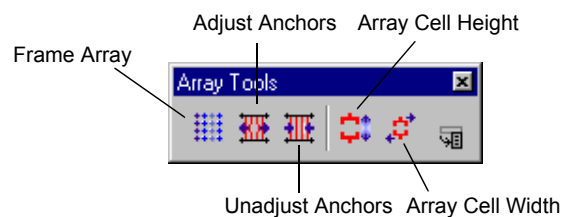


Fig.11-11. The Array Tools toolbar.

First, select Frame Array from the Lane menu or Array Tools toolbar.

A pop-up box will ask you to enter the number of columns in the array.

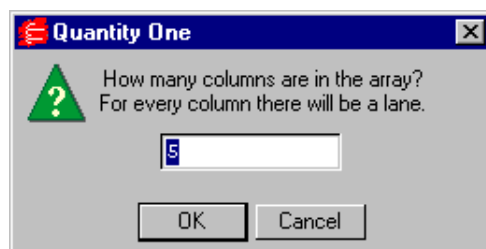


Fig.11-12. Setting number of array columns.

After you have entered the number of columns, click on OK. The next box will ask you the number of rows in your array.

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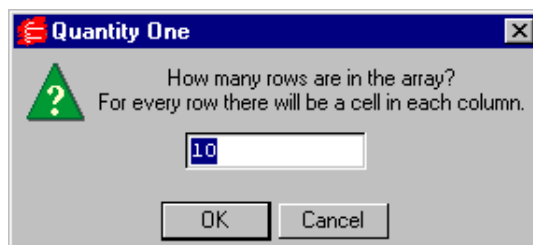


Fig.11-13. Setting the number of array rows.

Enter the number of rows and click on OK. Another pop-up box will ask you to confirm the creation of a array with the settings you specified.

If you choose to continue, an array matrix will appear on the image.

11.4.b. Adjusting the Array Matrix

Next, you will need to adjust the position of the array matrix so that it lines up with the regularly spaced cells on your dot/slot array. Select Adjust Anchors from the Array Tools toolbar or Lane menu to adjust the array frame (see section 11.1.b., Editing the Frame, for guidance on adjusting frames).

If necessary, the Adjust Lane (see section 11.1.d.) and Adjust Band (see section 12.4.b.) commands can be used to “fine-tune” the placement of columns and cells within the matrix.

11.4.c. Setting Array Cell Height and Width

You may want to adjust the width and /or height of the cells in your array to maximize the area being sampled.

The sample width and height should be set to encompass as much of the cells as possible without including background region. Including background will decrease the average pixel values in the quantitation.

Use the functions Array Cell Height and Array Cell Width (on the Array Tools toolbar or Lane menu) to change the height and width of the array cells.

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Note: When adjusting the cell height and width, it is useful to view your cells in “brackets mode.” To do so, select View > Band Attributes and select Brackets in the dialog box.

Array Cell Height sets the vertical distance between the brackets for every defined cell. When you select this function, a dialog box will ask you to specify the height in millimeters that should be used for all cells.

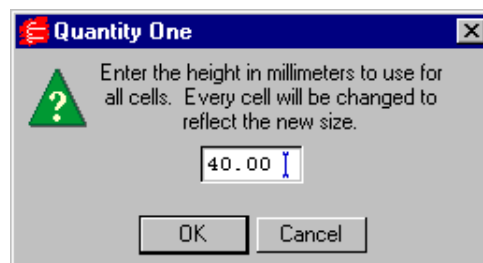


Fig.11-14. Setting the array cell height.

Changing this number and clicking on OK with your cells displayed as brackets will change your brackets in the vertical dimension.

Array Cell Width opens a dialog box in which you can set the width (in millimeters) of all the cells in your array.

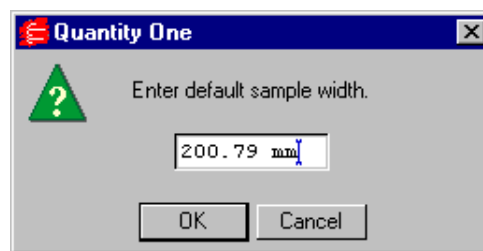


Fig.11-15. Setting the array cell width.

Changing this number and clicking on OK will change your brackets in the horizontal dimension.

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To change the width of individual columns in the array, use the Lane Sampling Width function. Select Lane Sampling Width from the Lane menu and click on the particular column in the array that you want to change. A dialog box will appear, asking you to enter the new sample width for the column you selected (see section 5.1.e for more details).

You can repeat this procedure with different sample widths to find the optimal width that includes the maximum area of the cells in the column without including background noise.

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12. Bands

Once you have defined the lanes on your gel image, you can automatically identify and quantitate the bands in those lanes using a set of adjustable parameters.

Note: You can also quantitate bands outside of lanes using the Volume Tools. See Chapter 14. for details.

The tools for band detection are located on the Band menu and on the Band Tools toolbar.

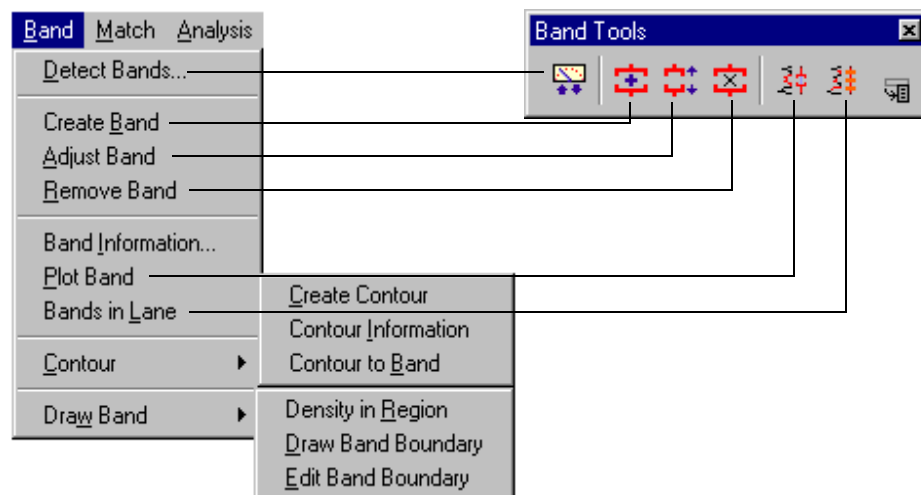


Fig.12-1. Band menu and Band Tools toolbar.

12.1. How Quantity One Identifies and Quantitates Bands

You can automatically identify all the bands in your image using the Detect Bands command, or you can mark them individually using the Create Band command.

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Each identified band is defined by brackets above and below the band. You can see these if you select the Brackets option button in the Band Attributes dialog box. (Select View > Band Attributes.) These brackets mark the boundaries of the band.

The width of each set of brackets is set by you before the bands are detected. This is the lane sampling width. The height of each set of brackets is determined automatically, using a band-finding formula together with parameters that you select.

When a band is quantitated, the average intensity value of each horizontal row of pixels within the brackets is calculated. Next, the number of pixel rows between the top and bottom brackets is determined. Taken together, these result in an intensity profile for the band.

Finally, the area under the profile curve to the baseline is integrated, resulting in units of intensity x millimeters. This is the “trace quantity” of the band.

Bands

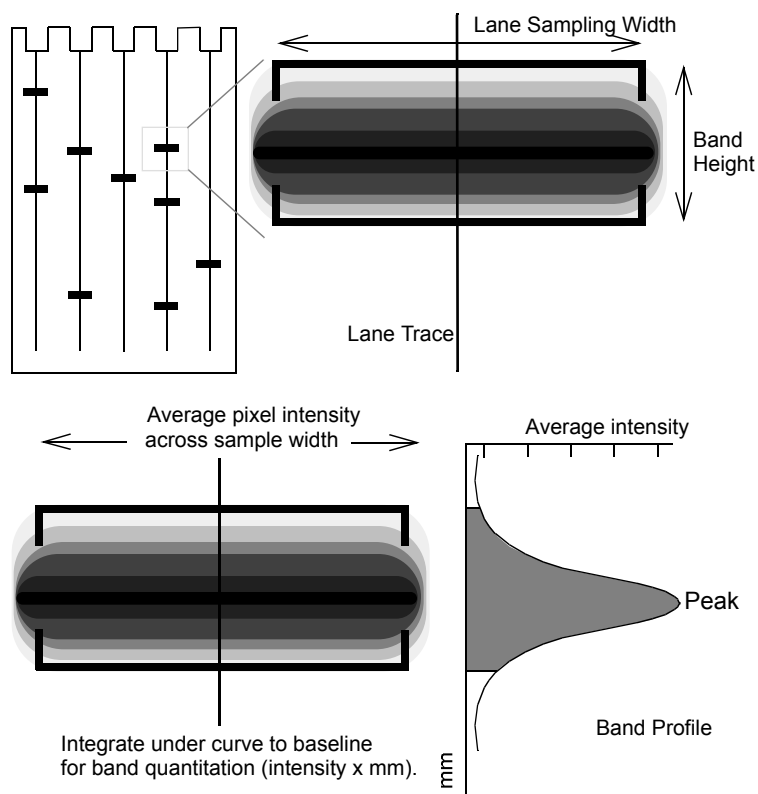


Fig.12-2. Illustration of bracket quantitation.

12.2. Automatically Identifying All Bands

To identify the bands in your image automatically, open the Detect Bands dialog box (Select Band > Detect Bands or click on the button on the Band Tools toolbar).

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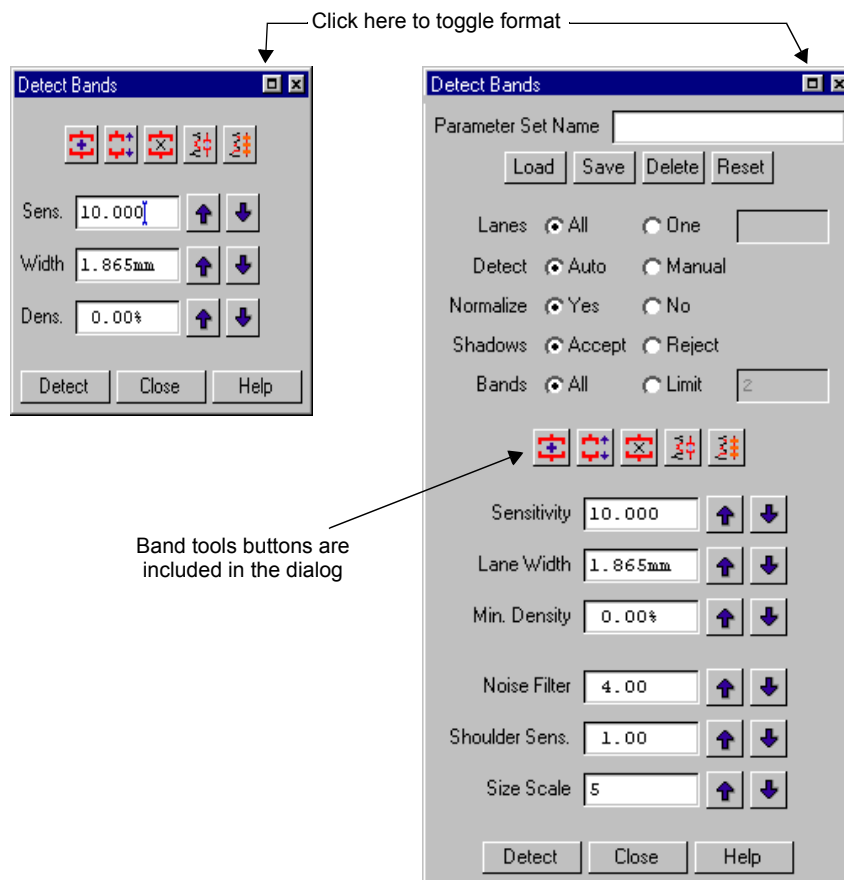


Fig.12-3. Detect Bands dialog box, short and long formats, with default values.

The Detect Bands function will find all of the bands in your defined lanes, based on parameters that you select.

Note: If you have already manually identified bands (using Create Band, Adjust Band, etc.), the Detect Bands function will overwrite your manual detection. For this reason, it is advisable to use Detect Bands first, and then manually add, adjust, or remove bands as needed.

Bands

The Detect Bands dialog box has both a short format and a long format, which you can toggle between by clicking the toggle box in the upper right corner of the dialog.

The long format allows you to set more of the parameters that affect band detection. Also in the long format, the detection parameters you select can be saved for use on future gel images.

To change any of the parameter settings, you can either type in a new value or use the arrows to increase or decrease the setting by 10 percent. Since the properties of your gel images are unique, we suggest that you experiment with various settings to find those best suited to your images.

12.2.a. Setting the Detection Parameters

Lanes to Detect

You can specify whether you want to detect the bands in all the lanes or in a single lane by selecting All or One next to the Lanes prompt. If you choose One, type the lane number in the field next to the One button.

Note: It may be necessary to detect one lane individually if the detection parameters applied to the whole image are not optimal for that lane (e.g., faint bands are not being detected or high background in that lane is being detected as bands). You can use the Detect Bands function to redetect bands in that lane using a different set of detection parameters.

When to Detect

If you select Auto next to the Detect prompt, band detection will occur immediately each time you change a detection parameter. You will not need to click on the Detect button located at the bottom of the form.

If you want to change more than one parameter before detecting, choose Manual. With the Manual option, you can change parameter settings first, and then apply them by clicking the Detect button.

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Normalization

Normalization is a way for band detection to compensate for differences in the darkness of the lanes on a gel image. *It does not normalize for band quantitation.*

If you select the No button next to the Normalize prompt, the band detection parameters are taken as absolute values and are applied in the same way to every lane on the image.

If you select Yes, the parameters are automatically adjusted according to the darkness of the lane. The darkness of the lane is determined by the darkest band in the lane. For example, suppose that all but one of the lanes on an image contain bands with an intensity of up to 50,000 counts. In the one light lane, the darkest band is only 25,000 counts. If normalization is turned on, band detection will be twice as sensitive when processing the light lane, improving the detection of faint bands.

Shadow Rejection

“Shadow bands” are common gel artifacts. Shadow bands are spaced at tandem repeat intervals and decrease in intensity as they progress further from a real band. The Shadows parameter is designed to limit the detection of shadow bands (see also Band Limit, below).

If you select Reject next to the Shadows prompt, the Detect Bands function will only find a band if it is darker than the one above it or if it is spaced further than one tandem repeat unit from the previous band. This greatly reduces the number of shadows identified as real bands.

If you select Accept next to the Shadows prompt, Detect Bands will not filter for shadows.

Band Limit

If you know that all the lanes in your gel image contain a specific number of bands, you can click on the Limit button next to the Bands prompt and type in the number of bands that you know are present. Only that number of bands will be detected in each lane, reducing the need for later editing.

Bands

Sensitivity

The Sensitivity setting determines the minimum signal intensity in the image that will be defined as a band. The higher the sensitivity value, the more bands will be detected.

If the sensitivity is set too high, background noise will be erroneously detected as bands. If the setting is too low, real bands may be missed.

The default sensitivity setting is 10.00. If your gel image has faint bands (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

Lane Width

The Lane Width determines the width along the lane lines that will be sampled for quantitative data. (You can also set the lane width for individual lanes using the Lane > Lane Width command.)

When a band is quantitated, an average intensity value for each horizontal row of pixels within the band brackets is calculated. You determine the number pixels in a row when you enter the Lane Width. The wider you set the Lane Width, the more pixels are included in the intensity average.

A Lane Width of two-thirds the actual lane widths in your gel is optimal for allowing high sensitivity with few false bands.

You should be careful not include background in the Lane Width, since this will decrease the average pixel values used in band quantitation. If you need to choose between including background versus leaving out part of the band, it is better to opt for the latter. Since you are taking the average intensity of all the pixels in a row, it's not critical if a few pixels are omitted from the band.

Minimum Density

When Normalization is turned off, this is the lowest signal intensity value that will be counted as a band.

Before entering a value for Min. Density, use the Lane > Plot Lane command to plot a trace of a lane that includes some faint bands. Then enter a value that is lower than the intensity of the peak of a faint band but is still above the background.

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If faint bands are still being missed after applying this parameter, double-check the intensity of faint bands on the lane trace and / or consider increasing the Sensitivity setting.

If Normalize is turned on, Min. Density is converted from an absolute value to a percentage. This percentage is the fraction of the signal intensity of the darkest band in the lane that will be detected as a band. For example, if the darkest band in a lane is 50,000 counts and the Min. Density is set to 25,000 counts, when you turn on Normalize, the Min. Density will switch to 50%. In other words, for a band to be detected it must be at least half as dark as the darkest band in the lane.

Noise Filter

The Noise Filter can be used to minimize the number of small fluctuations in the image (i.e., noise) that are called bands while still recognizing larger features (i.e., real bands). This distinction becomes increasingly critical the higher the Sensitivity parameter is set.

The Noise Filter value refers to the size of the filter in pixels (e.g., a value of 2.50 equals a filter size of 2.50 x 2.50 pixels). Features smaller than the filter size will not be recognized as bands. Entering a noise filter size of zero turns it off completely. The default value is 4.00.

If band detection calls a doublet a single band, decrease the Noise Filter setting and / or increase the Sensitivity.

You can also try decreasing the Size Scale parameter instead of the Noise Filter to improve the detection of closely-spaced bands. Be wary of decreasing both the Noise Filter and the Size Scale, as this may result in the fuzziness around bands being mistakenly detected as separate bands.

Shoulder Sensitivity

Normally, band detection tries to distinguish shoulders as separate bands. When looking at a lane trace, these bands appear as flat or gently sloping abutments to darker, better-defined bands (i.e., there is no dip on the trace between the two bands).

Bands

Increasing the Shoulder Sensitivity value will result in more shoulders being detected as bands. Changing this setting to zero will result in no shoulders being recognized as separate bands.

If band detection calls a doublet a single band, check the lane trace to see if there is a dip between the peaks of the two bands. If there is no dip, increasing the Shoulder Sensitivity value will help resolve the two bands.

Size Scale

The Size Scale field helps distinguish between trends in signal intensity and random intensity fluctuations. It is the number of pixels in a vertical column that are taken together to determine whether a band is present.

The Size Scale parameter is similar to the Noise Filter in that it uses the size of objects in the image to determine the nature of those objects. The default Size Scale setting of 5 pixels is optimal for most gel images. It can be set to any whole number greater than or equal to 3.

If a gel image has high levels of background noise, a larger Size Scale may be preferable. When noise permits, small values are preferred so that small features will be detected.

You may also choose to increase the Size Scale if your gel only has a small number of thick bands and it is scanned at high resolution.

12.2.b. Band Detection Parameter Sets

Saving Parameters

Once you have found a set of parameters that are optimal for detecting bands in your image, you can save them for future use on similar images. The Save button at the top of the Detect Bands dialog box allows you to do so.

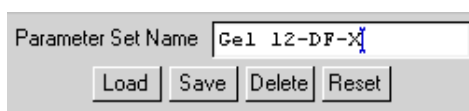


Fig.12-4. Parameter set controls in the Detect Bands dialog box.

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If you have not already entered a name for your parameter set in the Parameter Set Name field, you will be prompted to do so.

Loading Parameters

To load a previously saved set of parameters, click on the Load button.

Select the parameter set that you wish to load from the list. The values of that parameter set will be displayed in the Detect Bands form.

Deleting Parameters

To remove a set of parameters from your list, first load it, then press the Delete button. A dialog box will ask you to confirm the deletion.

Resetting Parameters

To return all detection parameters to their default values, click on the Reset button at the top of the form. Saved detection parameter files are not affected by this action, and can be reloaded at any time.

12.3. Band Attributes

Once you have identified bands in your image, you can display information about them using the Band Attributes command.

Select View > Band Attributes to open the dialog box.

Bands

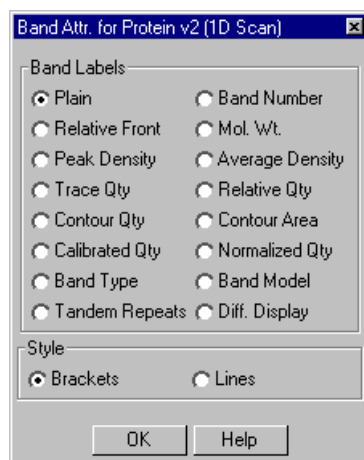


Fig.12-5. Band Attributes dialog box.

Click on any of the buttons in the Band Labels group to display that particular attribute next to each defined band on your image.

12.3.a. Summary of Band Attributes

Each band attribute is summarized below.

- **Plain**— No attribute is displayed.
- **Relative front**— The distance of a band from the top of a defined lane to the bottom, divided by the total length of the lane. This distance can be determined either by measuring a vertical line from the top of the lane to bottom or (if the lane is curved) by measuring along the length of the lane. Select Edit > Preferences to set the preferred measuring method.
- **Peak density** —The intensity of a band as determined by its peak intensity value.
- **Trace qty** —The quantity of a band as measured by the area under its intensity profile curve.
- **Contour qty** —The quantity of a band that has been identified using the Contour or Draw Band tools. It is the sum of the intensities of all the pixels within the band boundary multiplied by the area of each pixel.

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Units are intensity x mm².

- **Calibrated qty** —The trace quantity of a band that has been calibrated using quantity standards.
- **Band type** —The band type number of a particular band that has been matched and placed in a band set.
- **Tandem repeats** —The number of repeated base-pair units in a band that has been analyzed using the VNTR Calculations function.
- **Band number** —The sequential number of a band in lane, as counted from the top of the lane.
- **Molecular weight** —The molecular weight of a band in a lane, based on band position and the defined molecular weight standards in the gel.
- **Average density** —The total intensity of the rows of pixels used to generate the profile of a band, divided by the number of rows.
- **Relative qty** —The trace quantity of a particular band expressed as a percentage of the total quantity of all the bands in the lane. Alternatively, the trace quantity of a particular band expressed as a percentage of the total intensity of the entire lane. The calculation method (% of Lane / % of Bands) may be selected in Edit > Preferences.
- **Contour area** —The area (in mm²) inside the boundary of a band that has been identified using the Contour or Draw Band tools.
- **Normalized qty** —The trace quantity of a particular band expressed as a percentage of the quantity of a selected band type that is present in the same lane.
- **Band model** —Displays the modeling lines used by the matching algorithm to match bands.
- **Differential Display** —If your band types have been normalized, this displays trends in increasing or decreasing expression of a band type across a gel based on its normalized quantity.

12.3.b. Displaying Bands as Brackets or Lines

The options in the Style group of the Band Attributes dialog allow you to choose how your defined bands are marked—as brackets surrounding each band, or as a single line at the center of each band. Lines are usually easier to

Bands

read than brackets in images with closely packed bands, while brackets are better for displaying the boundaries of bands.

Note: When editing individual bands (see the following section), it is useful to display your bands as brackets. When you edit bands in brackets mode, a trace diagram will be displayed to help you determine band boundaries.

The boundaries of a band can be defined as either the troughs between peaks on the trace or as the inflection points on the curves defining the peaks. Most people looking at a gel will define a band by its inflection points, which correspond to the visual limits of the band. When using the trace profile of a gel image, they tend to include much more of the band's tail.

12.4. Identifying and Editing Individual Bands

To manually identify and edit individual bands in your image, you can use the Create, Adjust, and Remove Band commands on the Band menu and Band Tools toolbar. (These are also available as buttons in the middle of the Detect Bands dialog box.)



Fig.12-6. Create, Adjust, and Remove Bands buttons.

12.4.a. Identifying Individual Bands

In Brackets Mode

With your bands displayed as brackets, first select Create Band from the menu or toolbar, then click on either the top or bottom boundary of the band of interest. An intensity trace of the lane will pop up next to the band.

Drag the cursor until the area of the band that you want to define—represented by a peak on the intensity trace—has been completely enclosed. As you drag, lines extending from the band boundaries on the image to the corresponding boundaries on the intensity trace will appear.

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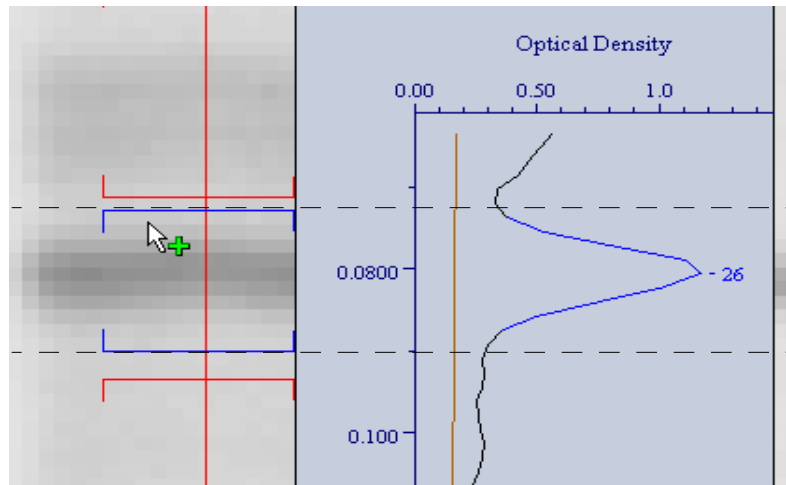


Fig.12-7. Creating a band in brackets mode.

When you release the mouse button, brackets will appear enclosing the region that you have called a band on the image. The corresponding section of the intensity trace will be highlighted.

In Lines Mode

With your bands displayed as lines, first select Create Band from the menu or toolbar, then click on the center of the band of interest. A line will appear at the center of the band at that point on the lane.

Bands

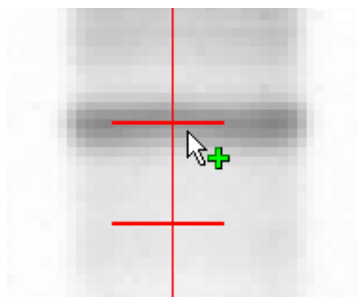


Fig.12-8. Creating a band in lines mode.

In lines mode, no intensity trace will pop up when you click on the image.

Note: After identifying several bands, be sure to renumber the bands in your image by selecting Sort and Recalculate from the Edit menu.

12.4.b. Adjusting Bands

After you identify the bands on your image—either automatically or manually—you may want to go back and change the placement of the band boundaries.

To reposition a band's boundaries, select Adjust Band from the menu or toolbar.

If you are in brackets mode, click on the upper or lower bracket of the band that you want to adjust. If you are in lines mode, click near the band you want to adjust.

If it is not already shown, an intensity trace will be displayed. The band being adjusted will be highlighted, and lines extending from the band boundaries on the image to the corresponding boundaries on the intensity trace will appear.

Drag the band boundary to the desired location, then release the mouse button.

12.4.c. Deleting Bands

You can use Remove Band to delete bands from lanes.

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Select Remove Band from the menu or toolbar, point the cursor at the unwanted band, and click the mouse button.

If your bands are displayed as brackets, a trace of the band will be displayed and a pop-up box will ask you to confirm removal of the band. Click on Yes to delete the band or No to cancel the procedure.

If your bands are displayed as lines, the band will simply be deleted.

Note: If your bands are displayed as lines, you can delete more than one band at a time. With Remove Band assigned to your mouse, drag a box around the bands to be removed. The bands in the box will be deleted when you release the mouse button.

If you delete a band, the region will no longer be counted as a band, but its intensity will still contribute to the total lane intensity.

Note: After removing bands, be sure to renumber the bands in your image by selecting Sort and Recalculate from the Edit menu.

12.5. Plotting Traces of Bands



Fig.12-9. Plot Band and Bands in Lane buttons.

Band > Plot Band (also on the Band Tools toolbar) displays an intensity trace of any band that you select on the image. First select the command, then click on the band of interest.

Band > Bands in Lane (also on the Band Tools toolbar) displays an intensity trace of any lane you select and highlights all the defined bands in that lane. First select the command, then click on the lane of interest.

Bands

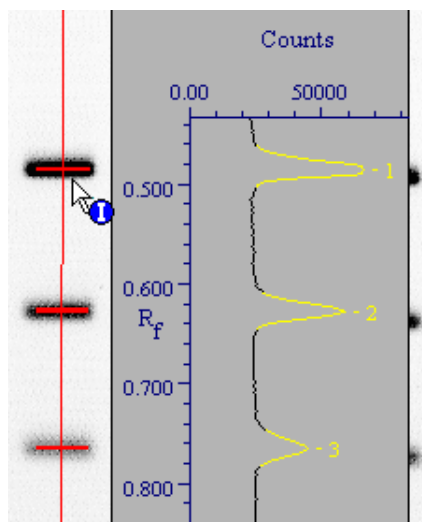


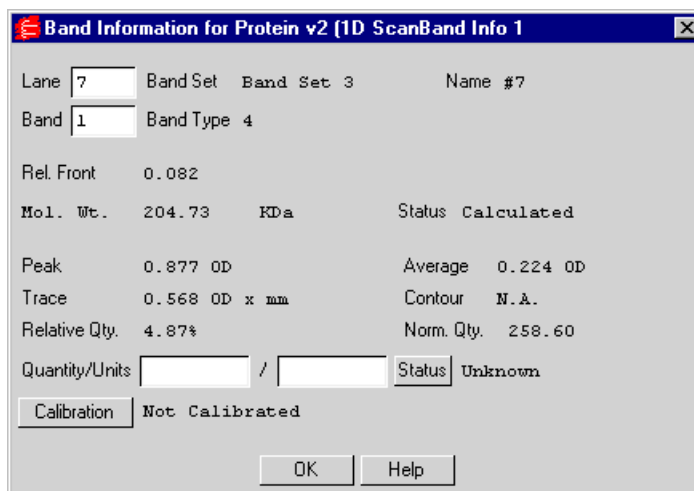
Fig.12-10. Bands in Lane command.

12.6. Displaying Band Information

You can display and enter information about individual bands using the Band Information dialog box. To open this dialog, select Band Information from the Band menu and click on a band of interest.

An intensity profile of the band's lane will be displayed, the selected band will be highlighted, and the dialog box will be displayed.

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The dialog box displays the following information:

Lane	7	Band Set	Band Set 3	Name	#7
Band	1	Band Type	4		
Rel. Front	0.082				
Mol. Wt.	204.73	KDa		Status	Calculated
Peak	0.877 OD			Average	0.224 OD
Trace	0.568 OD x mm			Contour	N.A.
Relative Qty.	4.87%			Norm. Qty.	258.60
Quantity/Units		/		Status	Unknown
Calibration	Not Calibrated				

Buttons: OK, Help

Fig.12-11. Band Information dialog box.

The Lane and Band fields at the top of the form allow you to type in the numbers of other bands on your gel image. The information displayed in the Band Information dialog box will update to reflect the new band, as will the trace overlay and band highlighting in the image window.

If the band is part of a Band Set and is of a known Band Type (see Chapter 12), this will be noted in the form.

If the quantity of the band you have selected is known, you can enter it next to the Quantity / Units prompt.

If you want to calibrate that band against known quantities, you can do so by clicking on the Calibration button. You will be asked to select the calibration curve you want to use. (See section 13.3. for information on calibration curves.)

12.7. Identifying Irregularly Shaped Bands

If the bands on your gel image are irregularly shaped, you can use the contour or drawing features to define them. These functions give you more control over defining your bands than either Detect Bands or Create Bands.

Note: The contour functions will create contours semi-automatically, whereas the drawing functions allow you create bands entirely manually. At least one lane must be defined on your image before you can create a band with the contour or draw band functions.

Contoured or hand-drawn bands are quantitated based on the signal intensity of all the pixels within the band boundary, using the following formula:

Quantity = Sum of (intensity of pixel x pixel size) for all the pixels in the boundary.

Quantity One first multiplies the intensity of a pixel by the area of the pixel. This is done for all the pixels that are within the contour or drawn boundary. The area of the pixel is determined by the scan resolution of the image.

The resulting values have units of intensity x mm².

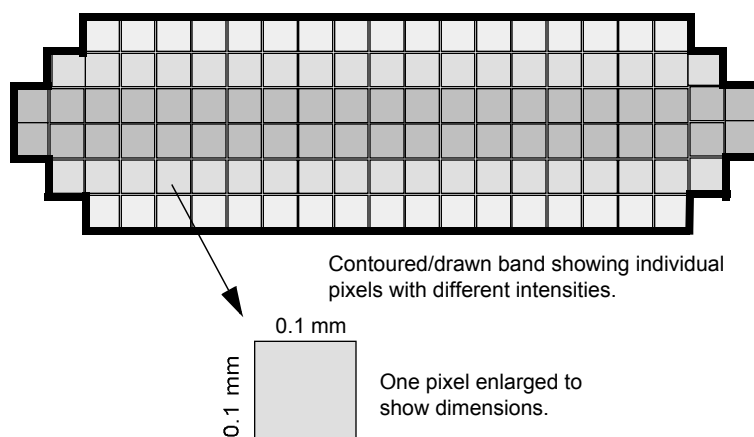


Fig.12-12. A contoured/drawn band scanned at 100 x 100 microns (micrometers).

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The functions needed to contour and draw bands are found on the Band > Contour submenu and the Contour Tools toolbar.

Before using any of these functions, we suggest that you magnify the image so that the individual pixels in the band are clearly visible. This allows you to position the cursor more accurately.

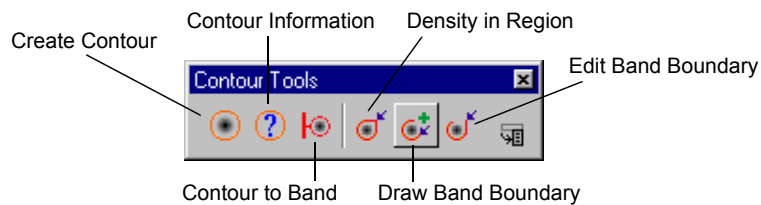


Fig.12-13. Contour Tools toolbar.

12.7.a. Contouring Bands

Creating Contours

Select Create Contour from the Band > Contour submenu or Contour Tools toolbar and click on a pixel at the edge of the band. This will display a contour that encloses pixels whose intensity is equal to or greater than that of the pixel at the cursor.

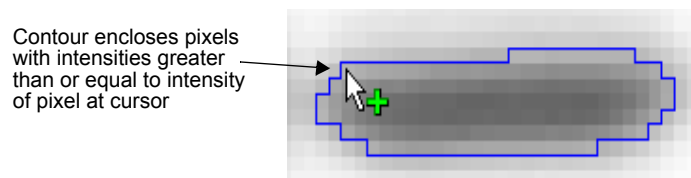


Fig.12-14. Creating a Contour

If the contour does not encircle the band, reposition the cursor and click again. A new contour will be drawn in place of the old.

Bands

Contour Information

To display information about the contour, select Contour Information while the contour is highlighted. A pop-up box will display the area, total intensity, and average intensity of the contour.

Converting a Contour into a Band

When you are satisfied with the contour, you can redefine the contour as a band with the function Contour to Band.

Note: Before you can convert a contour into a band, you must define at least one lane on your gel image. You can define a lane using the Create Lane tool while the contour is currently displayed.

Selecting Contour to Band from the menu or toolbar converts any currently displayed contour into a band and assigns that band to the nearest lane. The contour boundary will change color from yellow to red, and a band will appear on the nearest lane.

Once a contour is converted, all the functions in the Bands menu can be applied to it.

12.7.b. Calculating the Density in an Arbitrary Region

Use Density in Region to get intensity and area information about any region on an image.

Select Density in Region from the Band > Draw Band menu or Contour Tools toolbar. The cursor will change to an “information cursor.” Drag the cursor as if it were a pen to enclose a region of interest.

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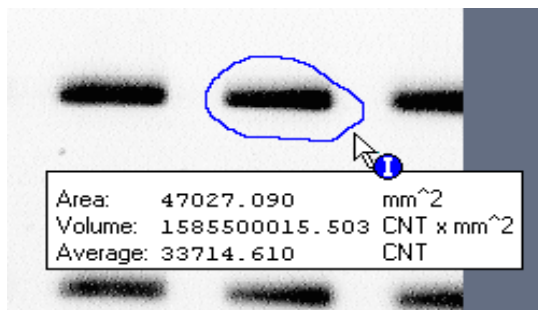


Fig.12-15. Density in Region tool.

When you close the border, it will change color and a pop-up box with information about the enclosed area will be displayed.

For very small regions, you will have to magnify the region before using this command.

12.7.c. Drawing Bands

Quantity One has tools for drawing bands directly, without having to detect them first.

Note: You must have a lane defined on your image before using the draw tools. The drawing tools work best if you first use the Zoom commands to magnify the region you want to draw in.

Drawing Band Boundaries

If you want to draw the boundary of a band manually, you can do so using Draw Band Boundary.

Select Draw Band Boundary from the Band > Draw Band menu or Contour toolbar. Your cursor will change to a pencil symbol.

Drag the cursor around the perimeter of the region you want to define as a band. A line will appear, defining the boundary.

Bands

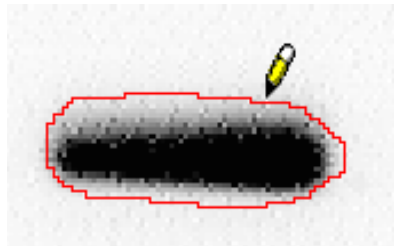


Fig.12-16. Draw Band Boundary tool.

If you make a mistake and need to retrace part of the band boundary, simply backtrack with the cursor; the previous path will be erased and you can redraw it.

Note: It is important to magnify the image before drawing a boundary. If you try to draw a very small boundary, the software will think that you are backtracking and erase the boundary.

When the cursor crosses over the line you are drawing, the drawn band is complete and the color of the line will change to indicate that it is a boundary.

If you keep drawing, each time the line crosses itself a new band will be created, replacing the old band.

Editing Band Boundaries

If your drawn band needs minor adjustments, you can fix it using Edit Band Boundary.

Select Edit Band Boundary from the Band > Draw Band menu or Contour toolbar. Your cursor will change to a pencil symbol.

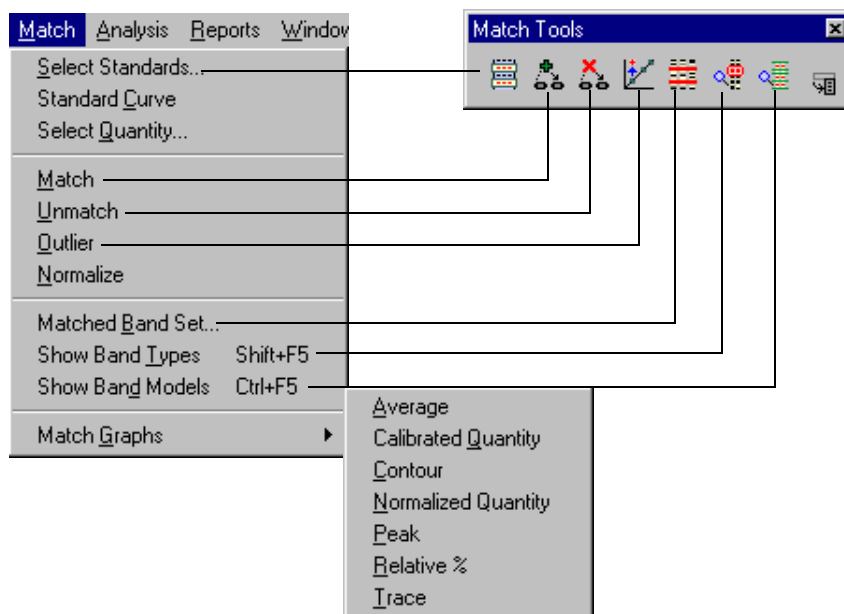
Drag the cursor across the previously defined boundary; a line will appear. When you recross the old boundary, the line will change colors and the new boundary will be created.

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13. Standards and Band Matching

After you have identified your lanes and bands, you can use Quantity One to define the molecular weight, band type, or other positional standards in your gel image and match the other bands using those standards. You can also identify bands of known quantity in your image and use these to generate a calibration curve for absolute quantitation of your unknown bands.

These tools are found on the Match menu and Match Tools toolbar.



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13.1. Creating and Applying Standards

If you have run a set of standards in one or more lanes on your gel, you can enter the values for those standards into Quantity One and use them to generate a standard curve and interpolate values for bands in other lanes.

Using the Standards command, you type in the values of your standard bands and apply this standard band set to the standard lane(s) on your gel image. Quantity One then automatically calculates the values for the remaining bands in the image, based on the standard band set.

If you have sets of standards that you regularly run on your gels, you only need to enter them once. You can then save them and access them every time you analyze a new gel image.

13.1.a. Creating a Set of Standards

To enter the values of your standards, choose Standards from the Match menu or Match Tools toolbar. A series of dialog boxes will guide you through the selection process.



Fig.13-1. Selecting a set of standards.

The first dialog box will ask you to select new standards or load previously created standards (if any have been saved).

Select New Standards to create a new set of standards. A dialog box will pop up in which you can specify the units.

Standards and Band Matching

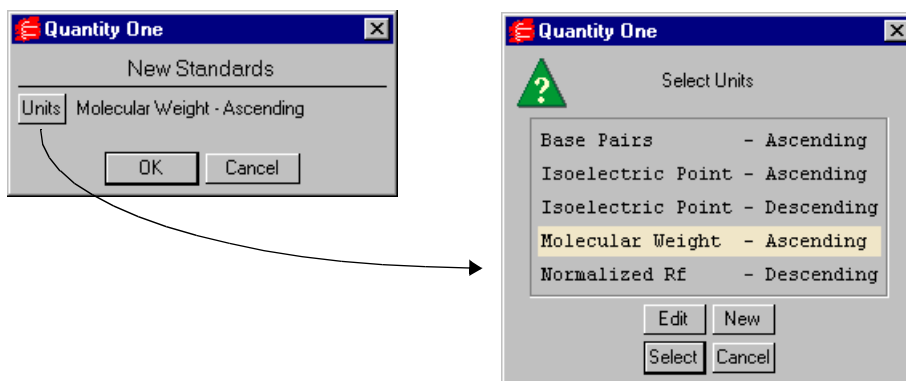


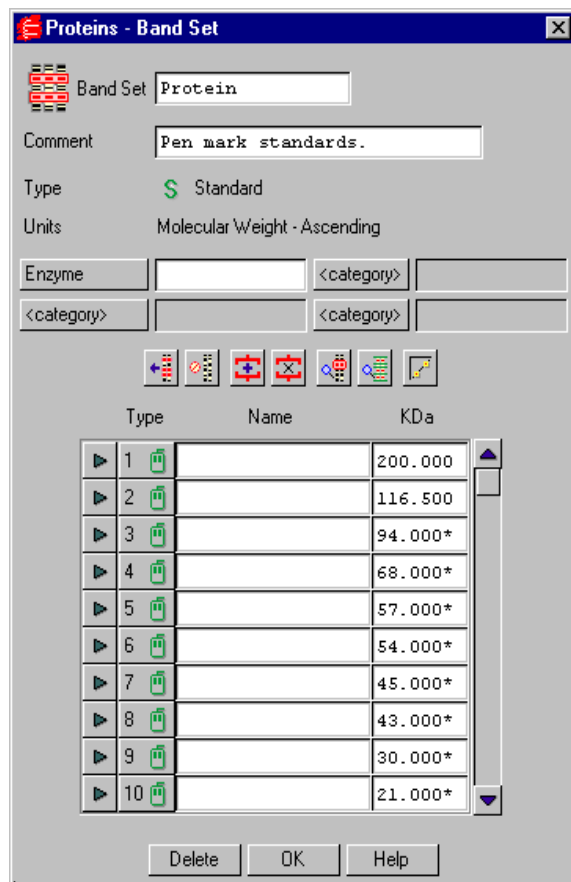
Fig.13-2. Specifying units.

Click on the Units button to specify your units. This opens a dialog box in which you can select from a list that includes Base Pairs, Isoelectric Point, Normalized Rf, and Molecular Weight. You can specify a set of units not on the list by clicking on the New button. If you have already defined a new set of units, you can edit them by clicking on Edit.

Note: “Ascending” means that bands of higher molecular weight or isoelectric point are at the top of the gel image, and bands of lower molecular weight or isoelectric point are at the bottom of the gel image.

After you select your units, the main Standards dialog box will appear. Here you can enter values for your standards, apply them to the bands on your image, and save them as a set for future use.

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The dialog box is titled "Proteins - Band Set". It contains the following fields and controls:

- Band Set:** A text field containing "Protein".
- Comment:** A text field containing "Pen mark standards."
- Type:** A dropdown menu showing "Standard" with a green "S" icon.
- Units:** A text field containing "Molecular Weight - Ascending".
- Enzyme:** A text field containing "<category>".
- Buttons:** A row of icons for adding, deleting, and applying standards.
- Table:** A table with 10 rows and 3 columns: Type, Name, and KDa. The KDa column contains values from 200.000 to 21.000*.
- Buttons:** "Delete", "OK", and "Help" buttons at the bottom.

Type	Name	KDa
▶ 1		200.000
▶ 2		116.500
▶ 3		94.000*
▶ 4		68.000*
▶ 5		57.000*
▶ 6		54.000*
▶ 7		45.000*
▶ 8		43.000*
▶ 9		30.000*
▶ 10		21.000*

Fig.13-3. Standards dialog box.

The Standards dialog box has fields for naming your standard band set, categorizing the standards, and entering values of your standard bands. It also has tools for applying the standards values to the lanes in your gel image.

Standards and Band Matching

Naming Your Standards

The Standards dialog box will open with a default name for your standards (e.g., Band Set 1). You can type a new name for the standards in the text box next to the Band Set prompt.

You can enter additional information about the standards in the text field next to the Comment prompt.

Categories and Attributes

The <category> buttons and fields can be used to cross-reference databased gel images using Diversity Database software. You can assign categories and attributes to gel images in Quantity One for future use in Diversity Database.

See Appendix C for more details

13.1.b. Entering and Applying Standard Values

Entering Standard Values

The next step is to enter values for your standard bands.

In the middle of the Standards dialog box is a table with a scroll bar on the right side. In this table, there are three columns labeled Type, Name, and the units you previously selected (e.g., KDa, pI, Rf).

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Type	Name	KDa
▶ 1		200.000
▶ 2		116.500
▶ 3		94.000*
▶ 4		68.000*
▶ 5		57.000*
▶ 6		54.000*
▶ 7		45.000*
▶ 8		43.000*
▶ 9		30.000*
▶ 10		21.000*

Fig.13-4. Entering the values of your standards.

In the “units” column, type a value for the first standard band and press RETURN. The cursor will skip to the field below, and you can enter a value for the second standard band. Repeat this process until all your standard values have been entered.

If you accidentally skip a standard value, enter it in the next available row and press RETURN. The values will sort themselves into descending order.

You can enter a name for each standard band in the Name column. This will appear in subsequent reports and print-outs.

To remove a standard value from the file, click on the arrow button under the Type header. A pop-up box will appear. Select Delete Band Type to remove the standard value from the template. The remaining standards will be renumbered.

Applying Standard Values to Lanes

The values you have entered in your Standards dialog box can now be applied to a lane or lanes in your gel image. This step will also automatically calculate the values of bands in your nonstandard lanes, based on the standards.

Standards and Band Matching

Note: The values in nonstandard lanes are calculated based on their proximity to the standard lanes that you assign. Generally, the greater the number of evenly spaced standard lanes in a gel image, the higher the accuracy of those calculations. We recommend a minimum of two standard lanes per gel.

The Apply Band Set button (in the Standards dialog box) applies all the standard values listed in the dialog box to the standard lane(s) in your image. Click on the button, then click on a lane.



Fig.13-5. Apply/Clear Band Set buttons in Standards dialog box.

The band values you have entered will be applied to the lane you select in the order in which they appear in the dialog box.

You can also click on the numbered button next to a band value to apply that value to a particular band in a lane on your image. Click on the button, then click on the band to be assigned the standard value.

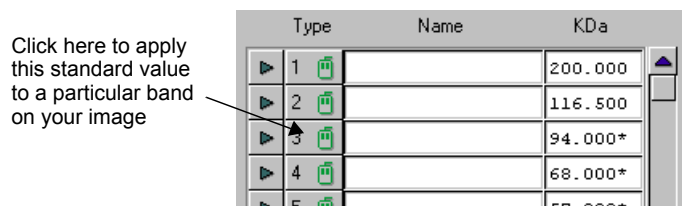


Fig.13-6. Applying standard values.

The subsequent bands in the lane will be numbered sequentially based on the initial assignment.

After you have applied your standard values to a lane, the bands in that lane will change color to green, indicating that they are now standards. The values of bands in non-standard lanes will be calculated automatically based on your standard lane.

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To view the new values of all the bands in your gel image, select Band Attributes from the View menu, then select the value to be displayed.

Removing Standard Values from Lanes

The Clear Band Set button removes all the standard band set values from the lane(s) you select. Click on the button, then click on the lane or lanes from which you want to delete the standard values.

13.1.c. Saving, Opening, and Deleting Standards

Values that you enter in Standards dialog box are automatically saved for future use. To use a standard band set on a different gel image, open the image, then choose Standards. A pop-up box will appear, allowing you to select an existing standard to apply to your image or create new standards.

To delete a standard band set, first open it, then click on the Delete button at the bottom of the dialog box. A pop-up box will ask you to confirm that you want to proceed with the deletion before completing the action.

13.1.d. Showing the Standard Curve

After you have defined your standards, you can display a regression curve based on those standards for any lane in your gel image.

Click on the Standard Curve button in the Standards dialog box, or select Standard Curve from the Match menu, then click on a lane to show the curve for that lane.

Standards and Band Matching

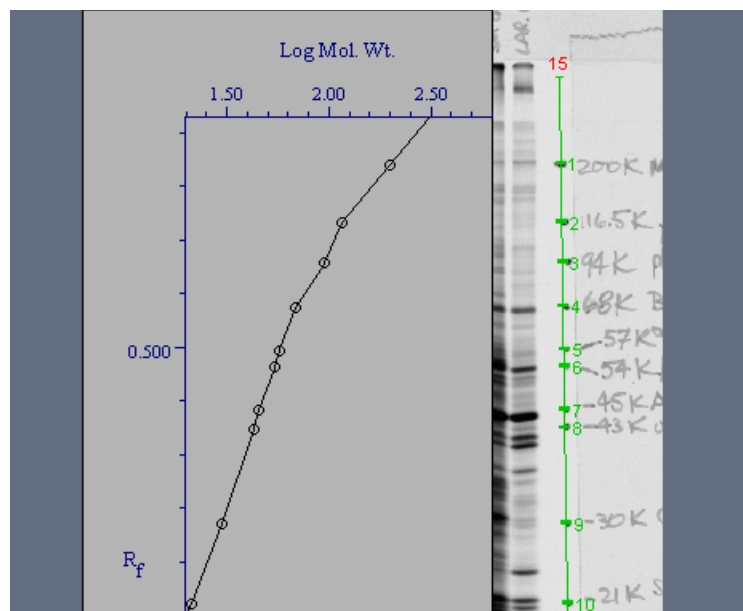


Fig.13-7. Example of the Standard Curve.

The X axis is the log of the standard value and the Y axis is the R_f value.

Clicking on a lane that includes your standards will display the curve with the standards marked as circles.

13.2. Matching

Once you have defined the standards on your gel image, you can use Quantity One to compare the bands in your experimental lanes and automatically match them with each other based on their position.

The functions you need to create, edit, and display sets of matched bands are found on the Match menu and Match Tools toolbar.

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13.2.a. Automatically Matching Bands

Select Match from the menu or toolbar, then click on the best experimental (i.e., non-standard) lane in your gel image to serve as the basis for automatically matching the rest of the bands.

Note: It is important that you pick a lane with most, if not all, of the bands you want to identify in your gel image, with minimal distortion and clean, easily identifiable bands.

This lane will be defined as the “First Lane” in your gel image. Each band in the lane will be designated as a different “band type,” and these will be used to match the bands in the other lanes. Matched bands must be in separate lanes (i.e., no two bands in the same lane can be the same band type).

The first time you click on an experimental lane with the Match command assigned, a pop-up box will ask you if you want to define a new band type for each band in the lane.

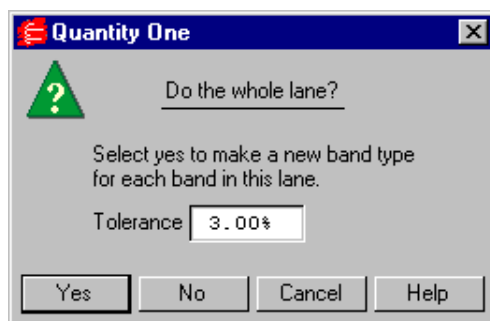


Fig.13-8. Query box: Apply matching to the whole lane?

Yes is the default selection and is the appropriate choice in most cases. Choosing Yes means that the software will use the lane you selected to automatically match all the bands in the gel image to the band types you initially identified.

Tolerance is the minimum spacing that the matching model expects to find between band types. It is expressed as a percent of lane height. You can enter

Standards and Band Matching

a number between 0.2 and 10. If your bands are very close together, use a Tolerance of 2% or less.

After you click on Yes, the bands in the lane you selected will change to green, indicating that they are known band types that have been identified by you. A band type number will appear next to each band.

Quantity One's automatic matching will correctly match most of the bands in your image to the known band types. Bands that the software can confidently match are labeled in red, with the number of the band type appearing next to each band. These matched bands are connected by modeling lines (to display these lines, see below).

Yellow bands are bands that the software cannot accurately match. The matching algorithm used by Quantity One is deliberately conservative to avoid incorrect labeling, so a number of yellow bands may appear on the image.

To summarize:

- Green bands are known band types, based on your identification.
- Red bands are bands that have been automatically matched with the known band types.
- Yellow bands are unknown and unclassified.

Displaying Band Types and Modeling Lines

To display the band type numbers, select Show Band Types from the menu or toolbar. This will display all the currently assigned band types.

Band type modeling lines reveal the path along the gel image that the software uses to match bands of the same type in different lanes. These lines are based on the identified bands and the matching tolerance.

To display the band type modeling lines, select Show Band Models from the menu or toolbar. The currently assigned band types will appear, as will the modeling lines connecting the band types.

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13.2.b. Manually Matching Bands

After you have matched the bands in your gel image automatically, you can “fine-tune” the results using the match tools. (These are on the Match menu and Match Tools toolbar.)

You can use these tools to manually change the band type of a band, create new band types, unmatch bands, or identify bands as outliers from your band set.

- Note:** Band types should only be added to areas of the gel image that are modeled well. The modeling lines are designed to give you guidance on adding new band types.
- To change a band to a particular band type, select Match and first click on the red or green band with the desired band type. This band type will be assigned to your mouse, and the status box for the cursor will change from Match to Match Band Type. Then click on the red or yellow band you want to change. The band will appear green (known) and the modeling line will change to reflect the new band type.
 - To change the status of a matched (red) band to known (green), select Match from the menu or toolbar and double-click on the red band.
 - To create a new band type from an unknown (yellow) band, select Match and click on the unknown band.
 - To create a new band type from a matched (red) band, select Match, hold down the SHIFT key, and click on the red band.
 - To change a band type to unknown (yellow), select Unmatch and click on that band.
 - To not include a red or green band in band type modeling, select Outlier and click on that band. An X will appear through the band, indicating that the software is ignoring it when modeling band types across the gel image.

13.2.c. Matched Band Set

The Matched Band Set dialog box displays the values of all the band types in your image. It has features that allow you to edit the results of your automatic matching and label and arrange all the band types in your gel image.

Standards and Band Matching

You can save a Matched Band Set under a user-defined name and apply it to lanes in other images.

Note: The Matched Band Set dialog box is similar in appearance to the Standards dialog box. However, the label next to the Type prompt will indicate that the band set type is Experimental (as opposed to Standard).

To display the Matched Band Set dialog box, select Matched Band Set from the Match menu or toolbar, then click anywhere on the image.

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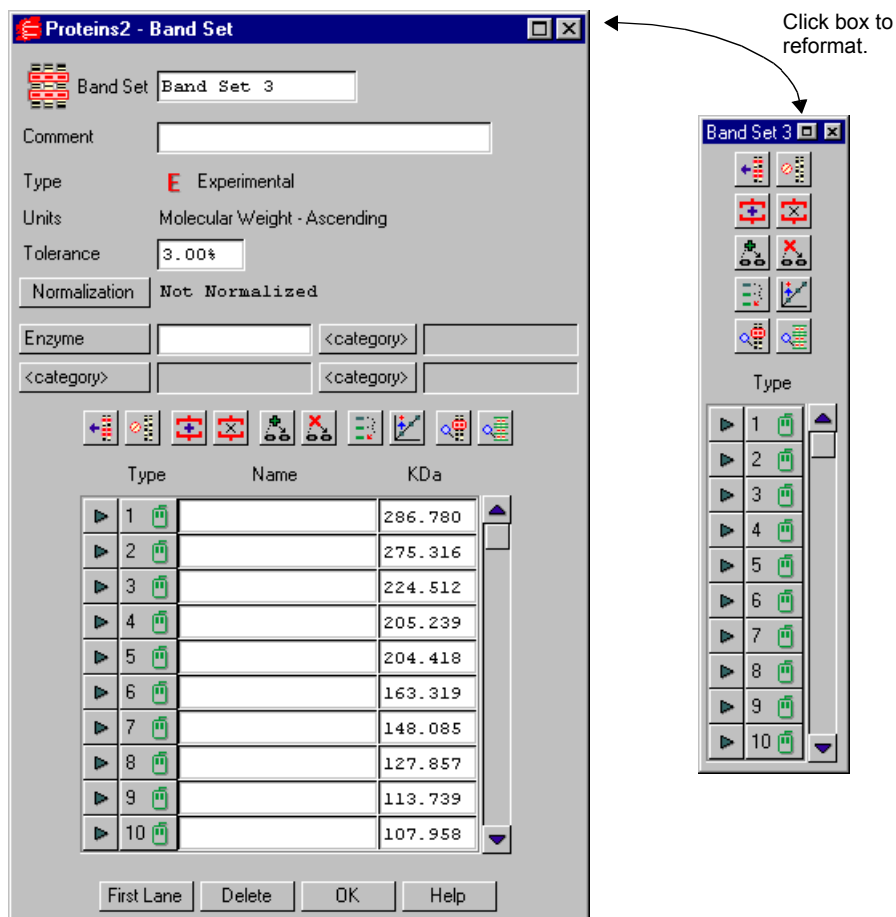


Fig.13-9. Matched Band Set dialog box.

The dialog box will open with a default name for the band set (e.g., Band Set 1). You can type a new name in the Band Set field at the top of the dialog, and add any comments or category / attribute information you want to associate with the band set.

Standards and Band Matching

The Normalization button allows you to pick a specific band type to normalize your other bands against (see section 16.1., Differential Display, for more information on band normalization).

Clicking on the resize button in the upper right corner will reconfigure the dialog box to its smaller, palette version, which displays only the tool buttons and band type buttons (see below).

To delete the entire Matched Band Set, click on the Delete button.

To save the Matched Band Set, click on OK.

List of Band Types

The values of the individual band types are listed within the dialog box in a table format. This is similar to the table of standard values in the Standards dialog box.

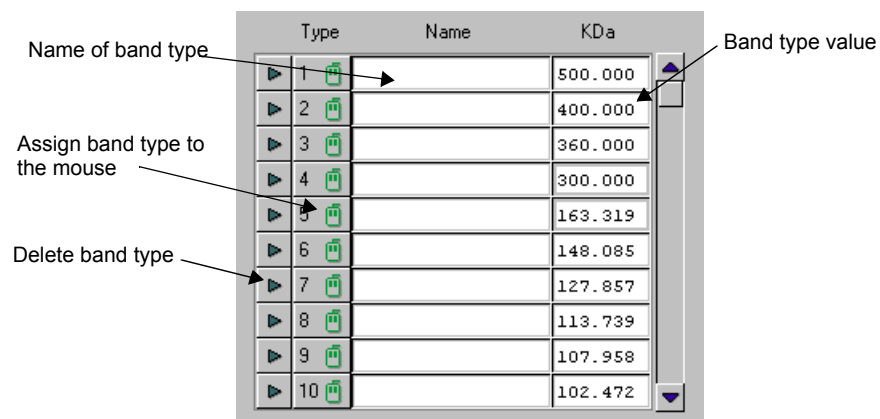


Fig.13-10. Applying and editing band type values.

If you change any of the band type values in the list, that will be reflected in the band type modeling on your image.

You can assign a particular band type to a band on your image. Click on the numbered button in the Type column to assign that band type to the mouse,

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then click on a band on your image to change that band to the assigned band type.

You can enter names for your band types in the Name column. These will appear in subsequent reports and print-outs.

To remove a band type from the set, click on the arrow icon associated with the particular band type. A pop-up box will ask for confirmation, and the remaining band types will be renumbered.

Matched Band Set Tool Palette

The Matched Band Set dialog has a tool palette that includes some of the tools on the Band Tools and Match Tools toolbars. It also includes tools that are specific to the dialog box.



Fig.13-11. Matched Band Set tool palette.

Apply Band Set applies the Matched Band Set to any lane in your gel image. Click on the Apply Band Set button, then click on the lane. The bands in that lane that can be matched will change to red.

Clear Band Set removes the Matched Band Set modeling from any lane in your gel image. Click on the Clear Band Set button, then click on the lane to be cleared.

Propagate Standards applies all the known band types to the bands in a lane, based on a few manually-assigned bands and the band set model. Click on this button, then click on the lane. The bands in that lane that can be matched will change to green to indicate their known status.

Standards and Band Matching

13.2.d. Displaying Graphs of Match Data

You can display graphs of different kinds of data associated with your matched bands. The commands for displaying these are located on the Match > Match Graphs submenu.

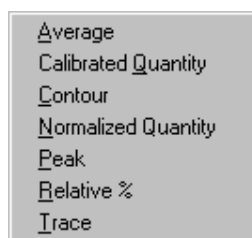


Fig.13-12. Match Graphs submenu.

From the Match Graphs submenu, select the match graph you want to display, then click on a matched band. The bands in the matched group will be displayed, as will the histogram that you chose.

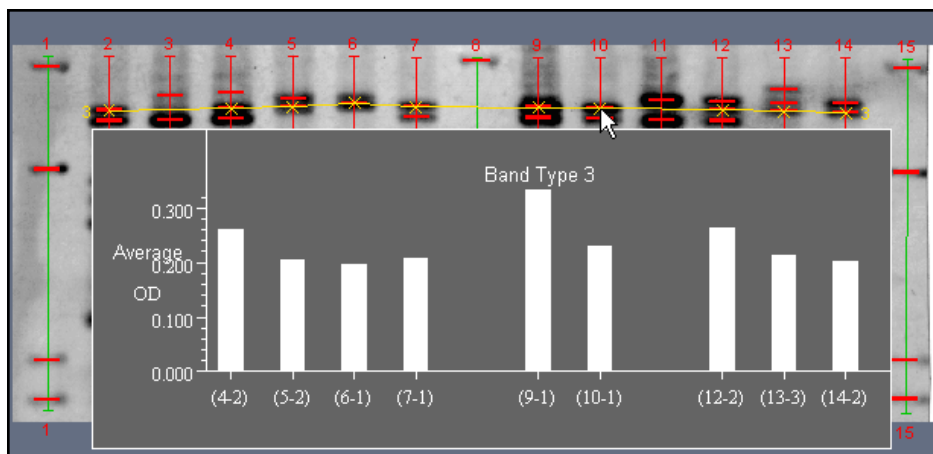


Fig.13-13. Example of a match graph.

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- **Average** displays a histogram of the average densities of the bands in a match group.
- **Calibrated Quantity** displays a histogram of the absolute quantities of matched bands based on a calibration curve (for more information on calculating absolute band quantities, refer to section 7.3).
- **Contour** displays a histogram of the density of each contoured band in a matched group. This function is only valid for contoured bands.
- **Normalized Quantity** displays a histogram of the normalized quantity of the bands in a match group.
- **Peak** displays a histogram of the peak densities of the bands in a match group.
- **Relative %** displays a histogram in which each bar represents the percentage that the bands in your match group represent out of either the total density of their respective lanes, or the sum density of the bands in their lanes (see section 2.5.c for more information on Relative % calculations).
- **Trace** displays a histogram of the integrated densities of the bracketed bands in a match group.

Each bar of the histogram will be labeled on the X axis with the lane number, and (where space permits) the band number. The Y axis is labeled with quantitative values.

The histogram display reserves space on the left side to label the Y axis and show its units. If the bands in the match group span the entire window from the left edge to the right, the histogram will not include bars for the bands in the left-most lanes because of the space required for labeling the axis.

To avoid this problem, decrease the magnification of the image using Zoom Out until there is blank space between the left side of the window and the first band of the match group.

Redisplay the histogram. If one or more bars are still not included, continue decreasing the magnification until you can see a bar lining up with each lane that contains a band in the match group in which you are interested.

Standards and Band Matching

13.3. Calculating Quantity Standards

You can identify standards and generate a calibration curve for determining the absolute quantities of all the bands in your lanes. (To quantitate bands outside of lanes, see Chapter 14, Volume Tools.)

Note: Relative quantity is derived from band intensity, as described in previous chapters. For example, from the band intensity, you can determine that there is twice as much of Band X in Lane 4 as there is in Lane 5. However, band intensity does not tell you how many micrograms there are of Band X in the gel. Absolute quantity is calculated based on standards and a calibration curve that you create.

Using the Quantity Standards function, you can identify standards of known quantity and use these to generate a calibration curve. You can apply this curve to unknown bands in the current image as well as other images.

To create a calibration curve, the absolute quantities of at least two bands must be known. The greater the number of known bands and the wider the range of their values, the more accurate the calibration curve will be.

13.3.a. Creating and Applying a Set of Quantity Standards

Choose Quantity Standards from the Match menu. A pop-up box will appear, asking you to create a new curve or load a previously defined calibration curve (if any has been saved).

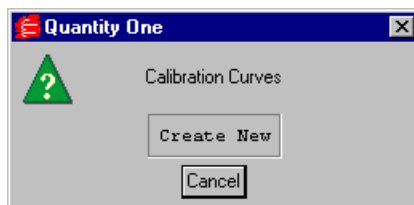
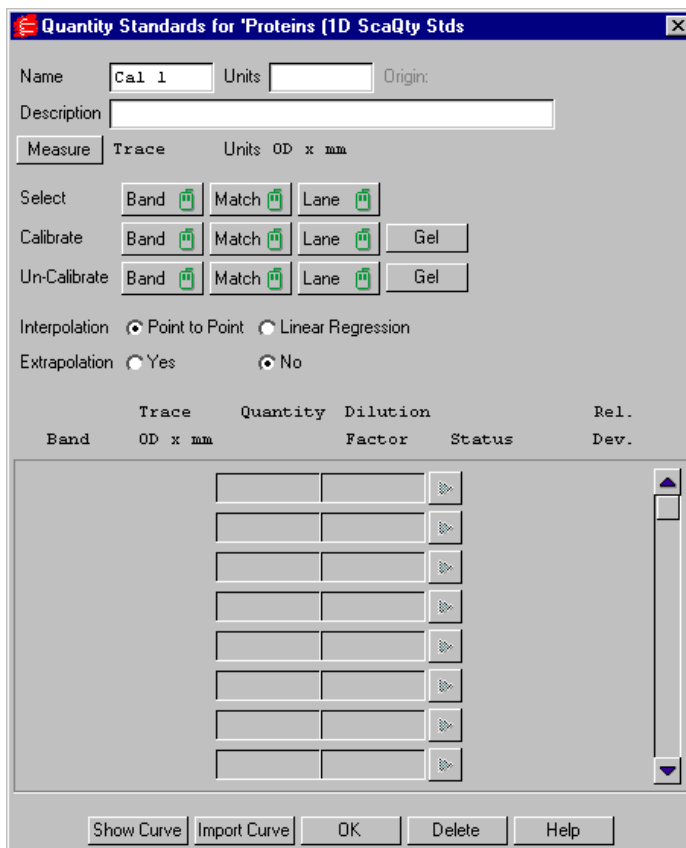


Fig.13-14. Loading a quantity calibration curve.

Selecting Create New will open the Quantity Standards dialog box.

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The dialog box is titled "Quantity Standards for 'Proteins (1D ScaQty Stds)". It contains several input fields and buttons. At the top, there are fields for "Name" (containing "Cal 1"), "Units", and "Origin:". Below these is a "Description" text area. A "Measure" button is located to the left of a "Trace" label. Below the "Trace" label is a "Units OD x mm" label. There are three rows of buttons: "Select" with "Band", "Match", and "Lane" buttons; "Calibrate" with "Band", "Match", "Lane", and "Gel" buttons; and "Un-Calibrate" with "Band", "Match", "Lane", and "Gel" buttons. Below these are radio buttons for "Interpolation" (Point to Point selected, Linear Regression) and "Extrapolation" (Yes, No selected). At the bottom, there is a table with columns: "Band", "Trace OD x mm", "Quantity", "Dilution Factor", "Status", and "Rel. Dev.". The table has 8 rows, each with input fields for the first five columns and a right-pointing arrow button in the "Status" column. At the very bottom are buttons for "Show Curve", "Import Curve", "OK", "Delete", and "Help".

Band	Trace OD x mm	Quantity	Dilution Factor	Status	Rel. Dev.

Fig.13-15. Quantity Standards dialog box.

The dialog box will open with a default name for the quantity standards (e.g., Cal 1). Enter a new name and specify the quantity value units (e.g., mg).

You can enter extra descriptive information next to the Description prompt.

The calibration curve is generated by plotting the known quantity values you provide against the intensity-based quantitation. There are several intensity-derived values that can be used for the Y axis of the plot. Click on the button labeled Measure to display a list from which to make your choice.

Standards and Band Matching

Selecting Bands of Known Quantity

The dialog box includes three buttons that are used to select bands on the gel image with which to generate the calibration curve. These buttons are located next to the Select prompt.



Fig.13-16. Buttons for selecting bands of known quantity.

- To select bands one at a time, click on the Band button, then click on each of the bands of known quantity.
- If all of the bands are part of a matched group (i.e., one band type), click on the Match button, then click on one of the bands in the matched group.
- If all of the bands are in one lane, click on the **Lane** button, then click on the lane with the known quantities.

After you have selected the bands, the bands will be highlighted.

In the lower part of the Quantity Standards dialog box, the lane and band numbers of the selected bands will appear in the Band column. The values derived from their signal intensities are also displayed.

Entering the Quantity Standards

After selecting the bands to be used as quantity standards, enter their quantities in the appropriate fields in the Quantity column.

After you have entered each quantity, the band status in the Status column changes from Unknown to Known.

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Band	Trace	Quantity		Dilution		Status	Rel. Dev.
	OD x mm			Factor			
6-12	4.042	4.04				Known	6%
10-20	3.869	3.87				Calc.	N.A.
1- 1	1.702	1.70				Known	1%
4-33	1.610	1.62				Known	1%
14- 6	0.4163					Unknown	N.A.
6-22	0.2879					Unknown	N.A.

Fig.13-17. Entering the known quantities.

After you have entered a few quantities, the status of the remaining bands in the list may change to O.R., meaning that the remaining bands are out of the current range of values (based on their intensity and what you have already entered).

Alternatively, the software may automatically calculate an unknown quantity (if it is between two known quantities) and enter a value for it; in this case the Status column will indicate that the quantity has been calculated (Calc.)

In either case, you can type a quantity directly into the Quantity column and the status will change to Known.

Relative Deviation of Known Quantities

After three values have been entered, relative deviations are automatically calculated and displayed in the Rel. Dev. column. These values are based on the difference between the known value that you entered and the values that would have been calculated from the calibration curve.

If the deviation value is too high, you may want to exclude a band from the calibration curve. Click on the arrow button next to the problem band. A pop-up box offers several different band status options from which to choose.

Standards and Band Matching

To remove a band from the quantity standards and omit it from calibration curve calculations, click on the Remove option. All the information regarding that band will be deleted from the quantity standards file.

Alternatively, you can select Outlier from the pop-up list. This retains the information about the band in the calibration file but does not include it when calculating the calibration curve.

13.3.b. The Calibration Curve

Calibration Curve Settings

In the Quantity Standards dialog box, select the method with which the calibration curve should be calculated. Next to the Interpolation prompt, choose either Point to Point or Linear Regression.

- **Point to Point** produces a curve in which each point is connected to the next, regardless of the shape of the resulting curve.
- **Linear Regression** (using the method of Least Squares) produces a curve which is the “best fit” of the values you provided.

Next, indicate whether the curve should be extrapolated. If you choose not to extrapolate, the highest and lowest values you entered will be the ends of the curve and you will be unable to calculate the estimated quantity for any bands that lie outside that range. If you want to extrapolate, you can calculate a quantity value for any band on the gel. But bands whose quantities are calculated from the extrapolated region of the curve may be less accurate.

Displaying the Calibration Curve

To display the calibration curve, click on the Show Curve button at the bottom of the Quantity Standards dialog box. A graph of known quantity versus measured density will be displayed in a separate window.

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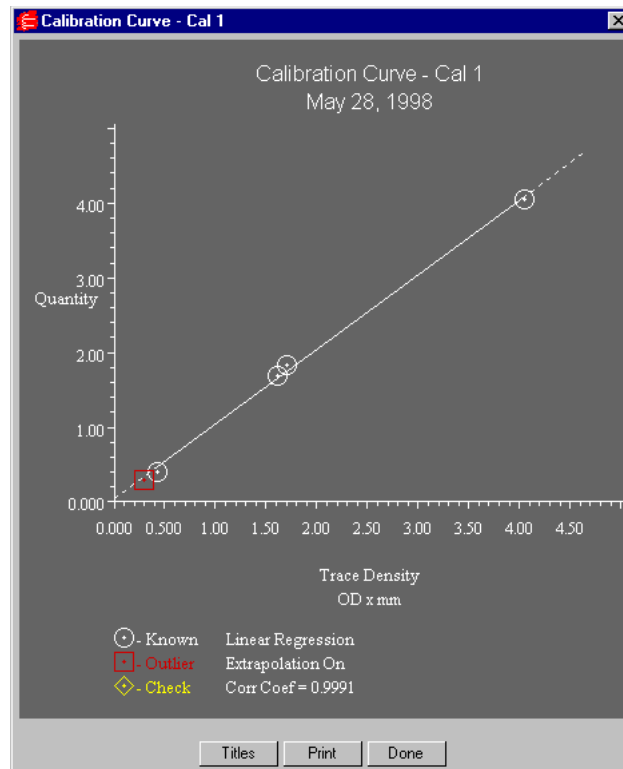


Fig.13-18. Calibration curve with titles displayed, linear regression, and extrapolation.

Points used to calculate the curve are enclosed in circles. Points that you identify as Outliers are enclosed in squares.

- To change the status of a point on the graph, click on it. The status will toggle between Known and Outlier.
- To see the legend for the different types of points on the graph, click on the Titles button.
- To print this graph, click on the Print button.
- To close the window displaying the graph, click on the Done button.

Standards and Band Matching

13.3.c. Applying the Calibration Curve to Bands on the Image

Once you have selected known bands and entered values for them, you are ready to choose the bands whose quantities you want to calculate with the calibration curve. This is done using the buttons found next to the Calibrate prompt.

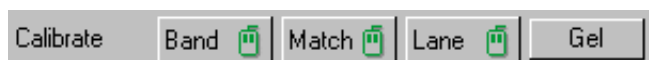


Fig.13-19. Buttons for calibrating bands of unknown quantity.

- To quantitate individual bands, click on Band, then click on the band of interest.
- To quantitate all the bands in a matched group, click on Match, then click on any of the bands in the matched group.
- To quantitate all the bands in a lane, click on Lane, then click on a lane of interest.
- To quantitate all the bands in the image, click on Gel.

Calibrated bands are highlighted after they are selected.

Unapplying the Calibration Curve

To undo the calibration of a band, match, lane, or gel, use the corresponding command next to the Un-Calibrate prompt.

13.3.d. Generating Standard Bands via a Dilution Series

One way to get a set of bands of known quantity is by starting with a stock solution and making several dilutions from it. A different dilution can be loaded onto each lane of a gel, resulting in a group of bands whose quantities can be calculated from the known quantity used to make up the stock solution and the dilution factor.

Quantity One will perform these calculations for you if you provide it with the known quantity and the dilution factor (e.g., for a dilution factor of 1 to 10, type 1 / 10 or 0.1).

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In the Quantity Standards dialog box, next to the band that came from the undiluted stock solution, enter the known quantity.

In the Dilution Factor column, type “stock.”

Next to each of the other bands in the dialog box, enter the dilution factor in the Dilution Factor column.

Trace		Quantity	Dilution	Rel.	
Band	OD x mm		Factor	Status	Dev.
10- 3	2.060	2.06	Stock	Known	6%
8- 5	0.7845	0.72	.35	Known	5%
11- 7	0.2401	0.21	.1	Known	27%

Fig.13-20. Entering a dilution series.

The quantity of each band will be automatically calculated.

13.3.e. Quantitating Bands Using Imported Calibration Curves

A calibration curve created on one gel image can be applied to bands on other images.

Make sure that both the image you want to quantitate and the image with the calibration curve are currently open.

Click on the image you want to quantitate. Select Match > Quantity Standards and click on “create new.”

In the Quantity Standards dialog box, click on the Import Curve button. From the list displayed, choose the calibration curve you want to use to quantitate bands on the image.

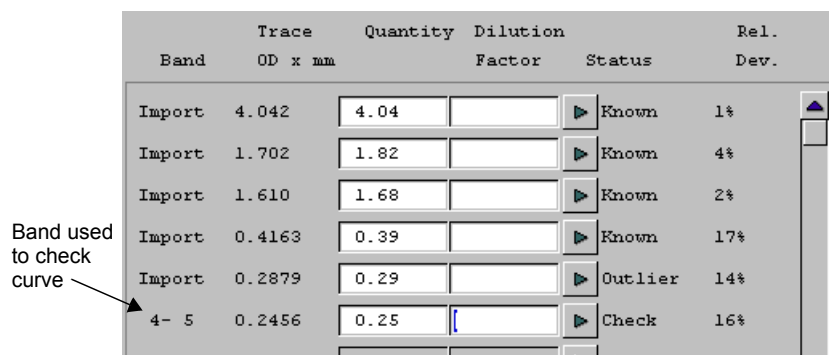
When you make your selection, the values associated with it will be displayed in the Quantity Standards dialog box. Each standard value will be labeled “Import” in the Band column.

Standards and Band Matching

Checking the Imported Curve

If the quantity of one or more bands in the image is known, you can use this information to double-check the imported calibration curve to ensure that it accurately quantitates the bands on this image.

To check these known bands against the calibration curve, go to the Select prompt, click on Band, then click on the known band. Its lane number, band number, and intensity will be displayed in one of the standard values fields.



	Trace	Quantity	Dilution		Rel.
Band	OD x mm		Factor	Status	Dev.
Import	4.042	4.04		Known	1%
Import	1.702	1.82		Known	4%
Import	1.610	1.68		Known	2%
Import	0.4163	0.39		Known	17%
Import	0.2879	0.29		Outlier	14%
4- 5	0.2456	0.25		Check	16%

Fig.13-21. Checking an imported calibration curve.

When you enter the band's value, its status will change to Check. This indicates that it is a check point and is not included in the calculation of the calibration curve.

If you display the graph by clicking on the Show Curve button, these check points are enclosed in diamonds. These check points should fall on or very near the calibration curve. If that is not the case, we recommend that you *do not use* the imported standards, since the selected calibration curve will probably not give you accurate quantity data for this gel image.

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14. Volume Tools

You can use the Volume Tools in Quantity One to quantitate and compare the intensity data in your images quickly and easily.

A volume is the data inside a defined boundary drawn on your image.

Volume = Sum of the intensities of the pixels within the volume
boundary x pixel area

Volume units = intensity units x mm²

Volumes are similar to contours (see section 12.7.a.), except that they are not dependent on lanes and bands.

To quantitate a particular object in your image, you first draw a box around that object using the Volume Tools. This creates a “volume object.” You can then classify your volume objects as unknowns or standards and compare them using the Volume Analysis Report and Volume Regression Curve features.

Note: The Volume Analysis Report and Volume Regression Curve are described in detail in sections 14.5 and 14.6, respectively.

To open the Volume Tools toolbar, select Analysis > Volume Tools, or click on the Volume Tools icon on your main toolbar. (The Volumes Quick Guide can be opened from the main toolbar or from the Help menu.)

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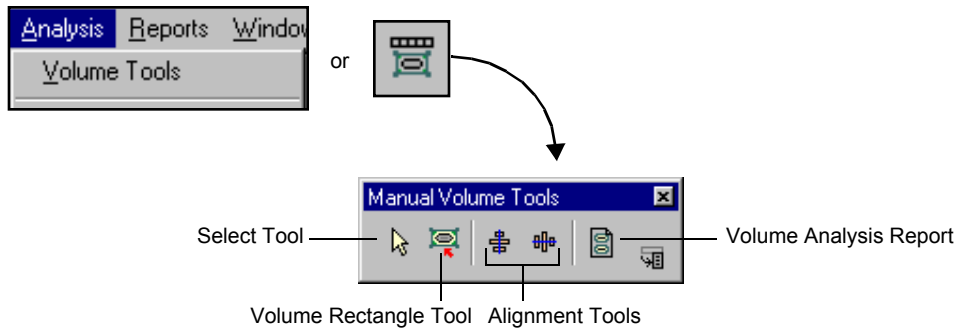


Fig.14-1. Volume Tools.

14.1. Creating a Volume Object

To create a volume object, click on the Volume Rectangle Tool button on the toolbar, then drag a box around the object to be quantitated.

The volume rectangle you draw should completely surround the data you want to quantitate. How big you make your box may depend on whether you are adjusting for background intensity (see section 14.3., Classifying a Volume Object, below).

You may want to experiment with several different rectangles drawn around the same object before selecting the volume that gives you the best quantity data.

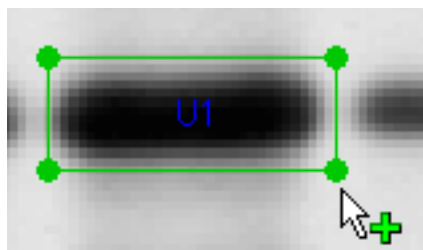


Fig.14-2. Creating a volume object.

Volume Tools

The box you draw will have a green border and anchor points (circles) at the corners. The green border indicates that the volume is selected.

If you click elsewhere on the image, the border will change to blue and the anchor points will disappear, indicating that the object is deselected.

If you move your cursor over the box—selected or unselected—the border will change to gold.

You can create as many volumes as you want.

14.2. Editing Volume Objects

Editing, Positioning, and Copying Individual Volume Objects

The following commands require that either the Select Tool or the Volume Rectangle Tool be assigned to your mouse. To assign one of these functions, click on the appropriate button on the Volumes Tools toolbar.

- To move a volume, position your cursor on it and drag.
- To resize a volume, click on it to select it, then position your cursor on one of the anchor points and drag.
- To copy a volume, position your cursor on it and hold down the **CONTROL** key while you drag.
- To rotate a volume, click on it to select it, then hold down the **SHIFT** key while dragging an anchor point. The volume object will pivot around its center. This is useful if the data object in your image is lying at an angle—for example, if your gel is smiling.
- To delete a volume, click on it to select it, then press the **DELETE** key.

Selecting and Editing Multiple Volumes

You can move, copy, or align multiple volumes. First you must first select them.

Click on the Select Tool button on your Volume Tools toolbar, then either drag a box around the objects you want to select or hold down the **SHIFT** key while you click on individual objects.

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Note: When dragging a box around objects to be selected, make sure that the box entirely surrounds all the objects.

Each object in the selected group will have a green border.

You can drag the selected objects to move them as a group, or click on the Alignment buttons on the Volumes Tools to align them.

Holding down the CONTROL key while you drag will copy all of the objects you have selected. Pressing the DELETE key will delete them.

14.3. Classifying a Volume Object

There are three types of volume objects: unknown, standard, and background. To classify a particular volume object, double-click on it with the Select Tool or Volume Rectangle Tool assigned to your mouse.

This will open the Volume Object Properties dialog box.

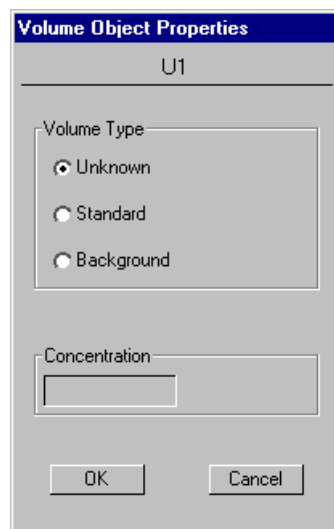


Fig.14-3. Volume Object Properties dialog box.

Volume Tools

Unknown

If you have drawn your volume around an object of unknown quantity, select the Unknown option button.

Known Standard

If you have drawn your volume around an object of known quantity, you can use it to calculate the quantities of your unknown volumes.

Select the Standard option button, then enter the quantity in the Concentration field. (Do not include units.)

After you have identified at least two standards, you can use the Volume Regression Curve feature under the Reports menu to calculate the quantities of your unknown volumes. The Volume Regression Curve and Volume Analysis Report are described in sections 17.6. and 17.5., respectively.

Background

When you draw a volume object, you will probably include some background pixels around the edge of your data. These background pixels will usually have an intensity value that you do not want to include in your quantitation. Quantity One has a mechanism for adjusting for this background intensity.

Note: You will have some residual background intensity even after you have performed background subtraction on your image. To delete background noise from your entire image, select Image > Subtract Background. To filter noise from your entire image, select Image > Filters > Filter Wizard.

To adjust for background intensity within your volumes, draw a volume object in a representative background region of your image (i.e., a region where there is no data and where the average pixel intensity appears to be the same as the background intensity surrounding your data).

Double-click on the object and select the Background option button in the Volume Object Properties dialog.

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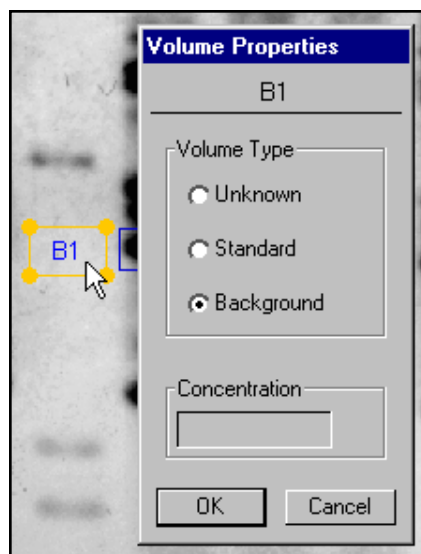


Fig.14-4. Defining a background volume object.

The average intensity value of the pixels in that background volume object will be calculated and subtracted from each pixel in your standard and unknown volumes. This will reduce the intensity of the pixels around the edge of your data in your volumes to close to zero.

The Volume Analysis Report will display both the unadjusted volume and the volume with background subtracted (adjusted volume) of your standards and unknowns, so you can see exactly how much intensity was subtracted. The Volume Analysis Report is described in section 17.5.

Note: If the region you identified as background has a higher average intensity value than your data object, then you will obtain a negative value for your adjusted volume in the Volume Analysis Report. If this happens, select a new background region that has less intensity than your data object.

15. Colony Counting

You can use Quantity One to automatically count the number of white, blue, or plaque colonies in a Petri dish image.

Note: For best results, when capturing the image of a Petri dish with an imaging device, the dish should fill the imaging window. Also, images with colonies should not have asymmetric pixels. (Asymmetric pixels can be generated by densitometers and the Reduce File Size command.) The colony counting function will not work properly on images with asymmetric pixels.

Select Colony Counting from the Analysis menu to open the Colony Counting dialog box.

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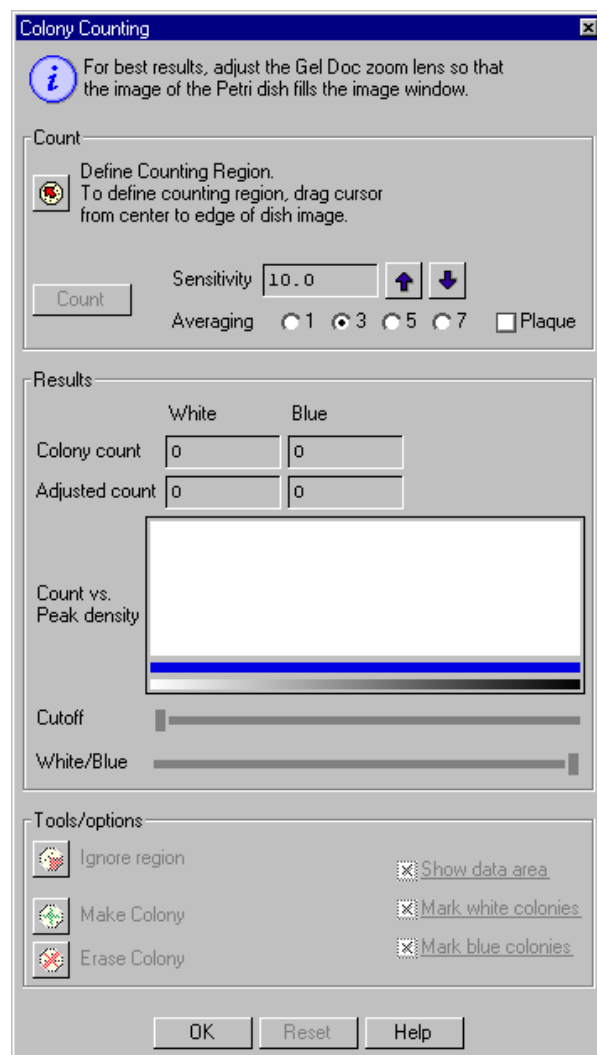


Fig.15-1. Colony Counting dialog box.

The Colony Counting dialog box has been arranged from top to bottom to guide you through the procedure.

Colony Counting

15.1. Defining the Counting Region

First, you must define the region you want to count in the Petri dish image.

Click on the Define Counting Region button in the dialog box and position your cursor at the center of dish image. Drag the cursor outward. As you drag, a blue circle will expand on the image—this defines the border of the counting region.

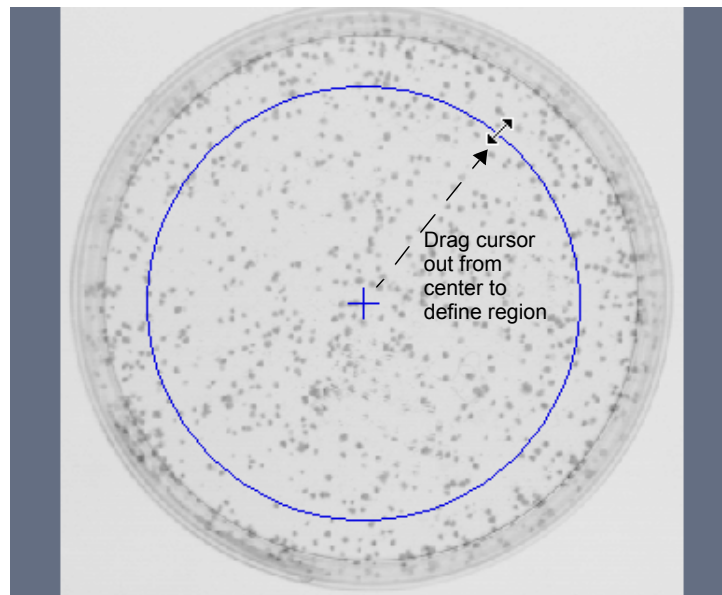


Fig.15-2. Defining a counting region.

If you make a mistake in defining your counting region, click on the Reset button at the bottom of the dialog box to start over.

Position the blue border until it is just inside the interior edge of the Petri dish.

Note: If your border disappears when you release the mouse button after dragging, check to make sure that the Show Data Area checkbox is checked. This checkbox is located at the bottom of the dialog box.

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If your circle is slightly off-center, you can reposition it by positioning your cursor on the center “target” of the circle. The cursor will change to a multidirectional arrow, and you can drag the entire circle.

To resize your counting region circle, position your cursor on the outer edge of the circle. The cursor will change to a bidirectional arrow and you can drag the border in or out.

15.2. Counting the Colonies

After you have positioned your circle, you are ready to automatically detect your colonies.

If you are counting plaques, click on the Plaque checkbox. (Because plaques appear as clear circles on a darker background, this checkbox must be selected for proper detection.)

Before counting, you may want to adjust the Sensitivity and Averaging parameters described below.

When you are ready to count, click on the Count button.

Sensitivity

The sensitivity setting determines the minimum signal intensity in the image that will be counted as a colony. The higher the sensitivity, the more colonies will be detected.

If the sensitivity is set too high, background noise will be erroneously detected as colonies. If the setting is too low, real colonies may be missed.

The default sensitivity setting is 10.00. If your image has faint colonies (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

Averaging

Averaging affects how well the software distinguishes individual colonies from each other as well as background noise. You should select the highest value that still results in good separation of colonies (default = 3).

Colony Counting

A low averaging value may result in background noise being detected as colonies. A high averaging value may result in two closely spaced colonies being counted as one.

15.3. Displaying the Results

After you click on Count, the number of detected colonies will appear in the Results section of the dialog box in the White column.

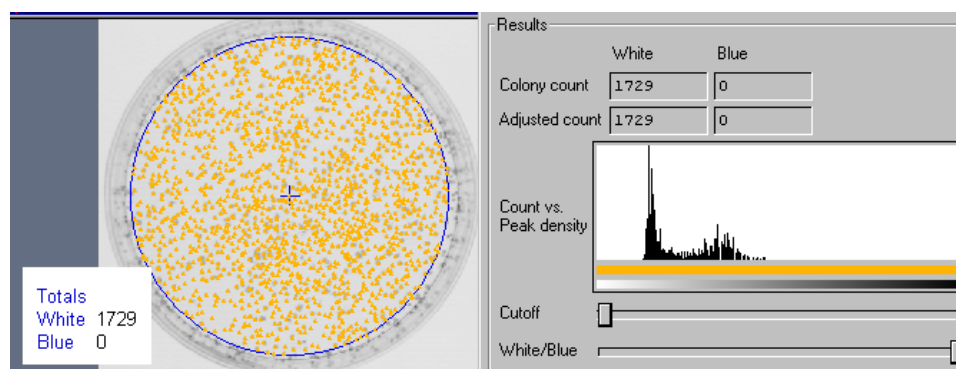


Fig.15-3. Example of a dish with white colonies.

The colonies will also appear marked as gold triangles on the image itself.

Note: If your colonies are not marked on the image, check to make sure that the Mark White Colonies checkbox at the bottom of the dialog is checked.

A text box on the image will indicate how many colonies were detected on the image.

Doing a Recount

If you want to recount using different parameters, simply change the Sensitivity and/or Averaging settings. This will erase your old count. Then click on Count again to recount.

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Redrawing the counting region circle or clicking on the Reset button will also erase the count.

15.4. Making and Erasing Individual Colonies

If automatic colony detection has missed or erroneously detected some colonies, you can manually mark or unmark them directly on the image using the buttons under Tools/Options.

To mark a colony, click on the Make Colony button, then click on the spot on the image that you want to identify as a colony.

To unmark a colony, click on the Erase Colony button, then click on the colony on the image that you want to unmark.

Your colony count will change accordingly.

15.5. Using the Histogram to Distinguish Colonies

The histogram in the Colony Counting dialog box is a graphical representation of the signal data in the image. You can use the histogram and associated sliders to reduce the number of incorrectly identified colonies and/or distinguish between white and blue colonies in the image.

Colonies Versus Background Noise

If there is a clear peak on the left end of the colony counting histogram, it is probably due to background noise in the image. (For information on subtracting background noise from entire images, see section 10.8.; for information on filtering noise from images, see section 10.9.)

If this background noise is being detected as colonies, you can use the histogram and Cutoff slider to correct this.

Drag the Cutoff slider to the right until it is centered on the right edge of the background peak.

Colony Counting

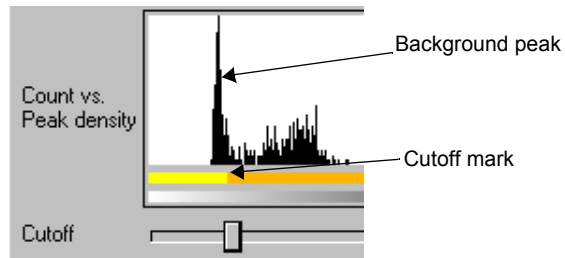


Fig.15-4. Using the Cutoff slider.

The yellow portion of the bar beneath the histogram marks the range of image data has been designated as background noise, and is not being considered for colony counting purposes. The gold portion of the bar marks white colony data range.

The colony count displayed in the dialog box and on the image should decrease. On the image, you should also see the incorrectly identified colonies disappear as you drag the slider.

White and Blue Colonies

If you know you have white and blue colonies in your image, and there are two clear peaks on the histogram to the right of the background peak, you can use the histogram to distinguish between these types of colonies.

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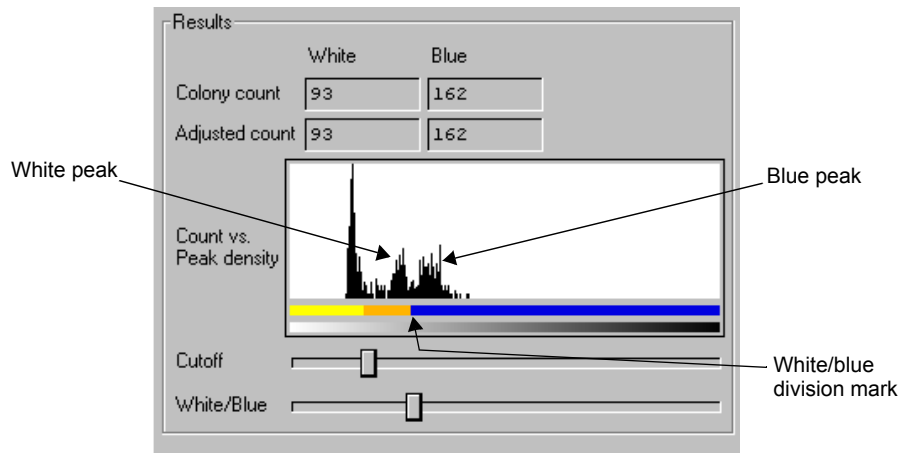


Fig.15-5. Using the White/Blue slider.

Drag the White/Blue slider to the left until it is positioned between the two peaks. The white colony data range is indicated by gold on the bar beneath the histogram, and the blue colony data range is marked with blue.

As you drag the slider, the numbers of white and blue colonies will change in the dialog box and in the text box on the image. Also on the image, you should see the marked white colonies (gold triangles) change to blue colonies (blue squares).

Note: If your blue colonies are not marked on the image, check to make sure that the Mark Blue Colonies checkbox at the bottom of the dialog is checked.

15.6. Ignoring a Region of the Dish

If a particular region of your Petri dish is damaged and you do not want to consider the colonies (if any) that appear there in your final count, you can use the Ignore Region function.

Colony Counting

Click on the Ignore Region button, then position your cursor on one edge of the region you want to ignore. Drag your cursor on the image, defining the full region you want to ignore.

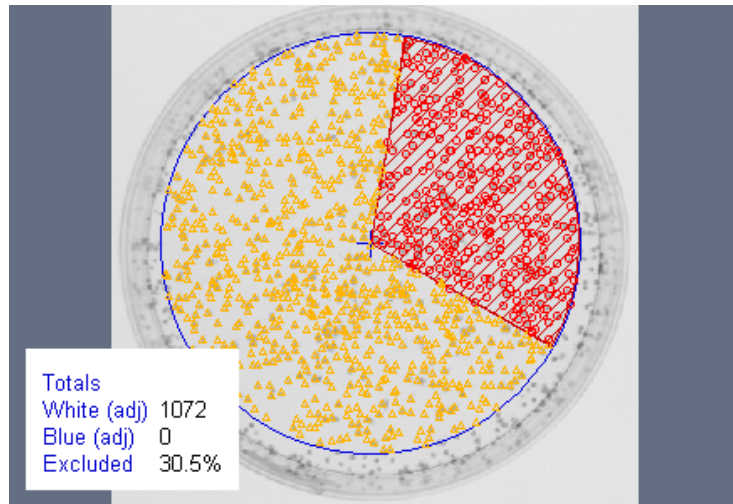


Fig.15-6. Marking a region to ignore.

As you drag, you will create a “pie slice” marked with red cross-hatching. Any colonies in this region will not be considered in the final count.

When you have defined the region, release the mouse button. If you want to change the size of the ignored region, position your cursor on the edge of the pie slice near the rim of the blue circle. Your cursor will change to a bidirectional arrow, and you can drag the edge of the pie slice.

Colony Count and Adjusted Count

After you have defined a region to ignore, two different counts will appear in the dialog box: the colony count and the adjusted count.

The colony count is the number of colonies that appear in the defined circle minus those in the ignored region.

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The adjusted count is an estimate of the total colony count in the Petri dish; it uses the known colonies to extrapolate the number of colonies that might have appeared in the ignored region if it had not been damaged. The adjusted count is calculated based on the area of the ignored region and the density distribution of colonies in the rest of the circle.

15.7. Saving/Resetting Your Count

To save the count you have performed on an image, click on the OK button in the dialog box. This will dismiss the dialog box and any image overlays. Then save the image as you normally would, using the save commands in the File menu.

To view your colonies and colony counting data again, open the image and open the Colony Counting dialog box.

The Reset button will clear the Colony Counting dialog box and any changes you have made to the image. This command cannot be undone.

16. Differential Display and VNTRs

This chapter describes the tools in Quantity One for studying sequence expression in PCR gels and counting VNTRs or other repeated elements in 1-D gels.

These analysis functions are located on the Analysis menu.

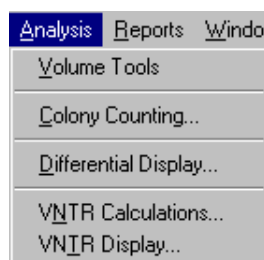


Fig.16-1. Analysis menu.

16.1. Differential Display

Differential Display is a technique using mRNA and PCR amplification to study gene expression.

Note: You must match the bands in your gel image before using this function. See section 13.2., Matching, for more information.

Differential display gels are analyzed just as any other gels in Quantity One would be, with one important distinction. Each gel image must be normalized for quantity to a specific band type that is present in all of your lanes. (A band type that is not present in a lane is equivalent to having a zero normalized quantity band of that type.)

16.1.a. Normalization

To assign a normalized band type, select Normalize from the Match menu and then click on the band you want to normalize against. That band type

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will be used as the normalizing type in all subsequent calculations and reports.

16.1.b. The Differential Display Dialog Box

After you have assigned a normalized band type, you can open the Differential Display dialog box

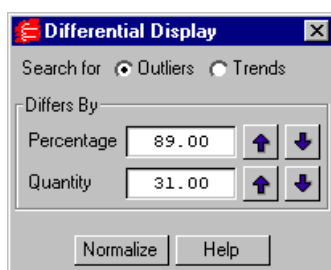


Fig.16-2. The Differential Display dialog box.

You can perform two types of searches on your gel image: an Outlier search or a Trend search. Both are determined by the Percentage and Quantity values that you specify in the Differs By field.

Select the Outlier button to search each band type for bands with normalized quantity that differ from the mean normalized quantity for its Band Type. An outlier is defined as a band with normalized quantity (n_q) that satisfies one of the following criteria:

$$\text{ratio} = 1.0 + (\text{percentage} / 100)$$

1. $n_q > \text{ratio} \times \text{mean}$ and $n_q > \text{mean} + \text{quantity}$
2. $n_q < (1 / \text{ratio}) \times \text{mean}$ and $n_q < \text{mean} - \text{difference}$

Select the Trend button to search for increasing or decreasing expression, represented by trends in normalized quantity across a Band Type. In Trend mode, a linear regression of (n_q) versus lane # is computed for each Band Type. The left most and right most lanes of a Band Set are determined. The

Differential Display and VNTRs

normalized quantities for these lanes are calculated from the regression model. If:

$\text{abs}(\text{left most_n}_q - \text{right most_n}_q) \geq \text{difference and}$
 $\text{Max}(\text{left most_n}_q, \text{right most_n}_q) / \text{Min}(\text{left most_n}_q, \text{right most_n}_q) \geq \text{ratio}$

then the Band Type is flagged as a trend.

Displaying the Results

Bands identified either as outliers or as belonging to a trend are displayed in white.

If a lane does not have a band assigned to a band type that is flagged as an outlier or a trend, an empty box will be drawn at the expected location of the band.

The Normalized Quantity command can be used to display a histogram of normalized quantities and the mean for any Band Type that you select.

16.2. Variable Number Tandem Repeats

If your experiments involve the use of microsatellites, VNTRs or other repeated elements, you can calculate the number of times a repeated element occurs in a band using Quantity One.

To calculate the VNTRs on your gel image, select VNTR Calculations from the Analysis menu. (You can also select the VNTR Quick Guide from the Help menu for guidance on analyzing gels with VNTRs.)

The Tandem Repeat Calculation dialog box will open.

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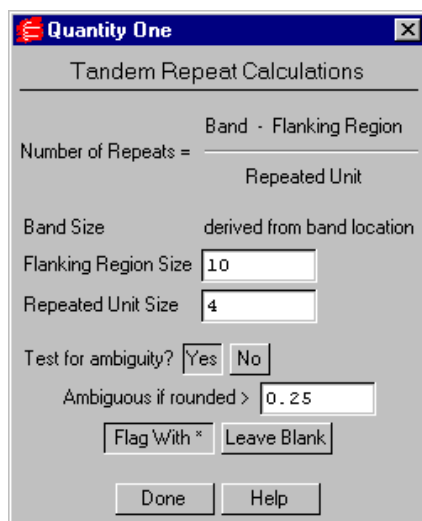


Fig.16-3. Tandem Repeat Calculations dialog box.

At the top of the dialog box is the equation used to calculate the number of times an element is repeated in a band.

The band size is determined by the position of the band on the gel image.

In the text box next to the Flanking Region Size prompt, enter the size in base pairs of the part of the fragment that does not include any repeated elements. This would include primer length and the length of any sequences that fall between the end of the primer region and the beginning of the stretch of repeats.

Differential Display and VNTRs

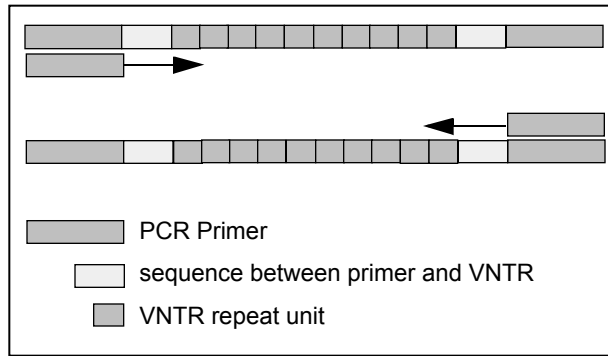


Fig.16-4. Diagram of components of a DNA fragment.

Next to the Repeated Unit Size prompt, enter the size in base pairs of the repeated element (e.g., if you are working with (CA) repeats, enter “2”).

Testing for Ambiguity

The number of repeats calculated by the software may not be a whole number, due to the limitations of exact band size determination. Since the number of times an element is repeated must be a whole number, the calculated value is rounded to the nearest whole number.

In the Tandem Repeat Calculations dialog box you have the option to “flag” those numbers that deviated from the nearest whole number. This flag serves to draw your attention to those values that may warrant further review.

Click on the Yes or No button next to the Test for ambiguity? prompt to indicate whether you want to test for ambiguity.

Next, specify what constitutes ambiguity. How much should the calculated value deviate from the nearest whole number for it to be flagged? Enter this value in the box next to the “Ambiguous if rounded >” prompt.

Finally, specify whether the tandem repeat numbers that meet the ambiguity criteria should be marked with an asterisk (*) or whether they should be left unmarked.

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When you have entered all the information in the dialog box, click on the Done button. The number of tandem repeats will be displayed next to the bands.

If the numbers have been concealed (e.g., by the Hide Overlays command), you can redisplay them by selecting VNTR Display.

17. Reports

Quantity One can display and print a variety of different analysis reports. You can format the reports to include different kinds of data.

The reporting features are located under the Reports menu.



Fig.17-1. Reports Menu.

17.1. The Report Window

All of the reports except Phylogenetic Tree share the same basic report window.

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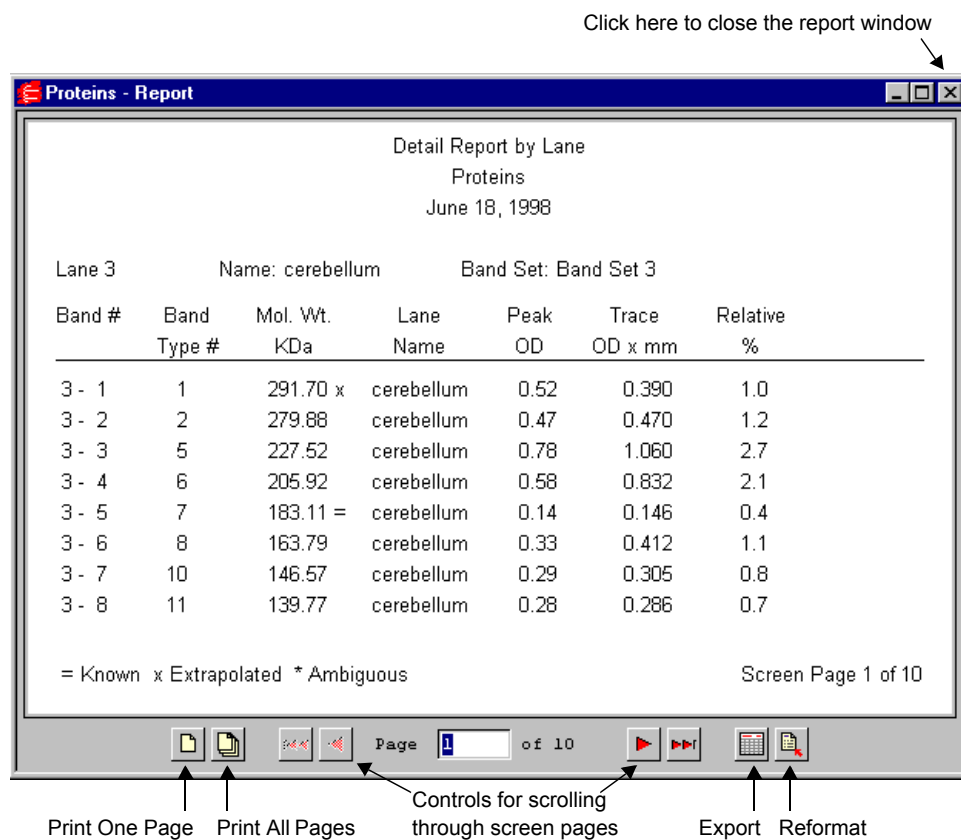


Fig.17-2. Example of a report window.

The standard report window has buttons for printing the report, scrolling through the screen pages of the report, and exporting the report to a spreadsheet application. Some report windows also have a Reformat button for displaying different report data.

To close a report window, click on the Close box in the upper right corner of the window.

Reports

Scrolling Through the Screen Pages

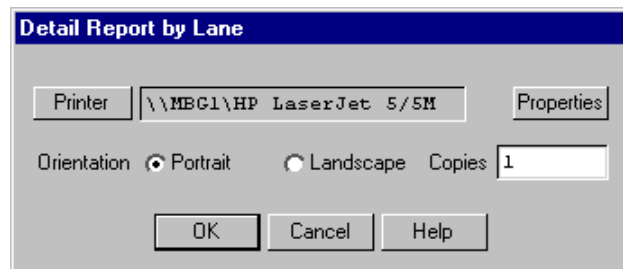
If your report has multiple screen pages, the scroll buttons in the report window will become active. You can use them to scroll to the next page, previous page, first page, or last page of the report. You can also enter a specific page number in the text field.

Printing Your Report

You can print your report using the Print One Page or Print All Pages commands in the report window.

Clicking on either of these buttons will open a smaller version of the standard print dialog (described in the next chapter).

Windows version:



Macintosh version:

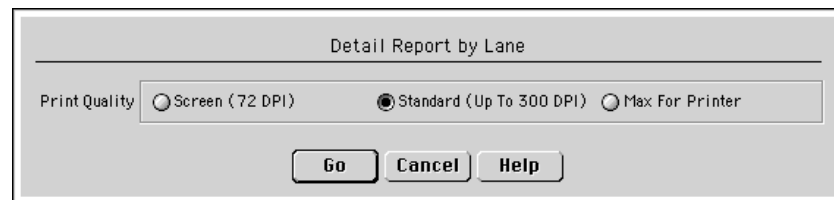


Fig.17-3. Print Report dialog box.

Print One Page sends only the current screen page to the printer. Print All Pages sends all the pages in the report to the printer.

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Note: If you select Print All Pages, the number of pages printed may be less than the number of screen pages listed in the report window. This is because Print All Pages reformats the data to make maximum use of the paper size.

Exporting Your Report Data

You can export your data to spreadsheet applications for further computation and analysis.

Click on the Export button to open the Export Report dialog box.



Fig.17-4. Export Report dialog box.

Different spreadsheet applications use different field separators when they import data. In the Export Report dialog, select Commas (for Lotus 1-2-3™) or Tabs (for Excel™) as your field separators.

You can save your data to a text file or to the clipboard by selecting the appropriate option button.

Click on the OK button to export the data.

Reports

Reformatting Your Data

If you change your mind about the data to display in your report window, click on the Reformat button (not available in all reports). This will reopen the options dialog box for your particular report.

17.2. Lane and Match Reports

There are four different lane and match reports: Lane Report, All Lanes Report, Match Report, and All Matches Report.

Lane Report generates a report on any one lane you select. First select Lane Report from the Reports menu, then click on the particular lane.

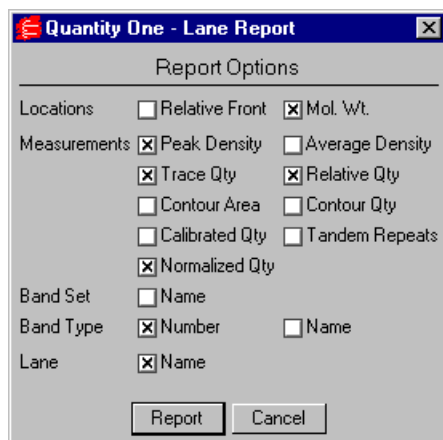
All Lanes Report generates a report on all the lanes in the current gel image.

Match Report generates a report on any band type that you select. First select Match Report from the menu, then click on a matched band.

All Matches Report generates a report on all the band types in the current gel image.

Selecting any of these commands will open the Report Options dialog box, where you can specify the type of data to be displayed in the report.

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The image shows a dialog box titled "Quantity One - Lane Report" with a close button (X) in the top right corner. Below the title bar is a section labeled "Report Options". The options are organized into several groups, each with a label and a list of checkboxes:

- Locations:** ☐ Relative Front, ☒ Mol. Wt.
- Measurements:** ☒ Peak Density, ☐ Average Density, ☒ Trace Qty, ☒ Relative Qty, ☐ Contour Area, ☐ Contour Qty, ☐ Calibrated Qty, ☐ Tandem Repeats, ☒ Normalized Qty
- Band Set:** ☐ Name
- Band Type:** ☒ Number, ☐ Name
- Lane:** ☒ Name

At the bottom of the dialog box are two buttons: "Report" and "Cancel".

Fig.17-5. Report Options form.

Select the data you want to display in the report by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

The report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

17.3. 1-D Analysis Report

The 1-D Analysis Report will display all the advanced analysis data (including band types, normalized quantities, amount of sample loaded, etc.) for all the lanes on your gel image. The lanes will also be ranked in similarity to the lane you initially select to generate the report.

Note: You must match the bands in your gel image before you can generate this report (see section 13.2., Matching, for more information).

Reports

Select 1-D Analysis Report from the Reports menu, then click on any experimental lane in your gel image. A dialog box will pop up, allowing you to select the report data to display.

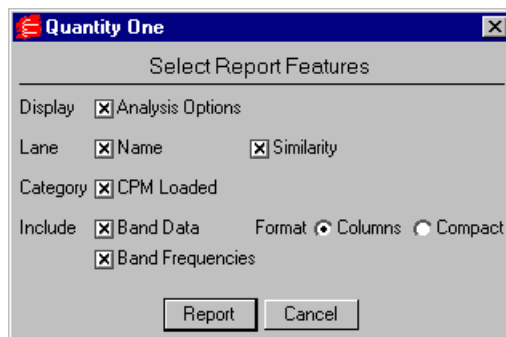


Fig.17-6. 1-D Analysis Report options.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

The report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

17.4. Similarity Comparison Reports

Quantity One has three reports that provide different ways of comparing the similarity of the lane-based samples in your gel image: Compare Lane Images, Phylogenetic Tree, and Similarity Matrix.

Note: You must match the bands in your gel image before you can generate these reports (see section 13.2., Matching, for more information).

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Comparison Options

Before opening any of these reports, select Comparison Options from the Reports menu to specify some settings for your reports.

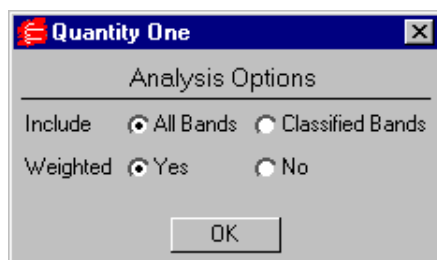


Fig.17-7. Comparison Options dialog box.

In the Comparison Options dialog box, select All Bands to include every band in your gel image, or selected Classified Bands to include only the matched bands in your gel image.

Band Weighting means that band intensity as well as position will be used when comparing the similarity of your samples. Select Yes to use band weighting when generating the report.

Select No to use only band position as the basis for comparing lane similarity.

17.4.a. Compare Lane Images

The Compare Lane Images command displays the lanes of your gel image in decreasing order of similarity to a lane that you select.

Select Compare Lane Images from the Reports menu, then click on the lane that you want to compare your other lanes to. A dialog box will pop up, in which you can select your report features.

Reports

:

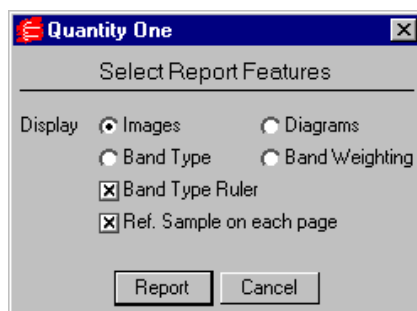


Fig.17-8. Compare Lane Images options.

If you select the Images option button, the lanes of your gel will be displayed as images. If you select Diagrams, schematic representations of the lanes will be displayed.

Select the data you want to display in the report by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

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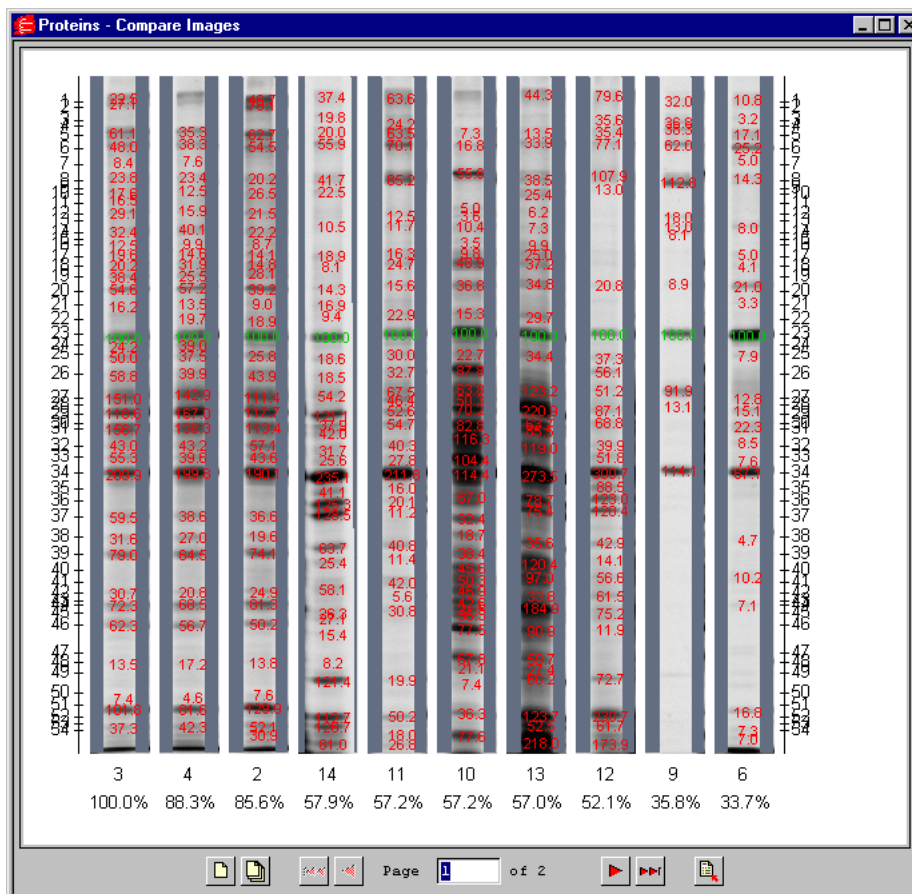


Fig.17-9. Compare Lane Images report window.

The Compare Lane Images report will be displayed in a standard report window (see section 14.1 above for information about report window options).

17.4.b. Phylogenetic Tree

Phylogenetic trees are schematic representations of lane similarity. To compare the similarity of your lanes in a phylogenetic tree format, select

Reports

Phylogenetic Tree from the Reports menu. (You can also use the Phylogenetic Tree Quick Guide under the Help menu to analyze your gel image.)

A dialog box will pop up displaying the different cluster methods for creating the tree.

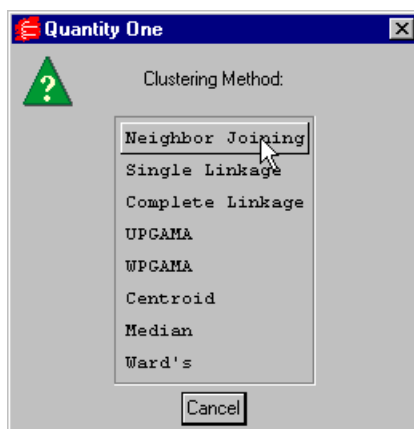


Fig.17-10. Selecting a Phylogenetic Tree clustering method.

Select a method to open the report window. See below for information about the different methods.

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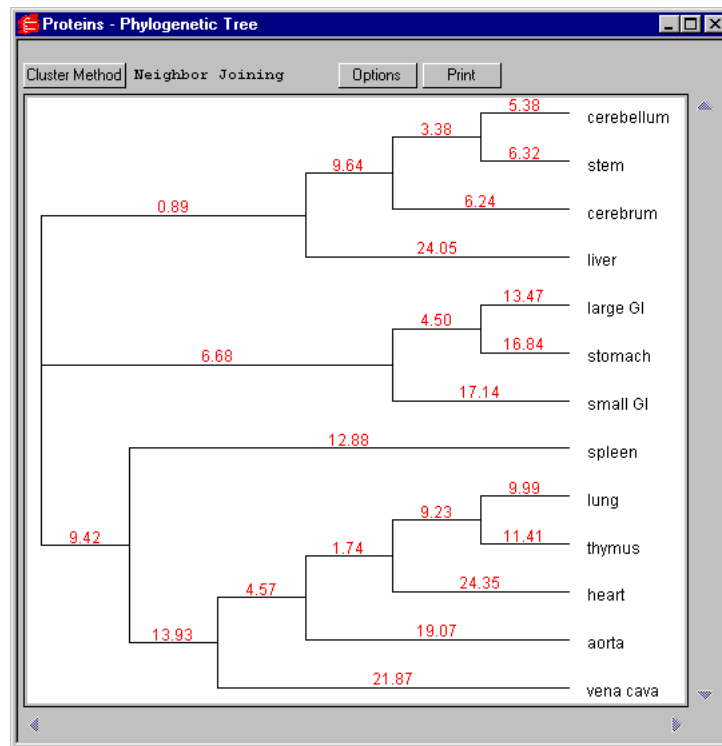


Fig.17-11. The Phylogenetic Tree report window.

The report window for phylogenetic trees is slightly different than the standard report window.

The tree appears in one window with scroll bars for moving up and down and left and right.

To print the tree, click on the Print button. This will open the standard Print Report dialog box.

To close the window, click on the Close box in the upper right corner of the window.

Reports

Clicking on the Cluster Method button allows you to select a different clustering method for displaying the tree. This will automatically reconfigure the tree displayed in the window.

Each method calculates the similarity of lanes in slightly different ways.

Neighbor Joining

This type of phylogenetic tree is computed is based on minimizing the total branch length at each stage of clustering. The method also finds branch lengths between nodes. The approximate distance between any two samples in this tree can be found by adding the branch lengths that connect the samples.

Other Methods^{1, 2}

The following methods are based on the algorithm below:

1. Begin with n clusters—one cluster for each sample.
2. Compute the similarity matrix for the samples.
3. Convert the similarity matrix into a distance matrix d using the appropriate distance formula.
4. Join the two clusters with the minimum distance into one cluster. Compute the similarity value for this cluster.
5. Recompute the distance matrix d using the cluster that was formed in Step 4.

Steps 4 and 5 are repeated until there is only one cluster. The difference between the methods below is based on the definition of minimum distance in Step 4, and on the method of computing the new distance matrix in step 5. In the discussion that follows, let:

1. For a complete explanation of the calculations and assumptions used to generate these dendrograms, please refer to Sneath and Sokal. *Numerical Taxonomy*, San Francisco: W. H. Freeman & Company, 1973.

2. Vogt and Nagel, *Clinical Chemistry* **38** (2): 182-198, (1992).

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p, q be indices indicating two clusters that are to be joined into a single cluster.

k be the index of the cluster formed by joining clusters p and q .

i be the index of any remaining clusters other than cluster p, q , or k .

n_p the number of samples in the p 'th cluster.

n_q the number of clusters in the q 'th cluster.

n the number of clusters in the k 'th cluster formed by joining the p 'th and q 'th cluster. $n = n_p + n_q$

d_{pq} the distance between cluster p and cluster q .

Single Linkage

This is also called Nearest Neighbor or Minimum Method.

$$d_{ki} = \min(d_{pi}, d_{qi})$$

Complete Linkage

This is also called the Furthest Neighbor or Maximum Method.

$$d_{ki} = \max(d_{pi}, d_{qi})$$

Single and Complete linkage are good algorithms for indicating outlier clusters.

UPGAMA

Unweighted pair group method using arithmetic averages. This is also called Weighted Average Linkage.

$$d_{ki} = \frac{\sum_{p \in n} n_p}{\sum n} d_{pi} + \frac{\sum_{q \in n} n_q}{\sum n} d_{qi}$$

Reports

WPGAMA

Weighted pair group method using arithmetic averages. This is also called Average Linkage.

$$d_{ki} = 0.5 \diamond d_{pi} + 0.5 \diamond d_{qi}$$

WPGAMA is a special case of UPGAMA that favors the most recent member clusters in forming new clusters.

Centroid

$$d_{ki} = \frac{\sum n_p}{\sum n} \diamond d_{pi} + \frac{\sum n_q}{\sum n} \diamond d_{qi} - \frac{(n_p \diamond n_q)}{n^2 d_{pq}}$$

Median

$$d_{ki} = 0.5 \diamond d_{pi} + 0.5 \diamond d_{qi} - 0.25 \diamond d_{pq}$$

Centroid and Median are similar to UPGAMA and WPGAMA, respectively, but the distance formula contains an additional third term. Centroid and Median methods are not monotonic hierarchical clustering algorithms. In other words, the similarity value between cluster k and any other cluster may be greater than the similarity between cluster p and cluster q. This condition occurs if the centroids (or medians) of the different clusters have approximately the same distance as the distances between the samples that make up the cluster.

Ward's

This method attempts to minimize the information by describing a set of N samples using a fewer number of clusters.

$$d_{ki} = \frac{n_p + n_i}{n + n_i} d_{pi} + \frac{n_q + n_i}{n + n_i} d_{qi} - \frac{n_i}{n + n_i} d_{pq}$$

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From our experience, we have found that Ward's method, UPGAMA, and WPGAMA give the most plausible clusters and are affected the least by samples that are outliers.

Options

Clicking on the Options button in the Phylogenetic Tree window will open the Options dialog box.

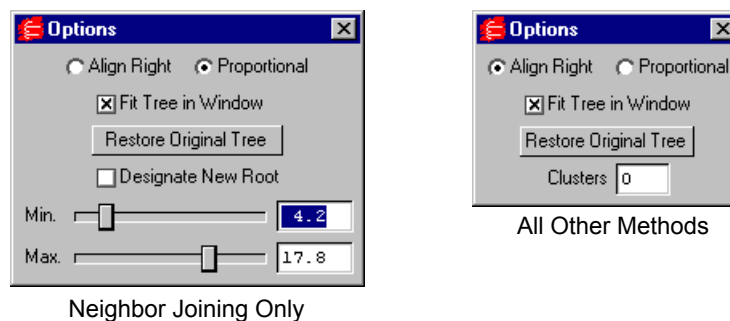


Fig.17-12. Two forms of the Options dialog box.

The box on the left has controls that are specific for the Neighbor Joining method. The box on the right is applicable to all other tree methods. Both boxes share certain features.

Each allows you to choose between aligning the tree to the right or proportionally aligning it. Right alignment uses a fixed branch length for displaying the distance between nodes; proportional alignment differs for Neighbor Joining and other tree modes.

In Neighbor Joining mode, proportional alignment shows branch length sizes proportional to the min./max. values (see below) set in the Options form. Other modes plot the nodes at locations determined by their similarity values.

The Fit Tree in Window option scales the tree so that the entire tree will fit onto a single printed page. With Fit Tree in Window turned off, the entire tree will appear with the correct distances between nodes preserved, and the printed tree can be tiled across multiple sheets of paper.

Reports

The Neighbor Joining Options form allows you to designate a new root node (the node at the very top of the tree), enabling you to look at the relationships between lanes from a different perspective. You do so by clicking on the Designate New Root button. This will highlight all the nodes of the tree on-screen. Choose the one you want to serve as the new root by clicking on it. The rearranged dendrogram will then be displayed. If you wish to return to the original form of the tree, click on the Restore Original Tree button.

The Min. and Max. sliders are active only for proportionally aligned Neighbor Joined trees. They allow you to adjust the Minimum and Maximum distance values between nodes on branches, so that you can adjust the display of your tree to highlight specific regions of data. Distances below the Min. or above the Max. values will be collapsed to fit on the form.

The Clusters option allows you to define the number of clusters (0 to 18) that will be identified in your tree for any non-Neighbor Joined tree. Entering a value greater than zero in this field will cause the tree to redisplay into separate clusters, identified by letters.

17.4.c. Similarity Matrix

The Similarity Matrix report displays the similarity values for all the lanes in your gel image. Select Similarity Matrix from the Reports menu.

The Similarity Matrix report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

17.5. Volume Analysis Report

The Volume Analysis Report can display all available information about your volumes.

To open this report, select Volume Analysis Report from the Reports menu or Volume Tools toolbar. The Volume Report Options dialog box will pop up, allowing you to specify the data that will appear in your report.

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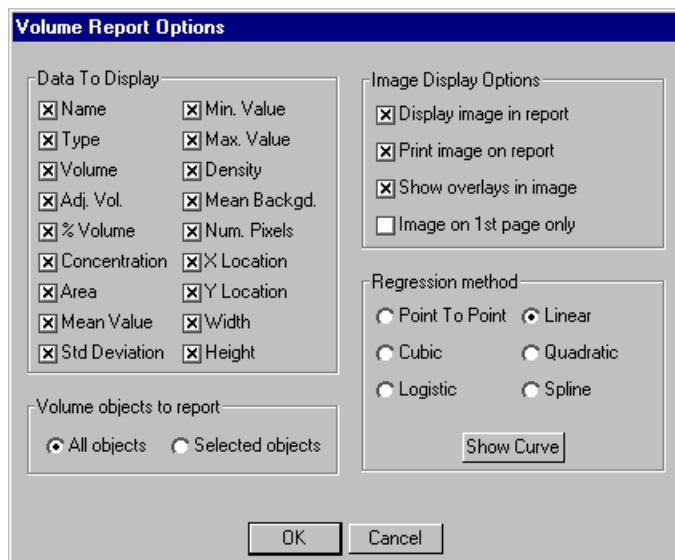


Fig.17-13. Volume Report Options dialog box.

Select the data you want to display and the image display option by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

In addition to the Data to Display and Image Display Options, you can choose whether to report on all your volume objects or only those objects you have selected using the Select tool. (See section 14.2., Editing Volume Objects, for information on selecting volumes.)

You can also select the regression method for displaying the Volume Regression Curve. See the following section for details. To display the curve, click on the Show Curve button.

When you have selected your report options, click on the OK button.

Reports

The report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

17.6. Volume Regression Curve

If you have defined at least two standard volumes on your image, you can display a regression curve for your volumes.

Select Volume Regression Curve from the Reports menu. This will open the Volume Regression Curve window.

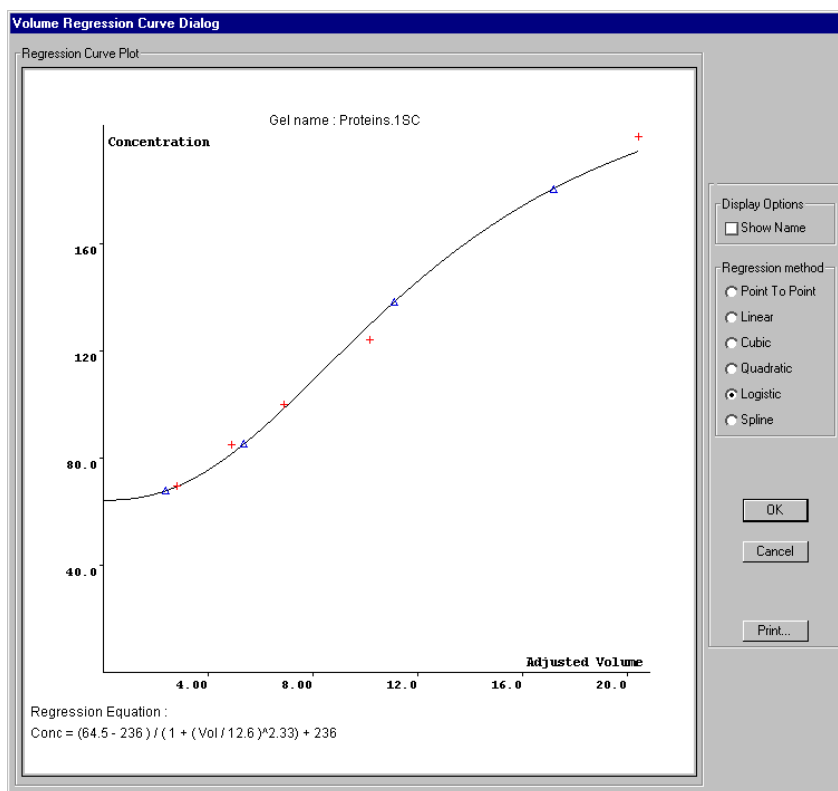


Fig.17-14. Volume Regression Curve window.

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Each standard volume is marked by a red X and each unknown is marked by a blue triangle.

The X axis is the adjusted volume and the Y axis is the concentration, based on the standards you have marked on the image.

To display the numbers and names (if any) of your volumes, click on the Show Name checkbox.

Select your preferred regression method from the list of option buttons. The regression equation for the selected method is displayed in the lower left of the window.

To print your curve, click on the Print button. This will open the standard Print Report dialog box.

To close the Volume Regression Curve window, click on the OK button.

17.7. VNTR Report

The VNTR Report displays your VNTR data.

Select VNTR Report from the Reports menu. A pop-up box will ask if you want to display the gel image in the report.

Reports

Tandem Repeat Report
Proteins: rat organs
June 18, 1998

Band	Mol. Wt. KDa	Raw Repeat #	Rounded Repeat #	Band Type Name
2-1	289	67.33	67*	
2-2	278	64.40	64*	
2-3	226	51.44	51*	
2-4	206	46.53	47*	
2-5	164	35.88	36	
2-6	148	32.02	32	
2-7	128	26.90	27	
2-8	114	23.40	23*	
2-9	108	21.99	22	
2-10	103	20.65	21*	
2-11	99	19.63	20*	
2-12	95	18.65	19*	
2-13	87	16.82	17	
2-14	81	15.23	15	
2-15	74	13.53	14*	
2-16	69	12.18	12	
2-17	64	11.00	11	
2-18	60	9.94	10	
2-19	56	8.92	9	
2-20	53	8.33	8*	

* Rounding changed the repeat number more than 0.25.

Screen Page 1 of 18

Fig.17-15. Example of a VNTR Report.

The report is displayed in a standard report window (see section 14.1 above for information about standard report window options).

The report displays the molecular weight, the raw repeat number, and the rounded repeat number for each band in the gel image.

The raw repeat number is the number of repeats calculated using the information that you provided in the Tandem Repeat Calculation dialog box, and is likely to include fractional values. The rounded repeat number is the raw repeat number rounded to the nearest whole number.

An asterisk (*) will appear next to some of the rounded repeat numbers if you selected Test for Ambiguity and Flag with * in the Tandem Repeat Calculations dialog box. The asterisk will appear next to those numbers that

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vary from the raw repeat number by more than the ambiguity value that you specified.

18. Printing and Exporting

The commands for printing and exporting images are located on the File menu.

Analysis reports (Volumes, VNTR, Phylogenetic Tree, etc.) are printed from within the individual report windows.

18.1. Printing

Quantity One offers four different printing options under File > Print:

1. Print Image prints a copy of the image and any overlays that are currently displayed.
2. Print Actual Size prints an actual-size copy of the image.
3. Scan Report prints the image and information about its scan history, number of pixels, data range, etc.
4. Video Print prints images and reports to a video printer.*

*Video printing requires installation of the video board and cable that come with the Gel Doc 1000/2000 gel documentation system. The video board and cable can also be ordered separately.

Macintosh Only

On the Macintosh, specify your printer settings—including paper size, page orientation, etc.—using the File > Print > Page Setup command. This will open the standard Macintosh Page Setup dialog box.

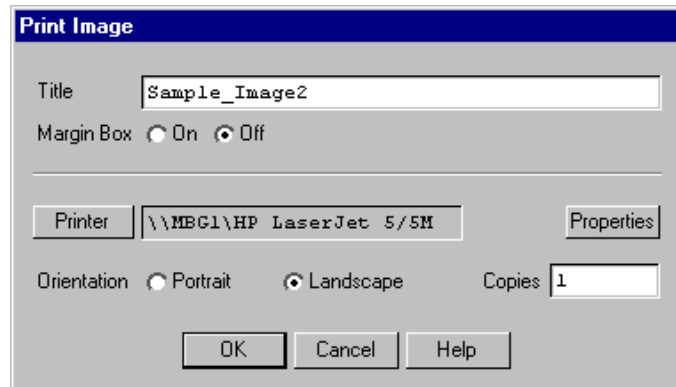
18.1.a. Printing Images

Print Image prints a copy of the active image window and any image overlays that are displayed.

File > Print > Print Image opens the Print Image dialog box.

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Windows version:



Macintosh version:

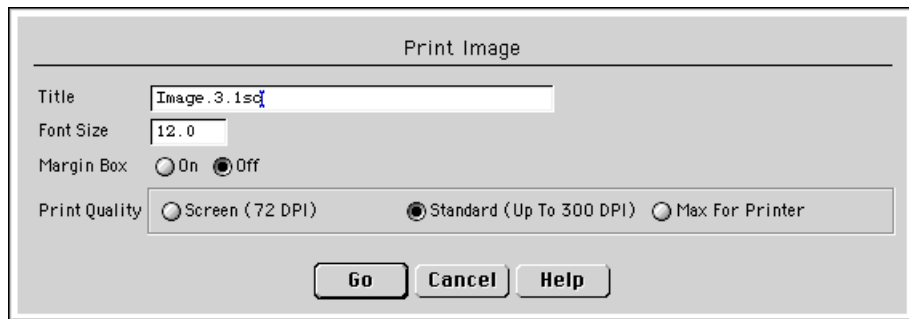


Fig.18-1. Print Image dialog box.

At the top of the dialog box, enter the title that you want to appear above the image when you print.

Choose whether or not a margin border should appear around the edge of the image by clicking either the On or Off button next to the Margin Box prompt.

Print Settings: Windows Only

Click on Printer to display a list of your available printers. After you have selected a printer, you can click on Properties to select the standard Windows print settings for that printer.

Printing and Exporting

Note: If you are using Windows 95 with certain printer drivers, you may experience difficulty printing some overlays and overlay symbols on images. While there is no single solution to this problem, you should try adjusting some settings in the Printer Properties dialog box, under the Graphics and Advanced tabs—specifically, the Dithering, DPI, Rasterizing, and RET settings.

Select the paper orientation by clicking on Portrait or Landscape next to the Layout prompt. (This can also be set in the Properties dialog box.)

Indicate how many copies should be printed in the text box next to the Copies prompt.

When you are satisfied with all the print parameters, click on the OK button to send the image to the printer.

Print Settings: Macintosh Only

Specify a font size for the image title in the text box next to the Font Size prompt. You can use font sizes ranging from 6 to 64, but we recommend you use the default setting of 12.0.

Select the printing resolution by clicking on one of the buttons next to the Print Quality prompt.

- Screen produces a printed image at the resolution of your computer screen (typically 72 dpi).
- Standard produces a printed image at the standard resolution of a typical laser jet printer (300 dpi, unless your printer prints at a lower resolution).
- Max for Printer prints at the maximum resolution of your printer.

When you are satisfied with all the print parameters, click on the Go button to send the image to the printer.

18.1.b. Print Actual Size

You can print your images at their actual size using the Print Actual Size command.

Note: If you are using the Gel Doc or the Fluor-S, you must specify the correct image area size when capturing your images to ensure accurate 1:1

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printing. You can specify the image area size in the acquisition window for the instrument. See the chapter on each imaging device for more information.

Select Print Actual Size from the File > Print menu. The standard Print Image dialog box will open (see previous section).

18.1.c. Printing the Scan Report

Scan Report allows you to print out a single-page report of an image and its associated information. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of an image for documentation purposes.

The report includes the following information:

- The image.
- The report date.
- The image title and a brief description.
- The directory location.
- The date the image was scanned.
- The type of imaging device.
- The imaging area and number of pixels.
- Pixel size.
- The intensity range, image color, and memory size.
- Image background information.
- Relevant lane and band information.

To print a scan report of a particular gel, select Scan Report from the File > Print submenu. This will open a smaller version of the standard print dialog box (see Print Image, above).

Note: TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the Scan Report may list this information as “Unknown” for imported TIFF files.

Printing and Exporting

18.1.d. Video Printing

The Video Print command allows you to print images and reports on a video printer. To create a video printout of the active window, select Video Print from the File > Print submenu.

Note: Video printing requires installation of the video board and cable that come with the Gel Doc 1000/2000 gel documentation system. The video board and cable can also be ordered separately.

Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W / P91W video printer. Set Contrast to zero, Brightness to zero, and Gamma to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

18.2. Exporting an Image in TIFF Format

To analyze your scanned image using Bio-Rad's Multi-Analyst or Molecular Analyst software, you must first save the image in TIFF format.

File > Export toTIFF Image opens a dialog box in which you can specify your export parameters.

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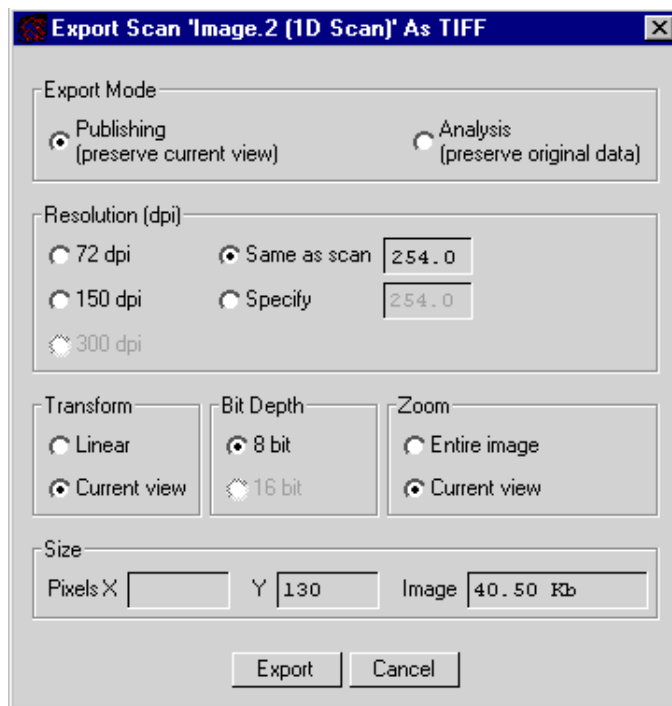


Fig.18-2. Export to TIFF Image dialog box.

Analysis Export Mode

Analysis mode exports the scan data, unmodified by Quantity One's viewing controls.

If you select Analysis mode, the other controls in this dialog box will become inactive.

Publishing Export Mode

Publishing mode exports a TIFF image that looks like the image as it is currently displayed in Quantity One (with whatever Transform adjustments you may have made).

Printing and Exporting

In Publishing mode, you can specify a resolution for the TIFF image by selecting 72 dpi (typical computer screen resolution), 150 or 300 dpi (standard printing resolutions), the same resolution as the scan file, or any resolution you want to specify (up to the resolution of the scan).

If you have log transformed the image, you can specify a linear transform, or preserve the current view.

If your image is 16 bits, you can compress it to 8 bits for export to TIFF.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the current view or export the entire image.

Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the Export button.

A Save As dialog box will open. The default file name will have a .tif extension, and the file type will indicate that this is a TIFF image. You can change the file name or select a different directory to save in.

Quantity One will generate a grayscale TIFF image compatible with Multi-Analyst and Molecular Analyst software.

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Appendix A. Keyboard Functions

Key	Normal	Shift	Command (Mac)	Control (PC)
Left	Pan Left			
Up	Pan Up			
Right	Pan Right			
Down	Pan Down			
F1	View Entire Image			Transform
F2	Zoom box	Grab		Zoom In
F3	Center image on cursor			
F4	Hide Overlays			
F5	Show Lanes and Bands	Show Band Types		Show Band Models
F6	Band Attributes			
F7				
F8	Gel Layout			
F12	Save	Save All		Save All
b			Transform	Transform
g			Grab	Grab
h			Help	Help
i			File Info	File Info
o			Open	Open
p			Print Window	Print Window
q			Exit	Exit
s			Save	Save
t			Plot Cross-Section	Plot Cross-Sect.ion
v			About Quantity One	About Quantity One
w			Close	

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Appendix B. Cross-Platform File Exchange

It is possible to move image data between Discovery Series software applications on different platforms. There are different protocols depending on the platform (PC or Macintosh) you are transferring from and to.

B.1. Macintosh to PC

To transfer a file from a Discovery Series application running on a Macintosh to a Discovery Series application running on a PC, you need to tag the file name with the suffix appropriate to the image file type (i.e., 1-D scan or DNA scan).

For 1-D gels (Diversity Database, Quantity One) use the suffix: .1sc

For database files (Diversity Database) use the suffix: .ddb

For DNA Scans (DNA Code) use the suffix: .dsc

B.2. PC to Macintosh

PC versions of Discovery Series files can be read directly by Macintosh applications, with no required modifications.

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Appendix C. Other Features

The following features are available in Quantity One, but have more utility in the databasing application Diversity Database. You can examine your gel images in Quantity One and then database them using Diversity Database.

C.3. Categories and Attributes

User-defined <category> buttons are available in the Standards dialog box, Matched Band Set dialog box, and Gel Layout dialog box. They allow you to categorize the characteristics of your particular gel or any related gel to which you might apply the same set of standards.

To define a new category, click on one of the <category> buttons. A dialog will pop up in which you can select from a list of categories or create a new one.

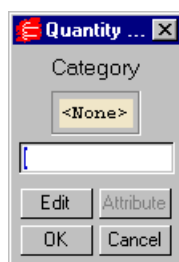


Fig.3-1. Category pop-up box.

To create a new category, click on the Edit button. This will open another pop-up box in which you can enter the name of your new category.

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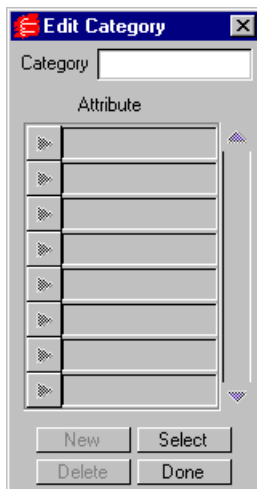


Fig.3-2. Edit Category dialog box.

4. Click **OK**

Type the name of the new category (e.g., “Color”) next to the Category prompt and list attributes of that category in the Attribute fields (e.g., “Red,” “Green,” “Blue,” etc.). The form will automatically sort your attributes alphabetically within the Attribute fields. Categories and attributes can be defined for any characteristics of your gel that would be useful to sort by. Typical categories might be “Enzyme,” “Primer,” “Probe,” “Type,” “Who,” etc.

Once you’ve created a category and attributes, you can use them in the dialog box. Once again, click on a <category> button.

Appendix C

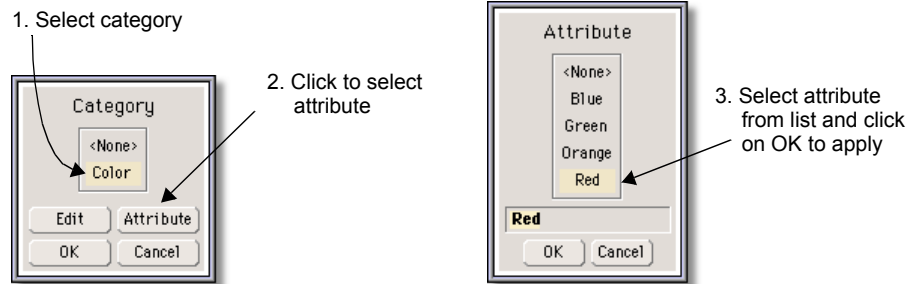


Fig.3-3. Selecting a category and attribute.

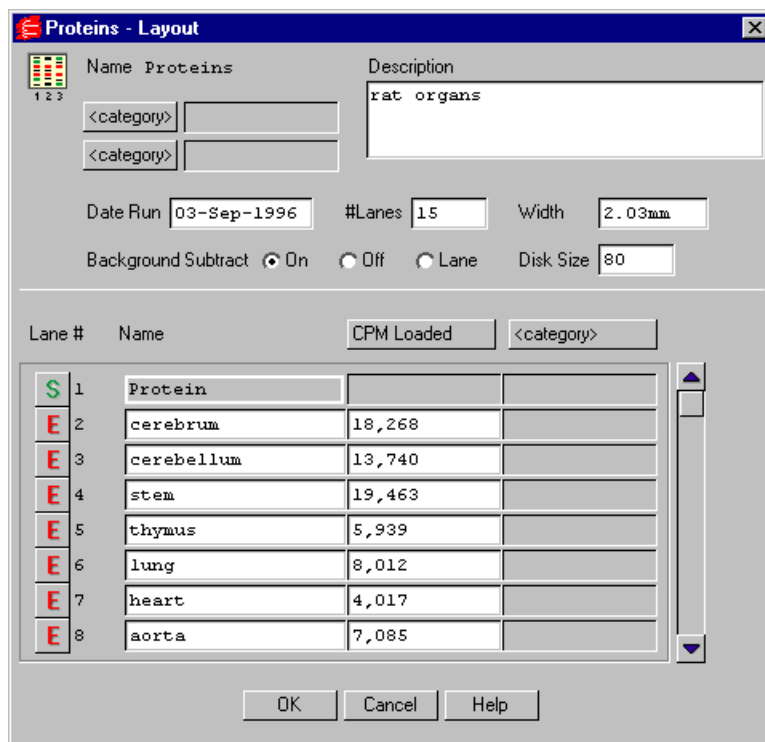
Select the category to be applied from the available categories on the list (or select <none>). Click on the Attribute button to specify an attribute. Click OK to apply your selection to the Standards box.

C.4. Gel Layout

In Diversity Database, you can use the Gel Layout dialog box to compare samples across multiple gel images. In Quantity One, you can use it to enter general information about your image.

To open the Gel Layout form, select Gel Layout from the Edit menu.

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The Proteins - Layout dialog box is used to configure gel layout parameters. It includes fields for Name Proteins, Description, Date Run, #Lanes, Width, Background Subtract, and Disk Size. A table lists lanes with their respective protein names and CPM values.

Lane #	Name	CPM Loaded	<category>
1	Protein		
2	cerebrum	18,268	
3	cerebellum	13,740	
4	stem	19,463	
5	thymus	5,939	
6	lung	8,012	
7	heart	4,017	
8	aorta	7,085	

Fig.3-4. The Gel Layout dialog box.

The following information about your gel can be specified at the top of the Gel Layout form:

- Category and attribute information (see above) for the whole gel.
- A brief description of the gel.
- Gel run date.
- Number of lanes.
- Lane width (see section 11.1.g.).
- Lane-based background subtraction and disk size (see section 11.2.).

Appendix C

Each lane in your gel has its own line on the Gel Layout form. The following information can be specified for a lane. (See section 13.2.c. for more information about band sets.)

- Band set type (standard or experimental).
- Lane number.
- Band set name.
- Sample name.
- Category / attributes information (see above) for individual lanes.

The pop-up buttons on the left side of the form offer a number of choices pertaining to the individual lanes on your gel. Clicking on one will open the Lane Choices list.

1. Click on a Band Set button...

	Lane #	Name
	1	Protein
	2	cerebrum
	3	cerebellum
	4	stem
	5	thymus
	6	lung
	7	heart
	8	aorta

2. The Lane Choices pop-up will appear.

Quantity One

Lane Choices:

- Lane Report
- Assign Band Set
- Un-Assign Band Set
- View Band Set

Cancel

Fig.3-5. Lane Choices pop-up list.

Choosing Lane Report will open a customizable lane report form that can be printed or exported (see section 17.2.).

Choosing Assign Band Set allows you to select the specific band set that is to be applied to that lane (see section 13.2.c.).

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Unassign Band Set allows you to remove the band set that is currently applied to that lane (see section 13.2.c.).

View Band Set displays the band set form for the band set that is currently applied to that lane (see section 13.2.c.).

View Sample Detail displays the sample detail form for the current lane (Diversity Database function).

View Sample opens an image window containing the current sample and displays it with the current band types overlaid on the image (Diversity Database function).

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